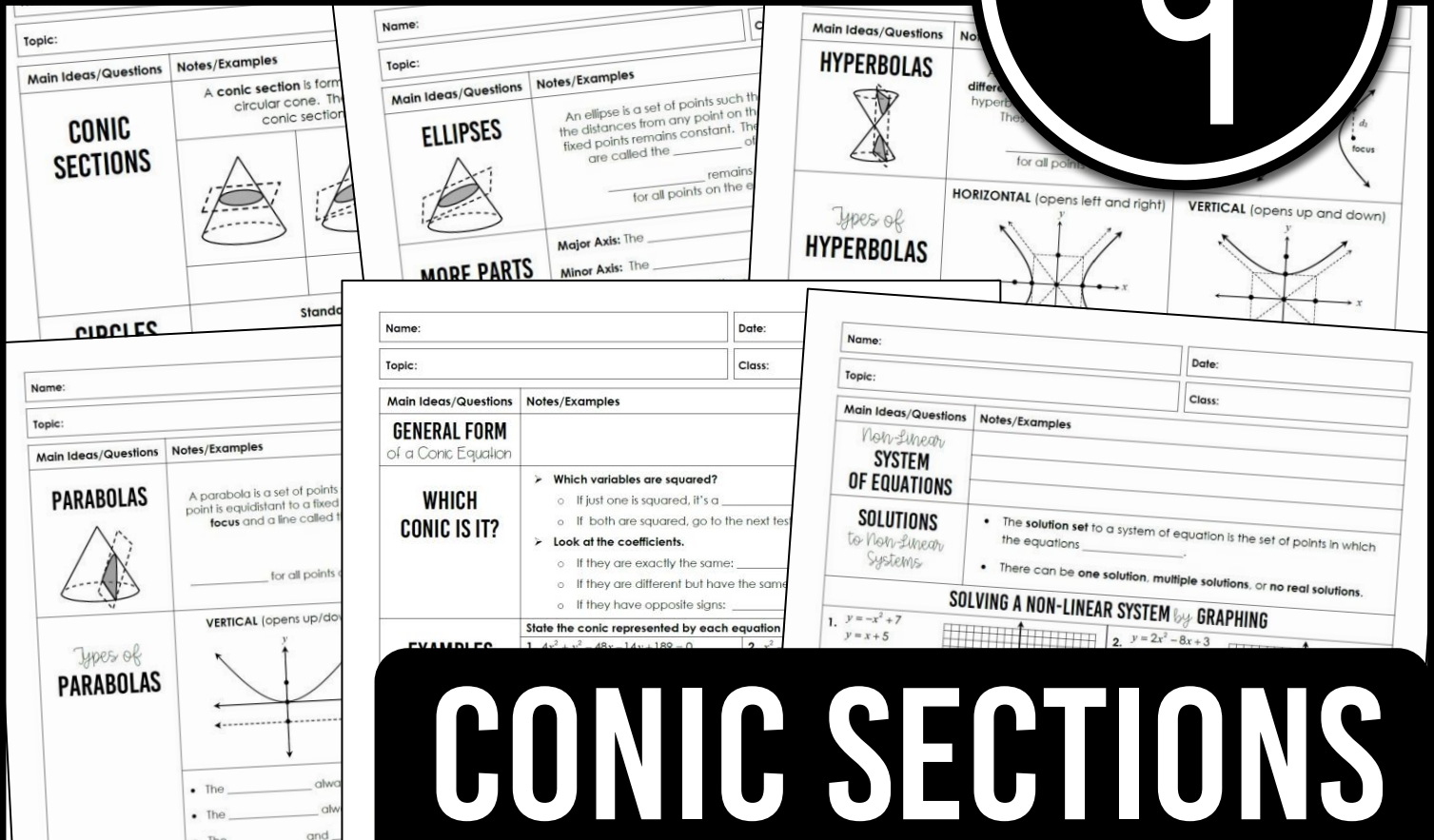


ALGEBRA 2

Unit 9



CONIC SECTIONS

NOTES • HOMEWORK • QUIZZES • TEST

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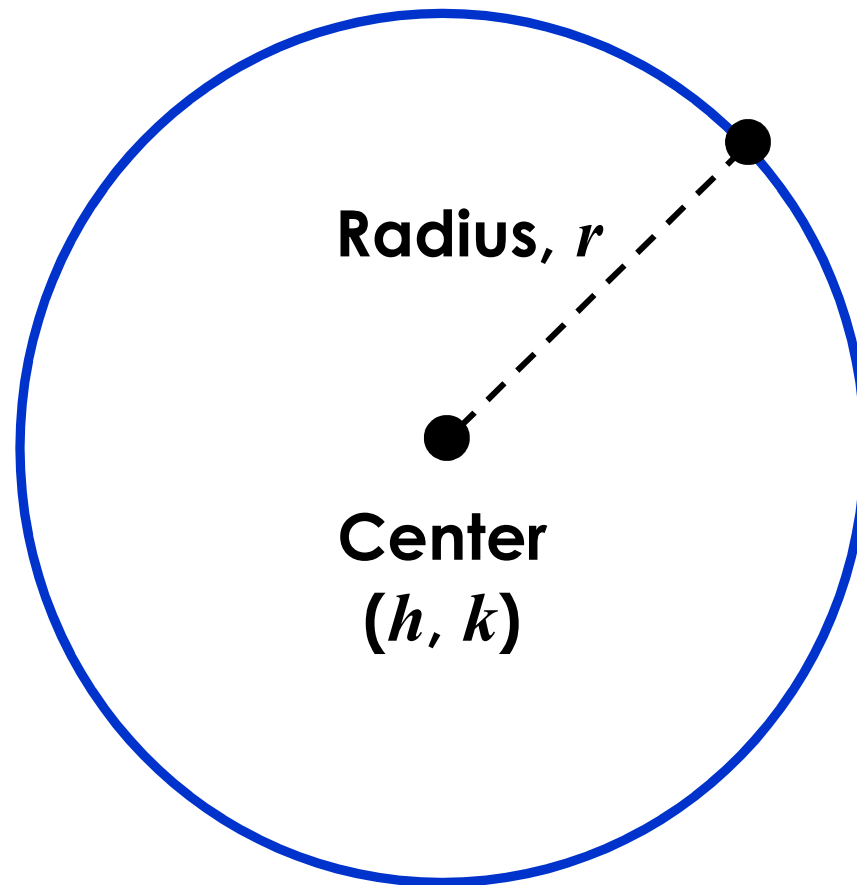
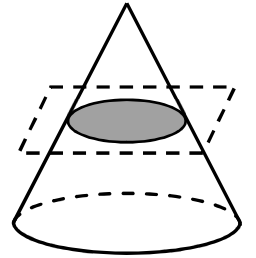
Gina



Unit 9 – Conic Sections: Sample Unit Outline

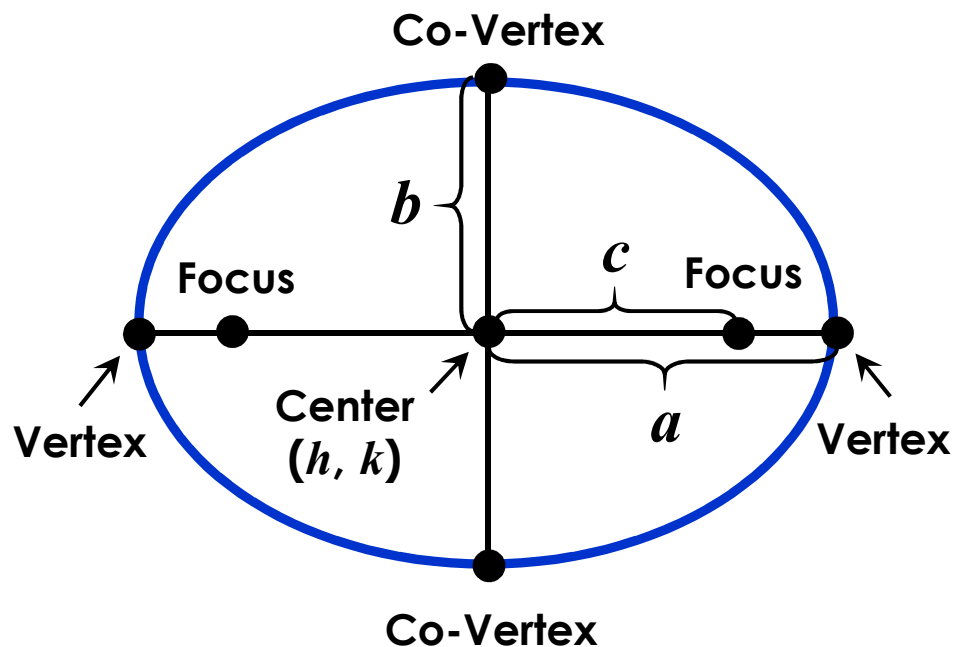
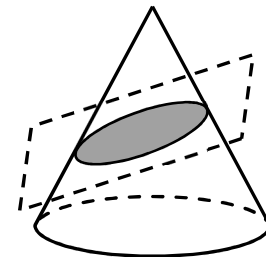
	TOPIC	HOMEWORK
DAY 1	Intro to Conic Sections (Slicing a Cone); Circles	HW #1
DAY 2	Ellipses (Day 1 – Graphing)	HW #2
DAY 3	Ellipses (Day 2 – Writing Equations)	HW #3
DAY 4	Quiz 9-1	None
DAY 5	Hyperbolas (Day 1 – Graphing)	HW #4
DAY 6	Hyperbolas (Day 2 – Writing Equations)	HW #5
DAY 7	Parabolas (Day 1 – Graphing)	HW #6
DAY 8	Parabolas (Day 2 – Writing Equations)	HW #7
DAY 9	Quiz 9-2	None
DAY 10	Identifying Conic Sections & Graphing Review	HW #8
DAY 11	Non-Linear Systems (Day 1 – Solve by Graphing)	HW #9
DAY 12	Non-Linear Systems (Day 2 – Solve Algebraically)	HW #10
DAY 13	Quiz 9-3	None
DAY 14	Unit 9 Review	Study for Test
DAY 15	UNIT 9 TEST	None

CIRCLE

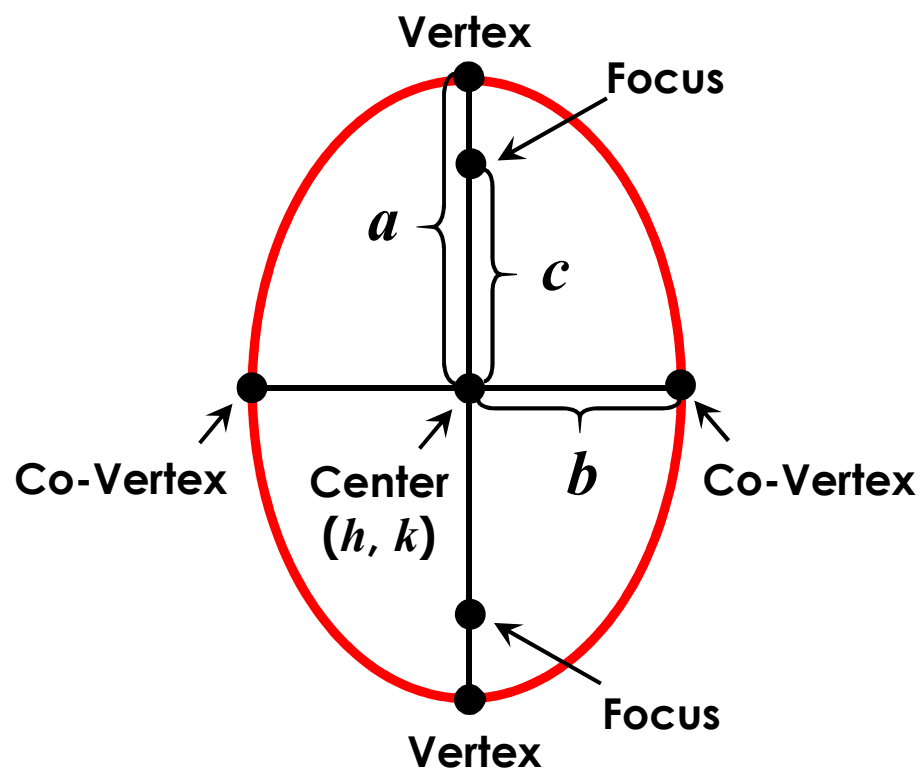


$$(x - h)^2 + (y - k)^2 = r^2$$

ELLIPSE



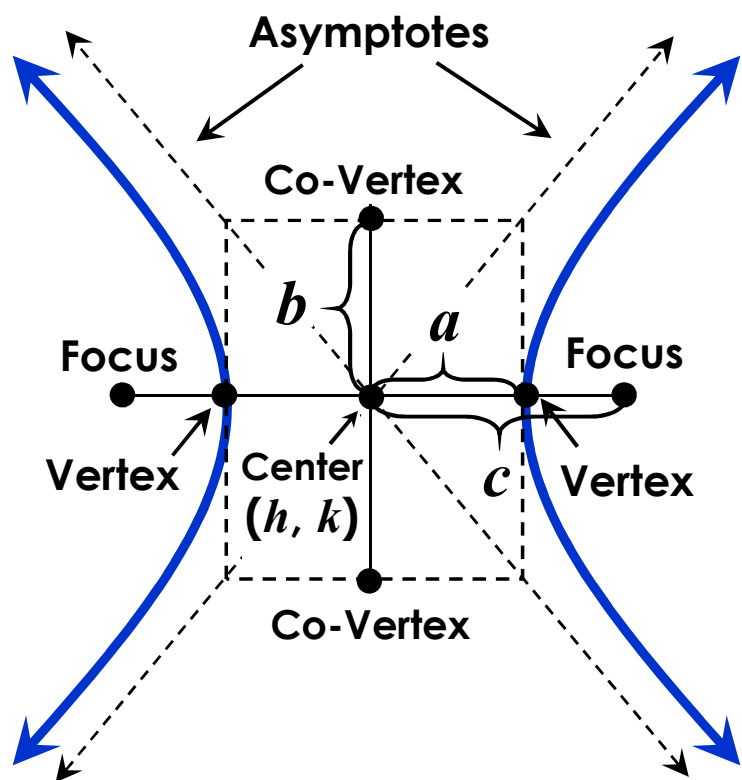
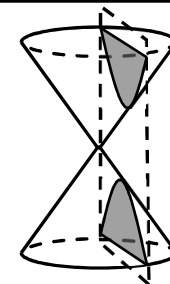
$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$



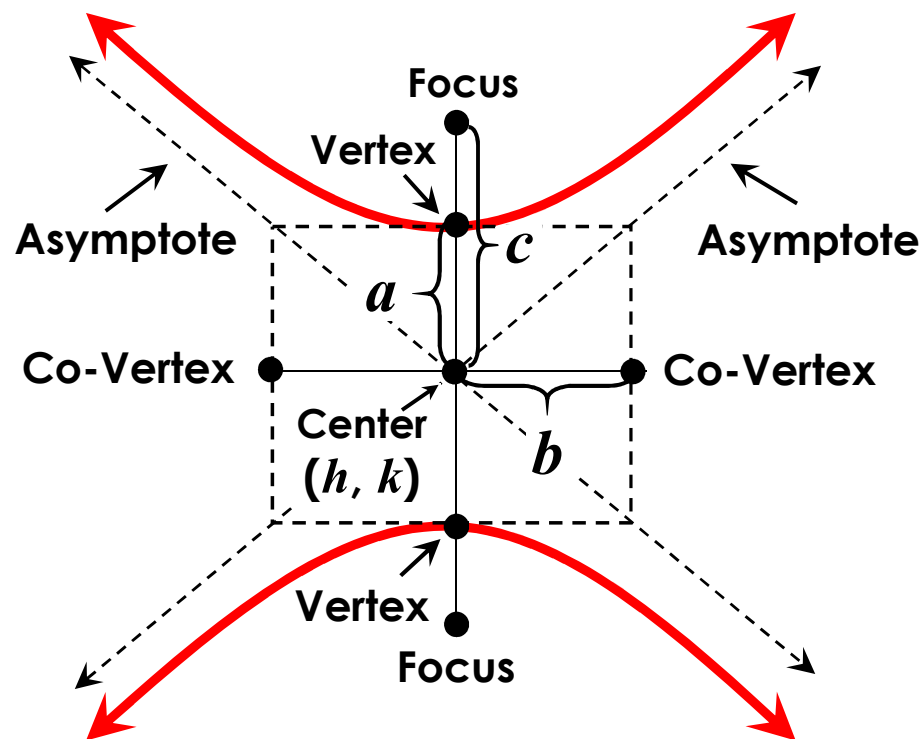
$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

FORMULA FOR C: $c^2 = a^2 - b^2$

HYPERBOLA



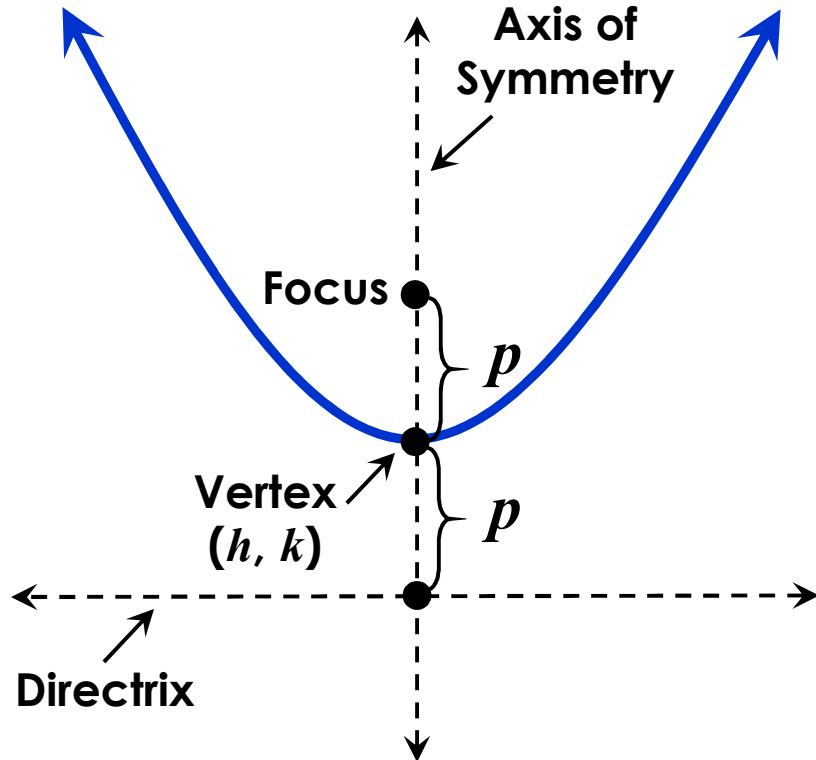
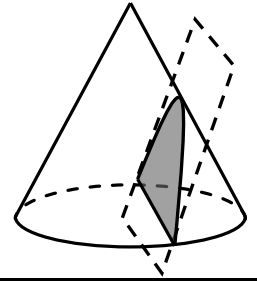
$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$



$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

FORMULA FOR C: $c^2 = a^2 + b^2$

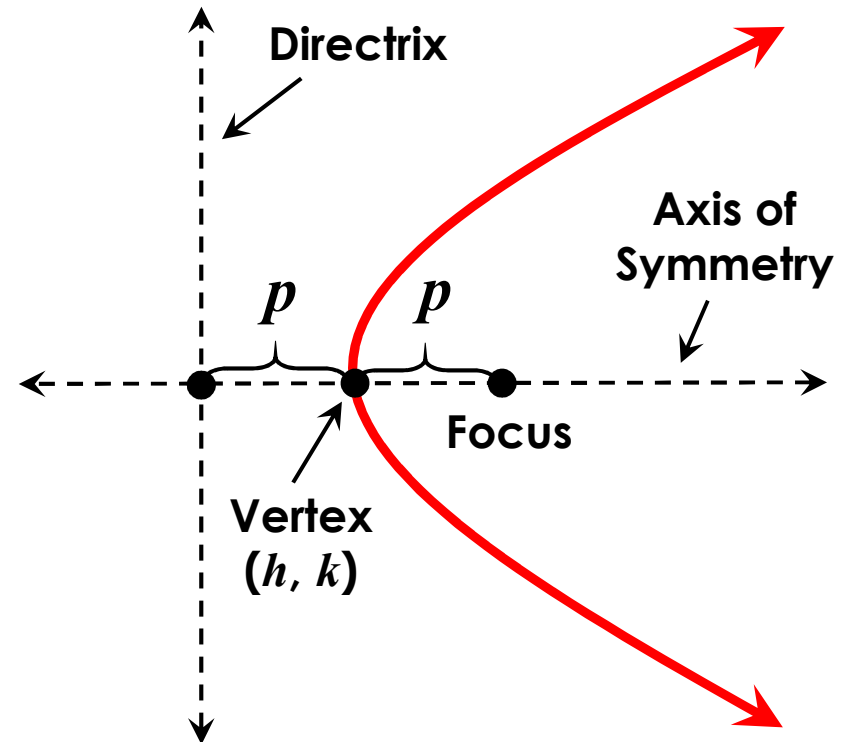
PARABOLA



$$(x - h)^2 = 4p(y - k)$$

Opens UP if $p > 0$

Opens DOWN if $p < 0$

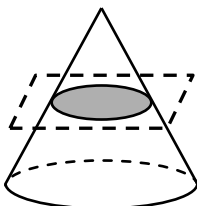
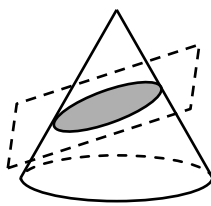
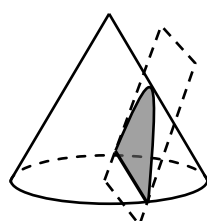
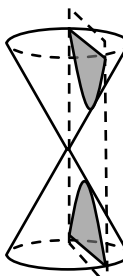
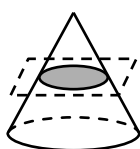
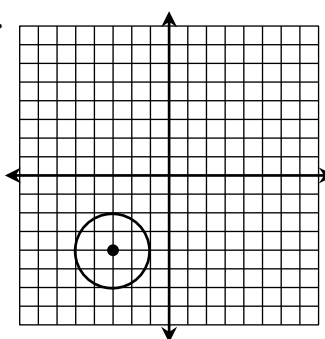
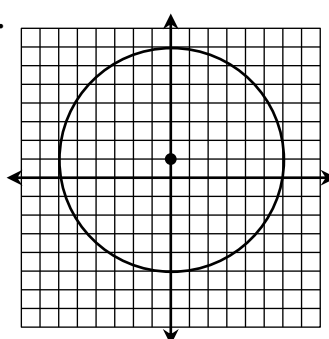


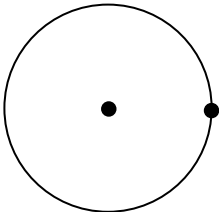
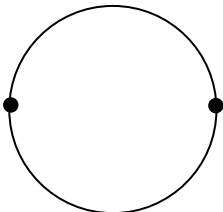
$$(y - k)^2 = 4p(x - h)$$

Opens RIGHT if $p > 0$

Opens LEFT if $p < 0$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples			
CONIC SECTIONS	A conic section is formed by the intersection of a plane and a right circular cone. The angle of the plane creates the type of conic sections. Label each conic section below.			
				
CIRCLES 	Standard Form of the Equation of a Circle: where _____ is the center and _____ is the radius			
WRITING EQUATIONS of Circles	Set 1: Given the center and radius.			
	1. Center: $(-4, 6)$, Radius: 4		2. Center: $(0, -2)$, Radius: 7	
	3. Center: $(1, -5)$, Radius: $\sqrt{10}$		4. Center: $(9, 0)$, Radius: $5\sqrt{3}$	
	Set 2: Given a graph.			
	5. 		6. 	

RECALL → 	Set 3: Given the center and an endpoint on the circle.	
	Distance Formula	Midpoint Formula
	7. Center: $(-2, 6)$, Endpoint: $(-4, 12)$	8. Center: $(-4, -7)$, Endpoint: $(-1, -5)$
	9. Center: $(0, 4)$, Endpoint: $(7, -2)$	10. Center: $(3, -3)$, Endpoint: $(-4, 10)$
	Set 4: Given the endpoints of a diameter.	
	11. $(-1, 8)$ and $(5, 6)$	12. $(0, -3)$ and $(-8, 3)$
	13. $(-4, 10)$ and $(2, 4)$	14. $(-16, -17)$ and $(-14, -13)$

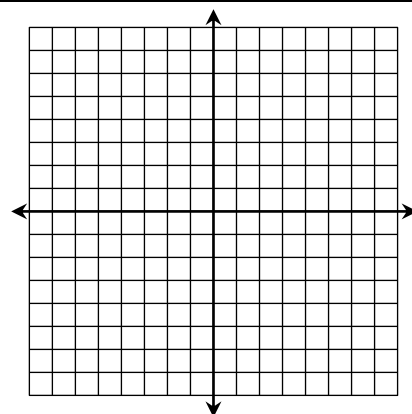
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Graphing CIRCLES	Standard Form of the Equation of a Circle: where _____ is the center and _____ is the radius	
	Identify the center and radius of each circle, then graph.	
	1. $(x + 2)^2 + (y - 5)^2 = 9$ Center: Radius:	2. $(x - 7)^2 + y^2 = 1$ Center: Radius:
WHAT ABOUT: Equations NOT in Standard Form?	Write the equation in standard form, identify the center and radius, then graph.	
	3. $5x^2 + 5y^2 = 80$ Center: Radius:	
	4. $(y + 3)^2 = 25 - x^2$ Center: Radius:	

$$5. 2(x-1)^2 - 8 = -2(y+6)^2$$

Center:

Radius:

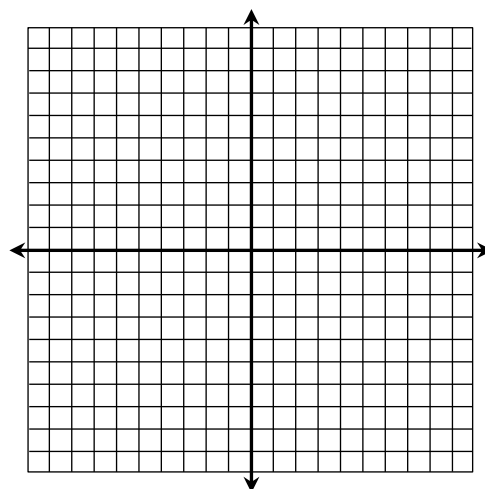


Examples with
**COMPLETING
THE SQUARE**

$$6. x^2 + y^2 + 6x = -5$$

Center:

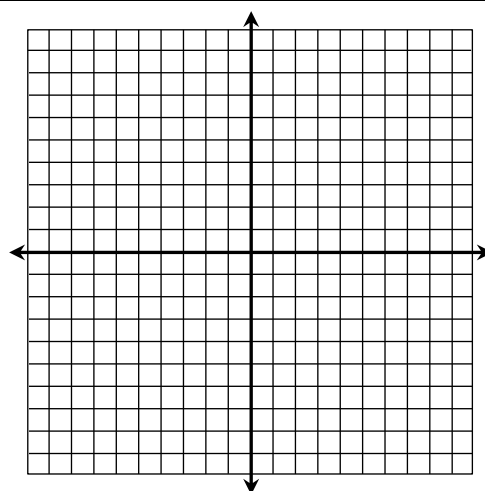
Radius:



$$7. x^2 + y^2 + 2y = 6y + 32$$

Center:

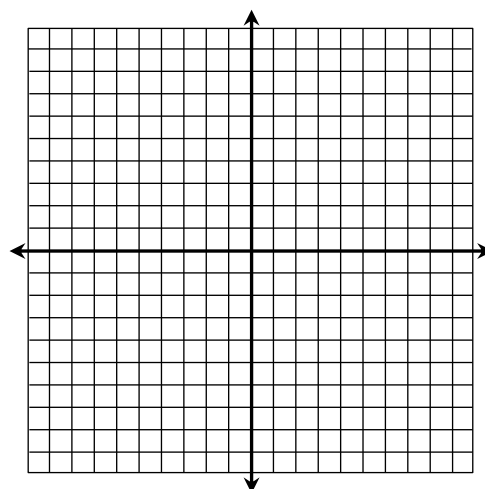
Radius:



$$8. x^2 + y^2 + 14x = 2y - 41$$

Center:

Radius:





Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 1: Circles

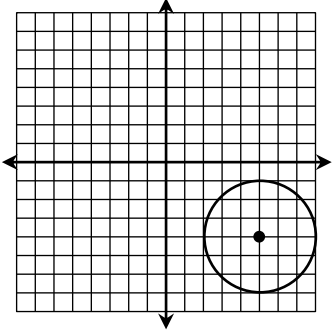
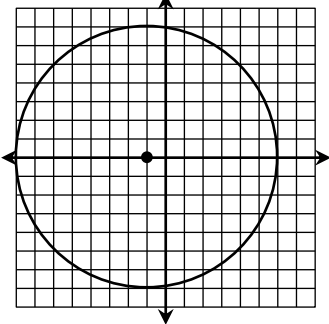
**** This is a 2-page document! ****

Write the equation of each circle with the given information.

1. Center: $(-9, 0)$, Radius: 1	2. Center: $(3, -3)$, Radius: $2\sqrt{6}$
----------------------------------	--

3. Center: $(4, 7)$, Endpoint: $(-1, -2)$	4. Center: $(0, -8)$, Endpoint: $(2, -5)$
--	--

5. Endpoints of Diameter: $(-6, 2)$ and $(-8, 10)$	6. Endpoints of Diameter: $(-13, -7)$ and $(11, 11)$
--	--

7. 	8. 
--	---

Identify the center and radius of each circle given the equations below.

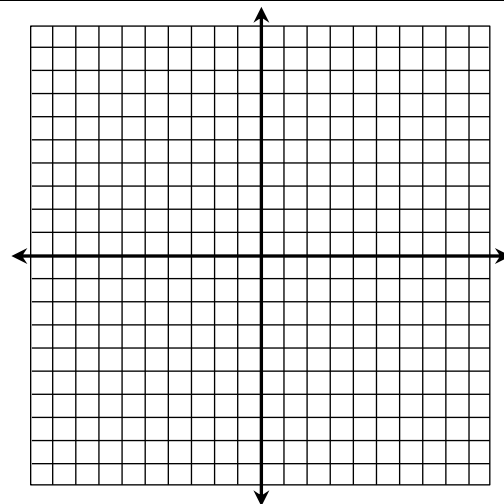
9. $(x + 1)^2 + (y + 4)^2 = 81$	10. $x^2 + (y - 5)^2 = 7$
---------------------------------	---------------------------

11. $(x - 7)^2 + (y + 2)^2 = 40$

12. $2(x + 6)^2 + 2y^2 = 144$

Write the equation in standard form, identify the center and radius, then graph the circle.

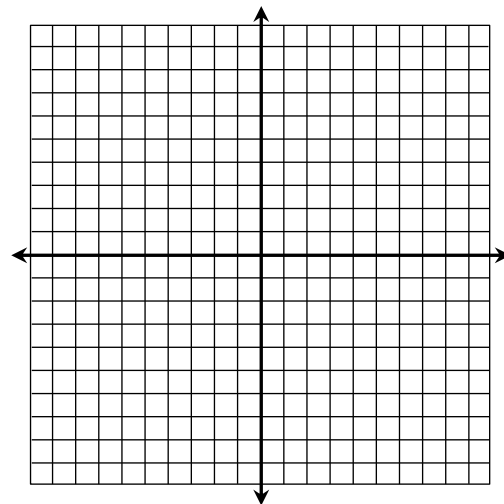
13. $3x^2 + 3y^2 - 5 = 142$



Center:

Radius:

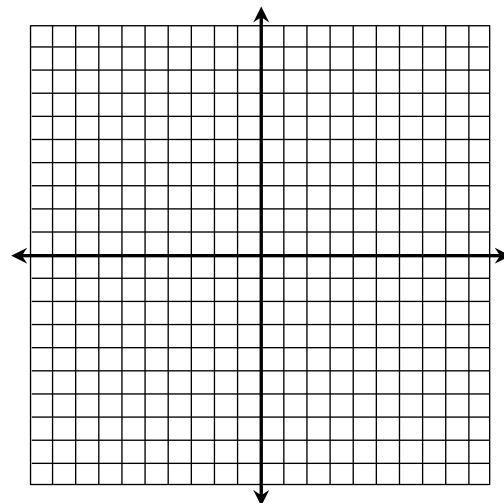
14. $x^2 + y^2 + 13y = y - 32$



Center:

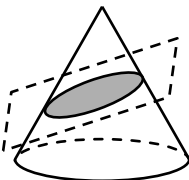
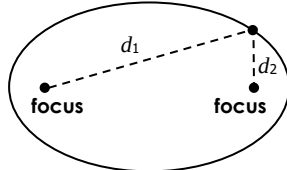
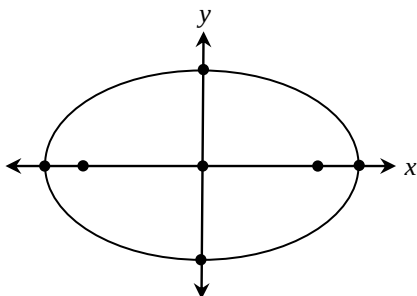
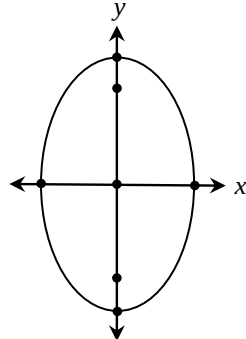
Radius:

15. $x^2 + y^2 + 7y = 10x + 3y - 13$



Center:

Radius:

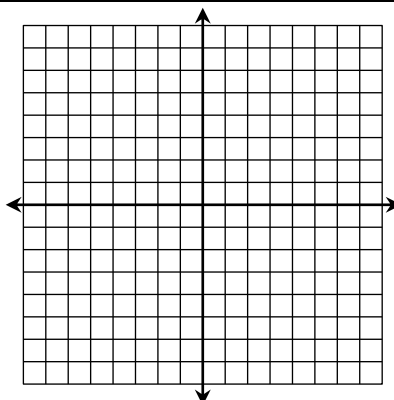
Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
ELLIPSES 		An ellipse is a set of points such that the sum of the distances from any point on the ellipse to two fixed points remains constant. These fixed points are called the _____ of the ellipse. _____ remains constant for all points on the ellipse! 	
MORE PARTS		Major Axis: The _____ axis of symmetry. Minor Axis: The _____ axis of symmetry. Center: The point at which the major and minor axis _____. Vertices: The _____ of the _____. Co-Vertices: The _____ of the _____.	
Types of ELLIPSES		HORIZONTAL ELLIPSE 	VERTICAL ELLIPSE 
		- The vertices are _____ units away from the center. - The co-vertices are _____ units away from the center. - Length of Major Axis: _____; Length of Minor Axis: _____ - The foci are _____ units away from the center. Formula to find c: _____	
STANDARD FORM Centered at (0, 0)			
STANDARD FORM Centered at (h, k)			

Graphing ELLIPSES

- ✓ Write the equation in **standard form**.
- ✓ Graph the **center**, (h, k) .
- ✓ Graph the **vertices**, using the larger denominator, a .
- ✓ Graph the **co-vertices**, using the smaller denominator, b .
- ✓ Use the formula $c^2 = a^2 - b^2$ to find and graph the **foci**.

Directions: Graph each ellipse. Identify the center, vertices, co-vertices, and foci.

1. $\frac{x^2}{4} + \frac{y^2}{36} = 1$



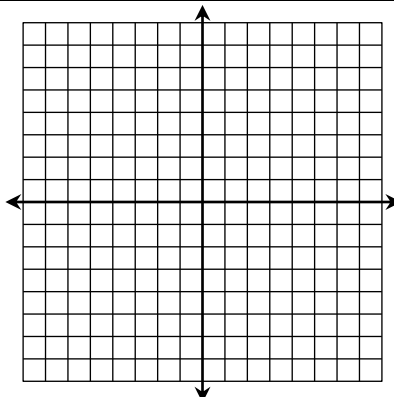
Center:

Vertices:

Co-Vertices:

Foci:

2. $\frac{x^2}{64} + \frac{y^2}{25} = 1$



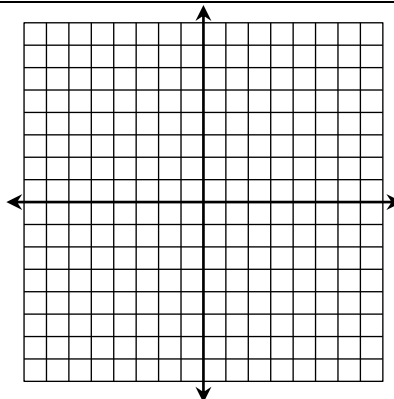
Center:

Vertices:

Co-Vertices:

Foci:

3. $\frac{(x+1)^2}{49} + \frac{(y-2)^2}{9} = 1$



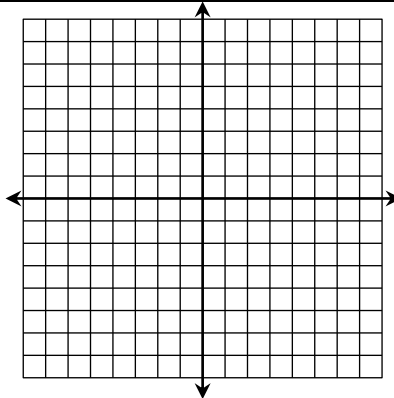
Center:

Vertices:

Co-Vertices:

Foci:

4. $\frac{(x-3)^2}{25} + \frac{(y-1)^2}{4} = 1$



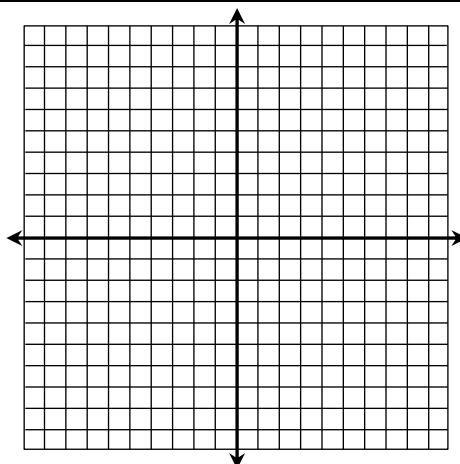
Center:

Vertices:

Co-Vertices:

Foci:

5. $\frac{(x+4)^2}{20} + \frac{(y+2)^2}{36} = 1$



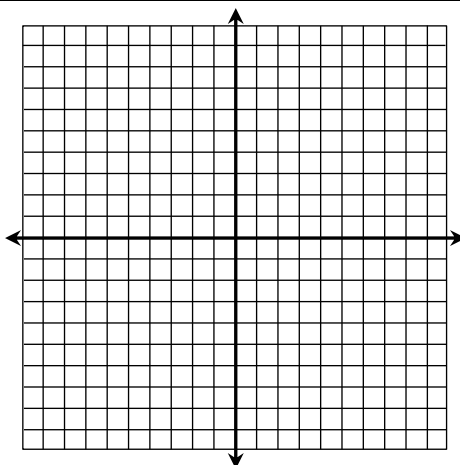
Center:

Vertices:

Co-Vertices:

Foci:

6. $9x^2 + 16y^2 = 144$



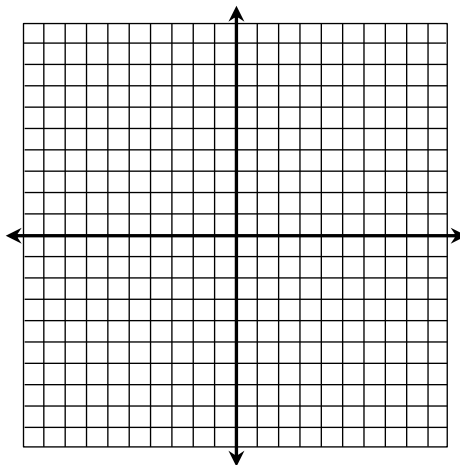
Center:

Vertices:

Co-Vertices:

Foci:

7. $x^2 + 4y^2 - 10x = -9$



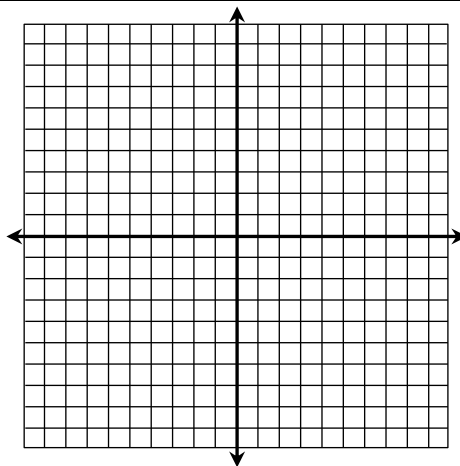
Center:

Vertices:

Co-Vertices:

Foci:

8. $4x^2 + y^2 - 32x + 8y + 76 = 0$



Center:

Vertices:

Co-Vertices:

Foci:



Name: _____

Unit 9: Conic Sections

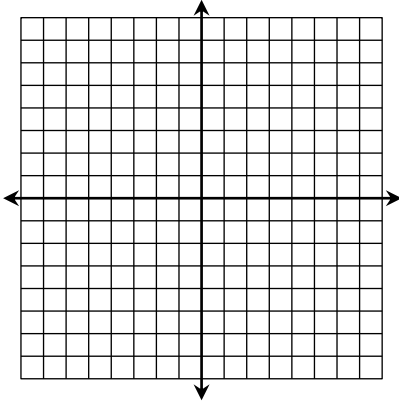
Date: _____ Per: _____

Homework 2: Graphing Ellipses

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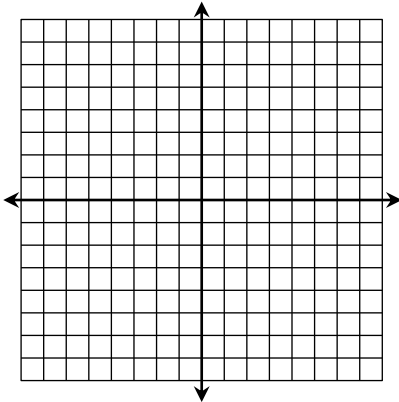
Graph each ellipse. Identify the center, vertices, co-vertices, and foci.

1. $\frac{x^2}{64} + \frac{y^2}{9} = 1$



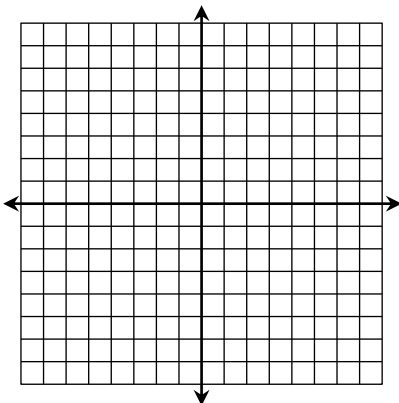
Center:
Vertices:
Co-Vertices:
Foci:

2. $x^2 + \frac{y^2}{36} = 1$



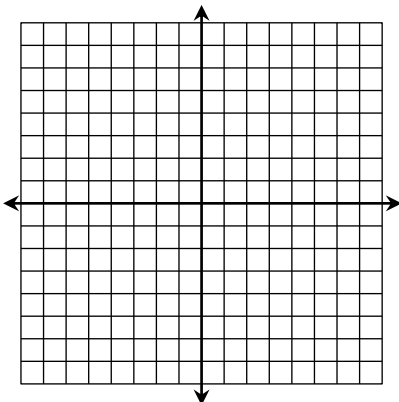
Center:
Vertices:
Co-Vertices:
Foci:

3. $\frac{x^2}{16} + \frac{y^2}{7} = 1$



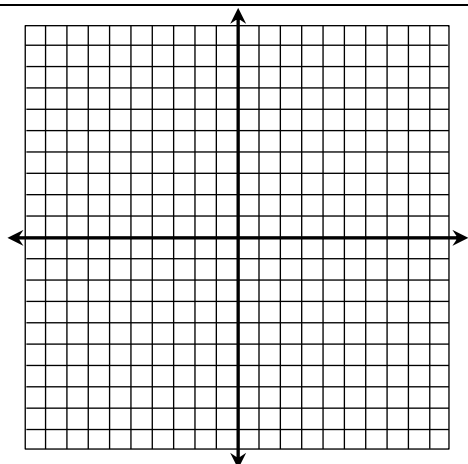
Center:
Vertices:
Co-Vertices:
Foci:

4. $\frac{(x+1)^2}{25} + \frac{(y+1)^2}{49} = 1$



Center:
Vertices:
Co-Vertices:
Foci:

5. $\frac{(x+3)^2}{45} + \frac{(y-1)^2}{81} = 1$



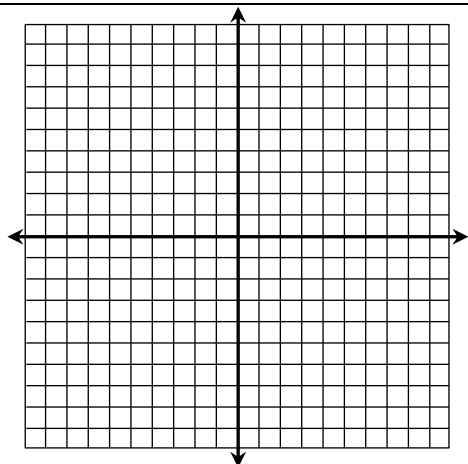
Center:

Vertices:

Co-Vertices:

Foci:

6. $9x^2 + 25y^2 = 225$



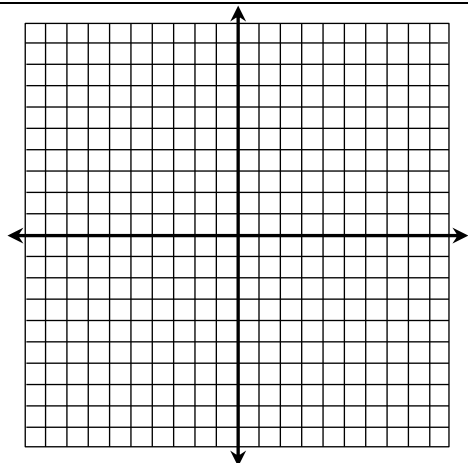
Center:

Vertices:

Co-Vertices:

Foci:

7. $27(x-1)^2 + 36(y+2)^2 = 972$



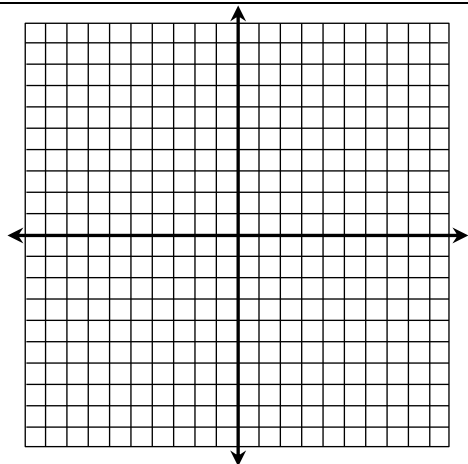
Center:

Vertices:

Co-Vertices:

Foci:

8. $9x^2 + y^2 + 144x = -567$



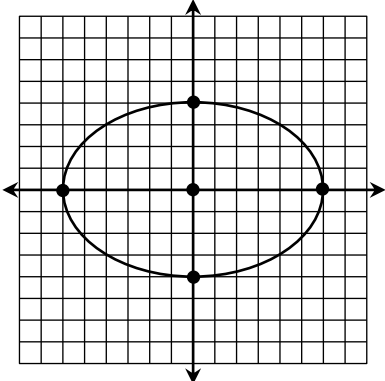
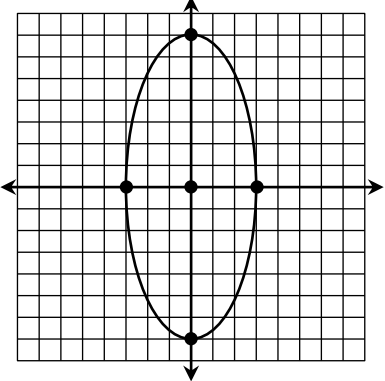
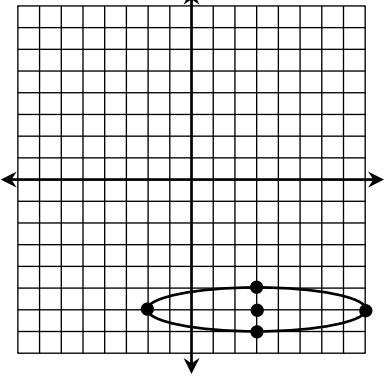
Center:

Vertices:

Co-Vertices:

Foci:

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ELLIPSE Equations	Standard Form for the Equation of an Ellipse at Center (h, k) :	
	HORIZONTAL ELLIPSE	VERTICAL ELLIPSE:
	_____ is the distance from the center to the vertices. _____ is the distance from the center to the co-vertices. - To find the foci use the formula, _____	
SET 1: Given a Graph	Write an equation for each ellipse. Then identify all points.	
	1. 	Equation:
		Center:
		Vertices:
		Co-Vertices:
		Foci:
	2. 	Equation:
		Center:
		Vertices:
		Co-Vertices:
		Foci:
	3. 	Equation:
		Center:
		Vertices:
		Co-Vertices:
		Foci:

SET 2:

Given Vertices,
Co-Vertices, & Foci

- ✓ Find the **center** (h, k). (Use the midpoint formula with the vert. or co-vert.)
- ✓ Use the **vertices** to find a ; Use the **co-vertices** to find b .
*If given the foci, use the formula $c^2 = a^2 - b^2$ to find the missing part.
- ✓ Determine the direction of the ellipse to write the equation.

Write an equation for each ellipse with the given information.

4. Vertices: $(-5, 0)$ and $(5, 0)$
Co-Vertices: $(0, -2)$ and $(0, 2)$

5. Vertices: $(0, -8)$ and $(0, 8)$
Co-Vertices: $(-6, 0)$ and $(6, 0)$

6. Vertices: $(-5, 1)$ and $(1, 1)$
Co-Vertices: $(-2, 3)$ and $(-2, -1)$

7. Vertices: $(5, 8)$ and $(5, -6)$
Co-Vertices: $(11, 1)$ and $(-1, 1)$

8. Vertices: $(0, 6)$ and $(-10, 6)$
Foci: $(-2, 6)$ and $(-8, 6)$

9. Vertices: $(-9, -2)$ and $(-9, -12)$
Foci: $(-9, -3)$ and $(-9, -11)$

10. Co-Vertices: $(5, 0)$ and $(-5, 0)$
Foci: $(0, 12)$ and $(0, -12)$

11. Co-Vertices: $(2, 7)$ and $(2, -9)$
Foci: $(-4, -1)$ and $(8, -1)$

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

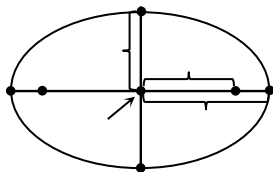
Homework 3: Writing Equations of Ellipses

**** This is a 2-page document! ****

Label a , b , c , h , and k on each diagram. Then give the standard form equation of each ellipse.

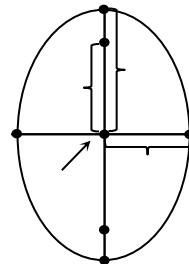
HORIZONTAL ELLIPSE

Standard Form:



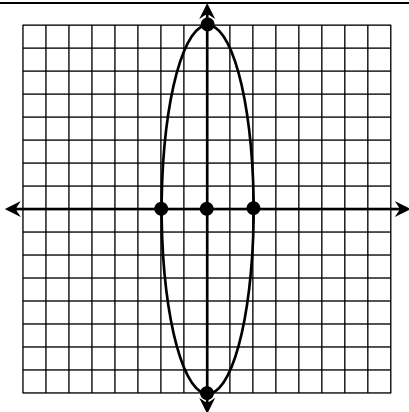
VERTICAL ELLIPSE

Standard Form:



Write an equation for each ellipse. Identify the center, vertices, co-vertices, and foci.

1.



Equation:

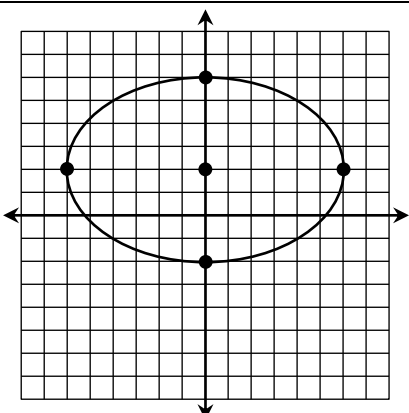
Center:

Vertices:

Co-Vertices:

Foci:

2.



Equation:

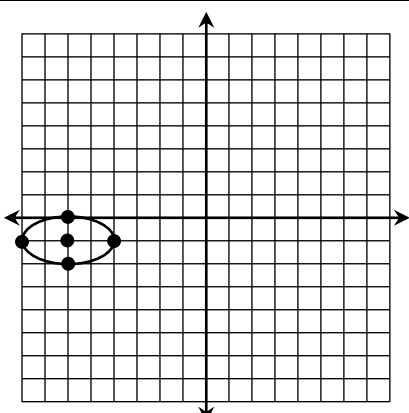
Center:

Vertices:

Co-Vertices:

Foci:

3.



Equation:

Center:

Vertices:

Co-Vertices:

Foci:

Write an equation for each ellipse with the given information.	
4. Vertices: $(0, -9)$ and $(0, 9)$ Co-Vertices: $(-8, 0)$ and $(8, 0)$	5. Vertices: $(-3, 0)$ and $(3, 0)$ Co-Vertices: $(0, -1)$ and $(0, 1)$
6. Vertices: $(12, -4)$ and $(-10, -4)$ Co-Vertices: $(1, 2)$ and $(1, -10)$	7. Vertices: $(-2, 17)$ and $(-2, 3)$ Co-Vertices: $(2, 10)$ and $(-6, 10)$
8. Vertices: $(12, 2)$ and $(2, 2)$ Foci: $(10, 2)$ and $(4, 2)$	9. Vertices: $(-4, -7)$ and $(-4, 7)$ Foci: $(-4, -\sqrt{13})$ and $(-4, \sqrt{13})$
10. Co-Vertices: $(16, -3)$ and $(-8, -3)$ Foci: $(4, 2)$ and $(4, -8)$	11. Co-Vertices: $(1, -13)$ and $(1, 3)$ Foci: $(-5, -5)$ and $(7, -5)$

Name: _____

Algebra 2

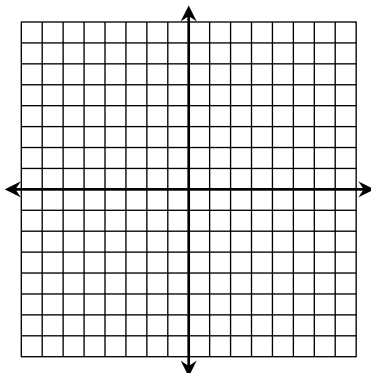
Date: _____ Per: _____

Unit 9: Conic Sections

Quiz 9-1: Circles & Ellipses

Graph each circle. Identify the center and radius.

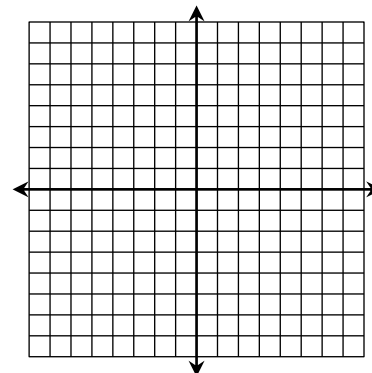
1. $x^2 + y^2 = 36$



Center: _____

Radius: _____

2. $(x - 4)^2 + (y + 1)^2 = 9$



Center: _____

Radius: _____

Write the equation of the circle in standard form. Identify the center and radius.

3. $7x^2 + 7y^2 + 3 = 31$

4. $x^2 + y^2 - 10x - 6y + 18 = 0$

Standard Form: _____

Center: _____; Radius: _____

Standard Form: _____

Center: _____; Radius: _____

Write the equation of the circle in standard form with the given characteristics.

5. Center: $(-7, -2)$; Radius: 9

6. Center: $(4, 0)$; Radius: $2\sqrt{6}$

Standard Form: _____

Standard Form: _____

7. Center: $(5, 9)$; Point on Circle: $(7, 8)$

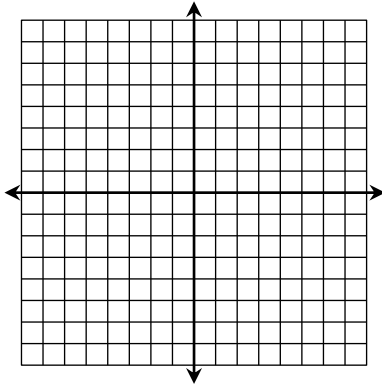
8. Endpoints of Diameter: $(-5, -8)$ and $(7, 2)$

Standard Form: _____

Standard Form: _____

Graph each ellipse. Identify the center, vertices, co-vertices, and foci.

9. $\frac{x^2}{9} + \frac{y^2}{49} = 1$



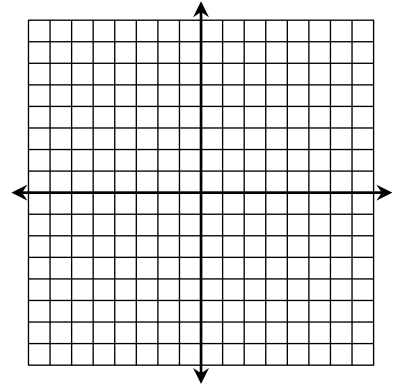
Center _____

Vertices: _____

Co-Vertices: _____

Foci: _____

10. $\frac{(x-3)^2}{16} + \frac{(y-5)^2}{4} = 1$



Center _____

Vertices: _____

Co-Vertices: _____

Foci: _____

Write the equation of the ellipse in standard form. Identify all characteristics.

11. $4x^2 + 64y^2 = 256$

12. $16x^2 + y^2 + 14y = -33$

Standard Form: _____

Center _____

Vertices: _____

Co-Vertices: _____

Foci: _____

Standard Form: _____

Center _____

Vertices: _____

Co-Vertices: _____

Foci: _____

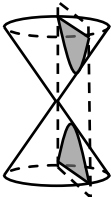
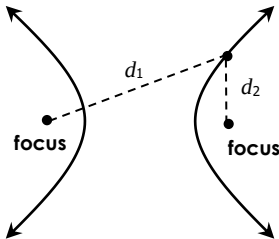
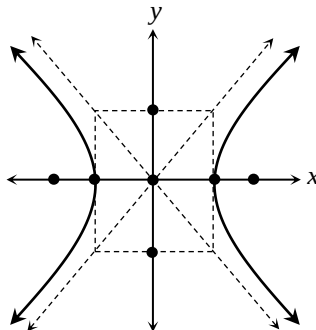
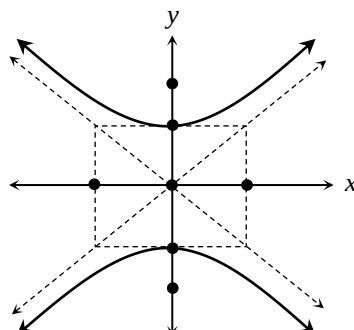
Write the equation of the ellipse in standard form with the given characteristics.

13. Vertices: (2, 1) and (2, 7)
Co-Vertices: (0, 4) and (4, 4)

14. Vertices: (-3, 4) and (5, 4)
Foci: (-1, 4) and (3, 4)

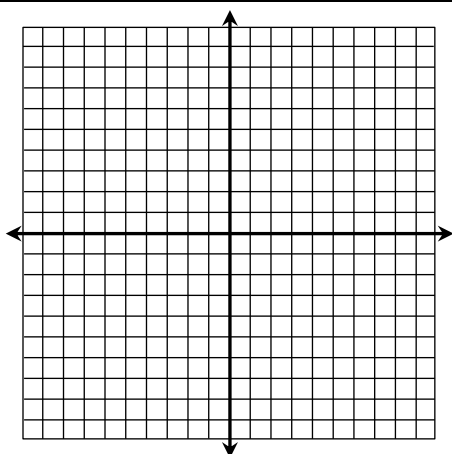
Standard Form: _____

Standard Form: _____

Name:		Date:
Topic:		Class:
Main Ideas/Questions	Notes/Examples	
HYPERBOLAS 	A hyperbola is a set of points such that the difference of the distances from any point on the hyperbola to two fixed points remains constant. These fixed points are called the foci. _____ remains constant for all points on the hyperbola! 	
<i>Types of</i> HYPERBOLAS	HORIZONTAL (opens left and right) 	VERTICAL (opens up and down) 
	• The center (h, k) is the _____ of the foci. • The asymptotes are the diagonal lines that “guide” the curves. • The vertices (turning points) are _____ units away from the center. • The co-vertices are _____ units away from the center. • The foci are _____ units away from the center. • Formula to find c: _____	
STANDARD FORM Centered at (h, k)		
ASYMPTOTE <i>Equations</i>		
<i>Graphing</i> HYPERBOLAS	✓ Write the equation in standard form. ✓ Graph the center, (h, k). ✓ Graph the vertices. Use the first denominator, a. ✓ Graph the co-vertices. Use the second denominator, b. ✓ Use the formula $c^2 = a^2 + b^2$ to find and graph the foci. ✓ Write equations for the asymptotes and graph them.	

Directions: Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{4} - \frac{y^2}{9} = 1$



Center:

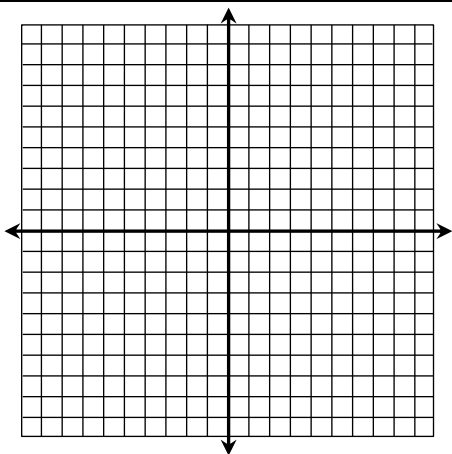
Vertices:

Co-Vertices:

Foci:

Asymptotes:

2. $\frac{y^2}{36} - x^2 = 1$



Center:

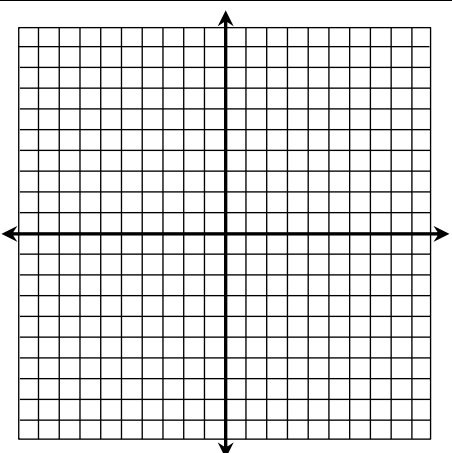
Vertices:

Co-Vertices:

Foci:

Asymptotes:

3. $\frac{(x-3)^2}{25} - \frac{(y-2)^2}{16} = 1$



Center:

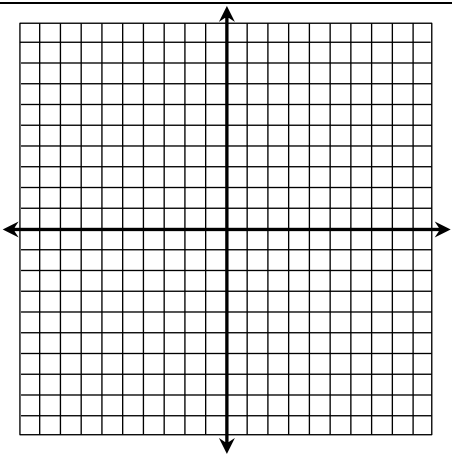
Vertices:

Co-Vertices:

Foci:

Asymptotes:

4. $\frac{(y-3)^2}{36} - \frac{(x+1)^2}{9} = 1$



Center:

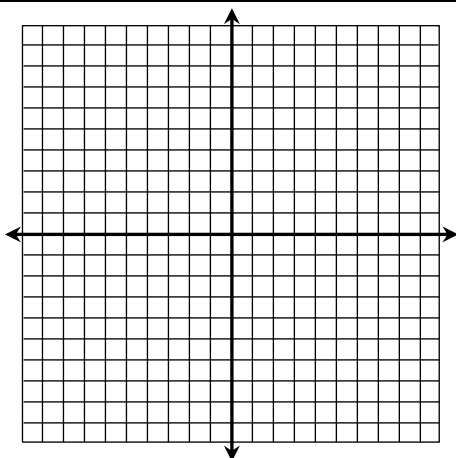
Vertices:

Co-Vertices:

Foci:

Asymptotes:

5. $4y^2 - 25x^2 = 100$



Center:

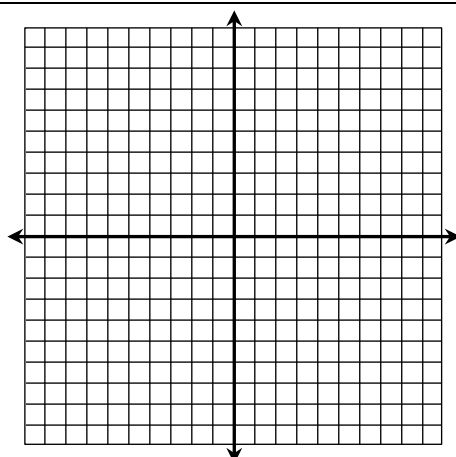
Vertices:

Co-Vertices:

Foci:

Asymptotes:

6. $x^2 - 9y^2 + 10x = -16$



Center:

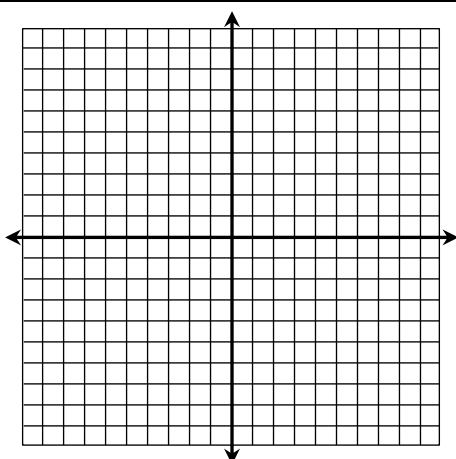
Vertices:

Co-Vertices:

Foci:

Asymptotes:

7. $-36x^2 + y^2 - 6y = -45$



Center:

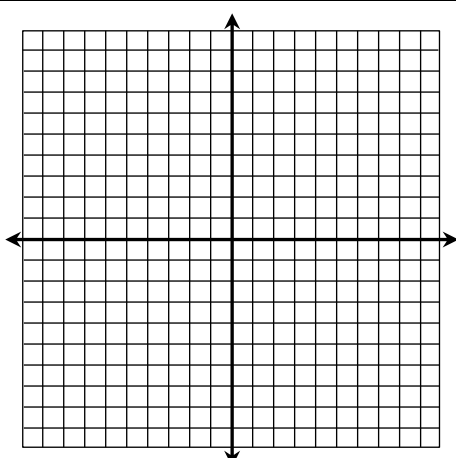
Vertices:

Co-Vertices:

Foci:

Asymptotes:

8. $-4x^2 + y^2 + 40x + 12y = 80$



Center:

Vertices:

Co-Vertices:

Foci:

Asymptotes:

Name: _____

Unit 9: Conic Sections

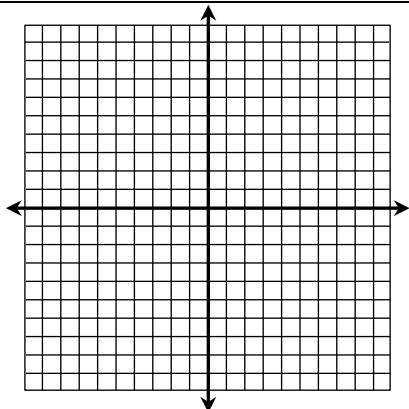
Date: _____ Per: _____

Homework 4: Graphing Hyperbolas

**** This is a 2-page document! ****

Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{64} - \frac{y^2}{16} = 1$



Center:

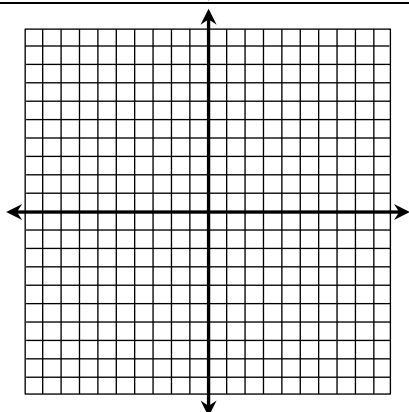
Vertices:

Co-Vertices:

Foci:

Asymptotes:

2. $y^2 - \frac{x^2}{25} = 1$



Center:

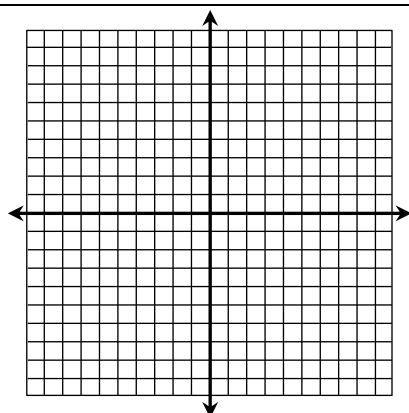
Vertices:

Co-Vertices:

Foci:

Asymptotes:

3. $\frac{(x+6)^2}{4} - \frac{(y-1)^2}{4} = 1$



Center:

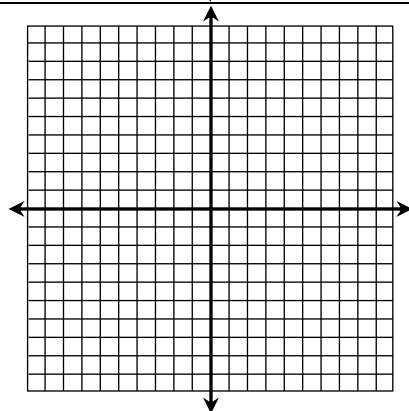
Vertices:

Co-Vertices:

Foci:

Asymptotes:

4. $\frac{(y-2)^2}{49} - \frac{(x+2)^2}{9} = 1$



Center:

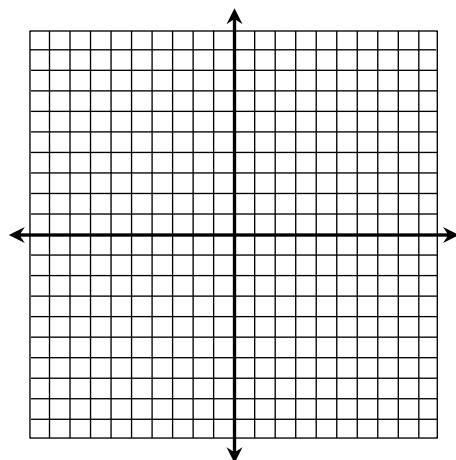
Vertices:

Co-Vertices:

Foci:

Asymptotes:

5. $9y^2 - 3x^2 + 1 = x^2 + 37$



Center:

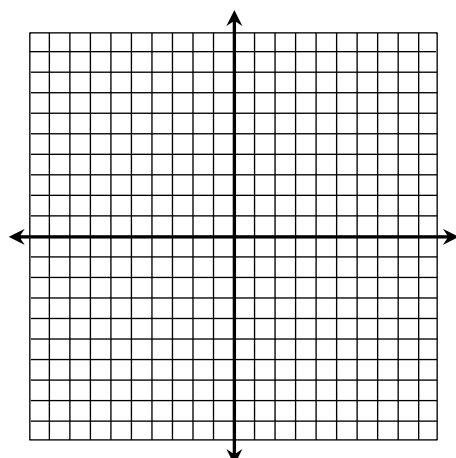
Vertices:

Co-Vertices:

Foci:

Asymptotes:

6. $4x^2 - y^2 - 6y = 25$



Center:

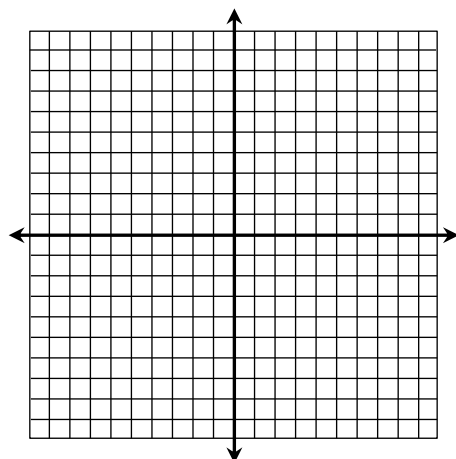
Vertices:

Co-Vertices:

Foci:

Asymptotes:

7. $x^2 - 16y^2 - 2x = 63$



Center:

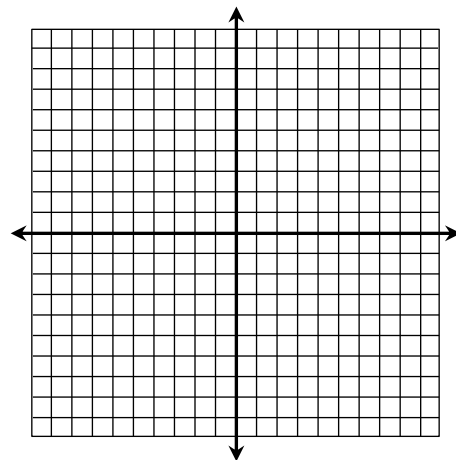
Vertices:

Co-Vertices:

Foci:

Asymptotes:

8. $-x^2 + 9y^2 - 4x + 72y = -131$



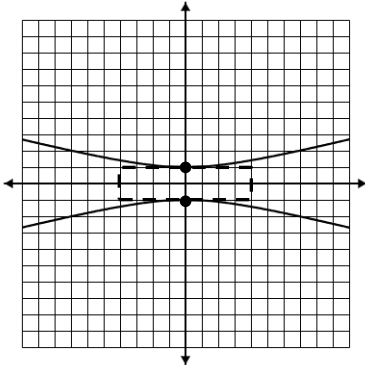
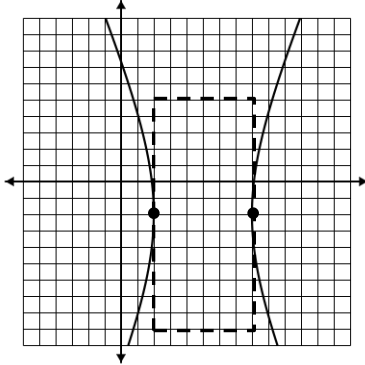
Center:

Vertices:

Co-Vertices:

Foci:

Asymptotes:

Name:		Date:	
Topic:		Class:	
HYPERBOLA <i>Equations</i>	Standard Form for the Equation of a Hyperbola at Center (h, k):		
	HORIZONTAL HYPERBOLA		VERTICAL HYPERBOLA
	_____ is the _____ _____ is the distance from the _____ to the _____. _____ is the distance from the _____ to the _____.		
SET 1: Given a Graph	1. 		2. 
	SET 2: Given Vertices, Co-Vertices, & Foci ✓ Find the center (h, k). (Use the midpoint formula with the vert. or co-vert.) ✓ Use the vertices to find a; Use the co-vertices to find b. *If given the foci, use the formula $c^2 = a^2 + b^2$ to find the missing part. ✓ Determine the direction of the hyperbola to write the equation.		
3. Vertices: $(\pm 5, 0)$ Co-Vertices: $(0, \pm 2)$		4. Vertices: $(4, 8)$ and $(4, -8)$ Co-Vertices: $(1, 0)$ and $(7, 0)$	

5. Vertices: $(12, -6)$ and $(-2, -6)$ Co-Vertices: $(5, -12)$ and $(5, 0)$	6. Vertices: $(-9, 3)$ and $(-9, -5)$ Co-Vertices: $(1, -1)$ and $(-19, 1)$
7. Vertices: $(0, \pm 3)$ Foci: $(0, \pm \sqrt{73})$	8. Vertices: $(2, 5)$ and $(6, 5)$ Foci: $(1, 5)$ and $(7, 5)$
9. Vertices: $(1, -9)$ and $(-7, -9)$ Foci: $(2, -9)$ and $(-8, -9)$	10. Vertices: $(3, -2)$ and $(3, 10)$ Foci: $(3, 4 \pm 2\sqrt{10})$
11. Co-Vertices: $(5, -3)$ and $(-7, -3)$ Foci: $(-1, 7)$ and $(-1, -13)$	12. Co-Vertices: $(-2, 0)$ and $(-2, 10)$ Foci: $(-2 \pm \sqrt{51}, 5)$



Name: _____

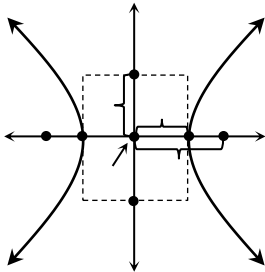
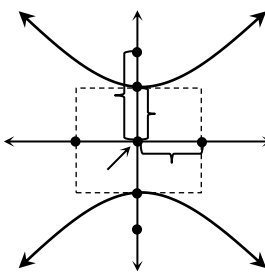
Unit 9: Conic Sections

Date: _____ Per: _____

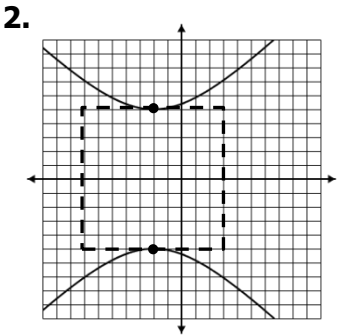
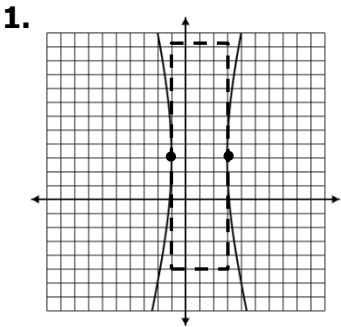
Homework 5: Writing Equations of Hyperbolas

**** This is a 2-page document! ****

Directions: Label a , b , c , h , and k on each diagram. Then give the standard form for the equation of a hyperbola.

HORIZONTAL HYPERBOLA	VERTICAL HYPERBOLA
Standard Form: 	Standard Form: 

Directions: Write an equation for each hyperbola.



Directions: Write an equation for each hyperbola with the given information.

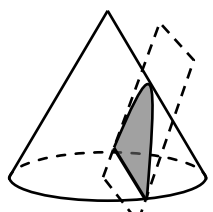
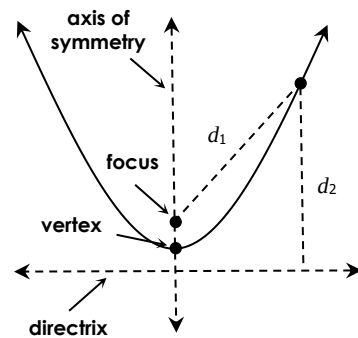
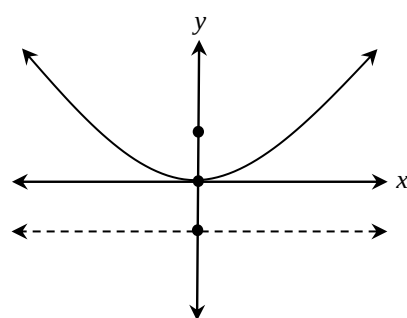
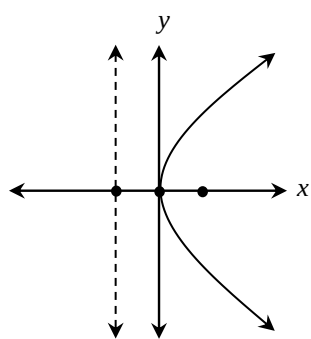
3. Vertices: $(0, \pm 3)$
Co-Vertices: $(\pm 1, 0)$

4. Vertices: $(14, -4)$ and $(-4, -4)$
Co-Vertices: $(5, 1)$ and $(5, -9)$

5. Vertices: $(4, -6)$ and $(-6, -6)$
Co-Vertices: $(-1, 3)$ and $(-1, -15)$

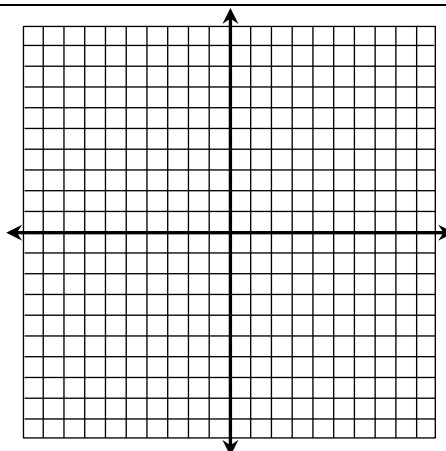
6. Vertices: $(-2, -8)$ and $(-2, 8)$
Co-Vertices: $(-6, 0)$ and $(2, 0)$

7. Vertices: $(\pm 1, 0)$ Foci: $(\pm\sqrt{37}, 0)$	8. Vertices: $(-4, -9)$ and $(-10, -9)$ Foci: $(-2, -9)$ and $(-12, -9)$
9. Vertices: $(4, 11)$ and $(4, -13)$ Foci: $(4, 12)$ and $(4, -14)$	10. Vertices: $(0, 9)$ and $(0, 5)$ Foci: $(0, 7 \pm \sqrt{29})$
11. Co-Vertices: $(-2, 3)$ and $(-8, 3)$ Foci: $(-5, 8)$ and $(-5, -2)$	12. Co-Vertices: $(8, -1)$ and $(8, -5)$ Foci: $(8 \pm \sqrt{5}, -3)$
Challenge! Hint: A graph may be helpful to write the equations of the hyperbolas below.	
13. Vertices: $(\pm 6, 0)$ Asymptotes: $y = \pm \frac{1}{3}x$	14. Co-Vertices: $(1, -1)$ and $(5, -1)$ Asymptotes: $y = 2x - 7$ and $y = -2x + 5$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
PARABOLAS 		A parabola is a set of points such that each point is equidistant to a fixed point called the focus and a line called the directrix. _____ for all points on the parabola! 	
Types of PARABOLAS		VERTICAL (opens up/down) 	
		HORIZONTAL (opens left/right) 	
		• The _____ always lies inside the parabola. • The _____ always lies outside the parabola. • The _____ and _____ are perpendicular. • The vertex is _____ units away from the focus and directrix.	
STANDARD FORM Centered at (h, k)			
DIRECTION OF OPENING		Opens UP if _____ Opens DOWN if _____	Opens RIGHT if _____ Opens LEFT if _____
Graphing PARABOLAS		✓ Write the equation in standard form. Determine the type of parabola. ✓ Graph the vertex, (h, k). ✓ Solve for p to determine the direction of the opening. ✓ Use p to graph the focus and directrix. ✓ Sketch the parabola.	

Directions: Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

1. $x^2 = -8y$



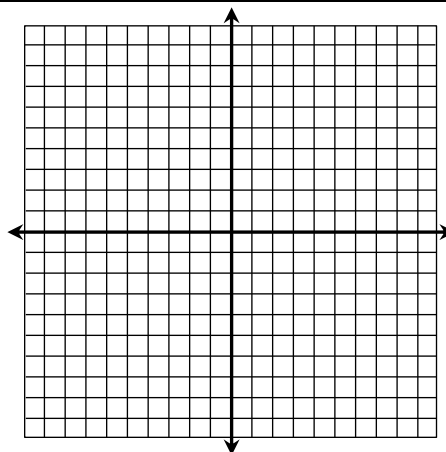
Vertex:

Axis of Symmetry:

Focus:

Directrix:

2. $y^2 = 6x$



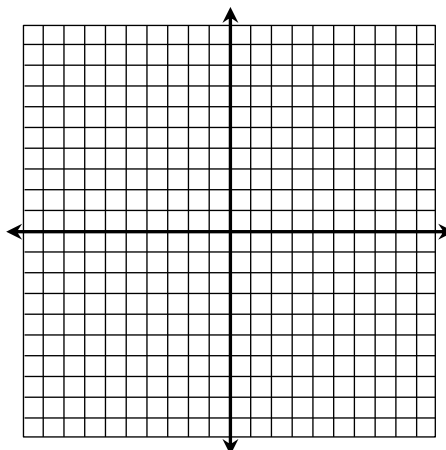
Vertex:

Axis of Symmetry:

Focus:

Directrix:

3. $(y - 2)^2 = -4(x - 3)$



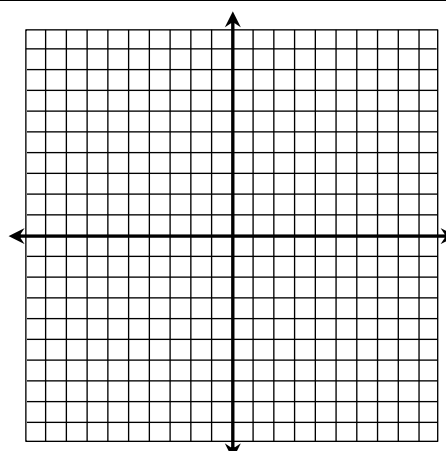
Vertex:

Axis of Symmetry:

Focus:

Directrix:

4. $(x + 1)^2 = 14(y + 2)$

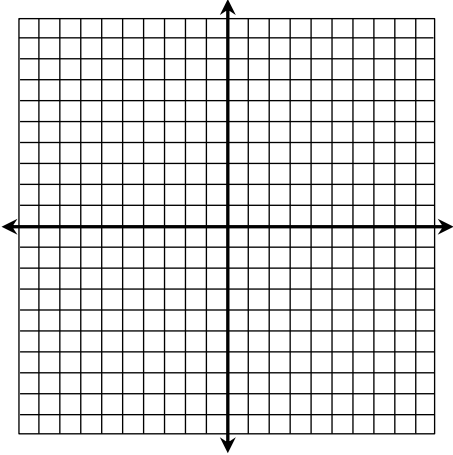
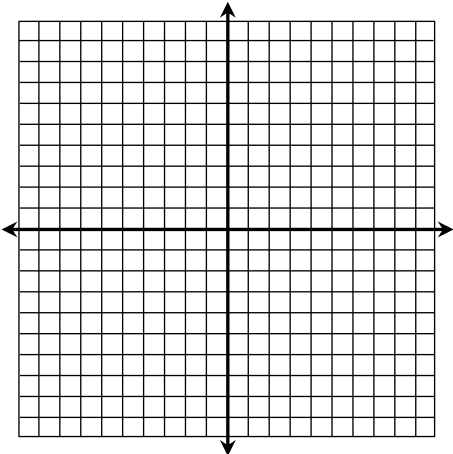
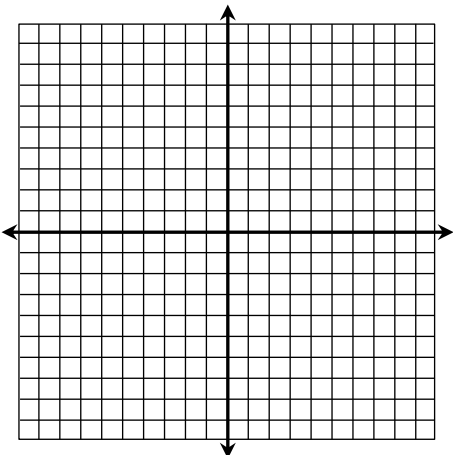
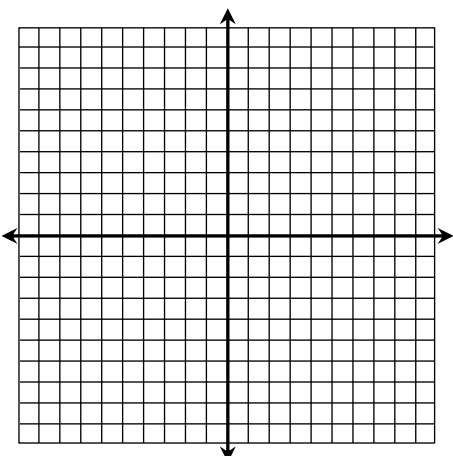


Vertex:

Axis of Symmetry:

Focus:

Directrix:

5. $\frac{1}{4}x^2 = y$ 	Vertex: Axis of Symmetry: Focus: Directrix:
6. $-\frac{1}{10}(y+2)^2 - 3 = x$ 	Vertex: Axis of Symmetry: Focus: Directrix:
7. $y^2 - 16x - 2y = -33$ 	Vertex: Axis of Symmetry: Focus: Directrix:
8. $x^2 + 10x + 4y + 1 = 0$ 	Vertex: Axis of Symmetry: Focus: Directrix:

Name: _____

Unit 9: Conic Sections

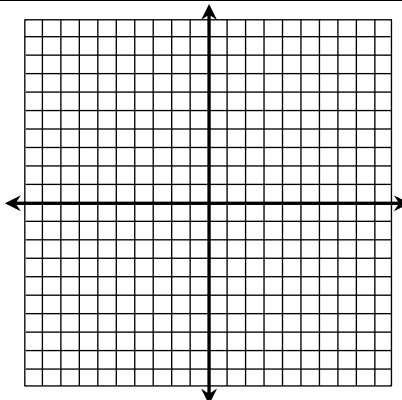
Date: _____ Per: _____

Homework 6: Graphing Parabolas

**** This is a 2-page document! ****

Directions: Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

1. $y^2 = 4x$



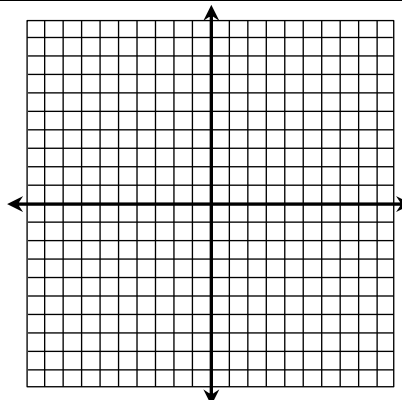
Vertex:

Axis of Symmetry:

Focus:

Directrix:

2. $x^2 = -12y$



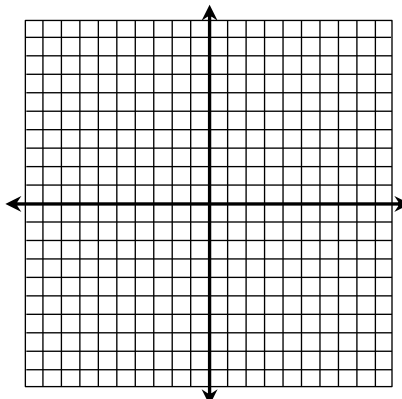
Vertex:

Axis of Symmetry:

Focus:

Directrix:

3. $(x - 2)^2 = 6(y + 4)$



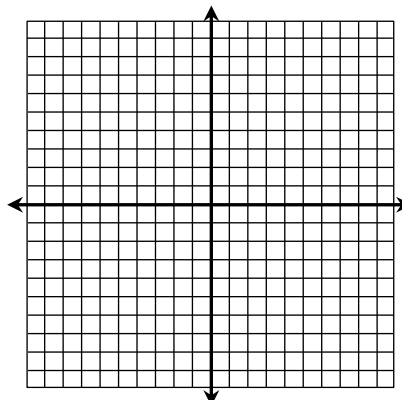
Vertex:

Axis of Symmetry:

Focus:

Directrix:

4. $(y - 5)^2 = -16(x - 1)$



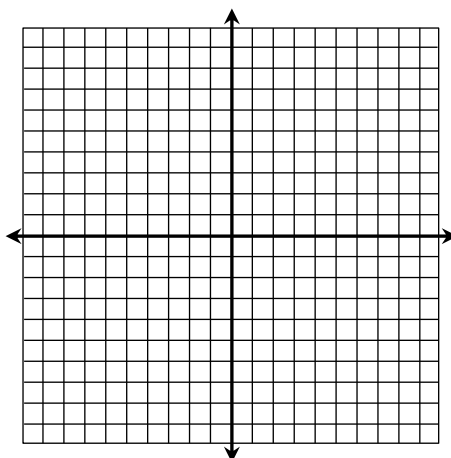
Vertex:

Axis of Symmetry:

Focus:

Directrix:

5. $\frac{1}{20}x^2 + 6 = y$



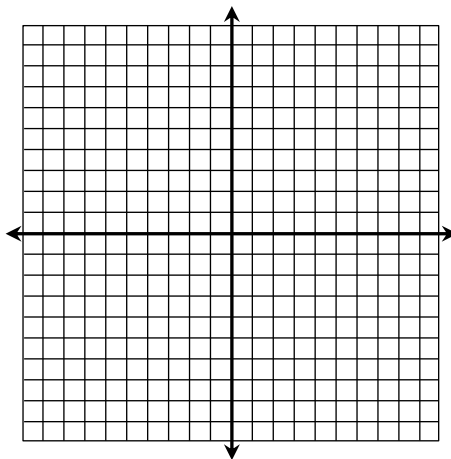
Vertex:

Axis of Symmetry:

Focus:

Directrix:

6. $-\frac{1}{8}(y+3)^2 = x-3$



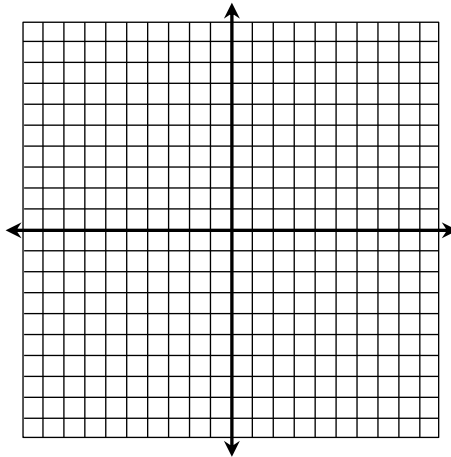
Vertex:

Axis of Symmetry:

Focus:

Directrix:

7. $y^2 + 4x + 10y + 33 = 0$



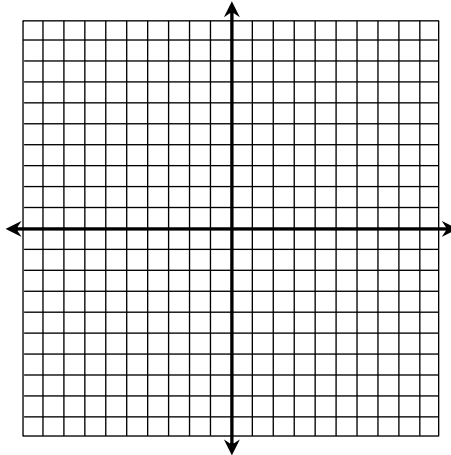
Vertex:

Axis of Symmetry:

Focus:

Directrix:

8. $x^2 - 6x - 18y - 27 = 0$



Vertex:

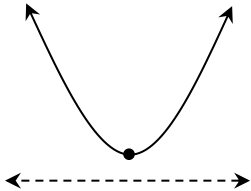
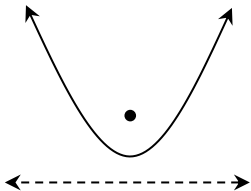
Axis of Symmetry:

Focus:

Directrix:

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
PARABOLA Equations	Standard Form for the Equation of a Parabola at Vertex (h, k) :	
	VERTICAL PARABOLA	HORIZONTAL PARABOLA
	_____ is the distance from the vertex to the focus and the directrix.	
SET 1: Given a Graph	1.	2.
	3.	4.
	5. Vertex: $(0, 0)$; Focus: $(0, 4)$	6. Vertex: $(0, 0)$; Focus: $(\frac{5}{2}, 0)$
SET 2: Given the Vertex & Focus		

	7. Vertex: $(-7, 1)$; Focus: $\left(-7, \frac{3}{4}\right)$	8. Vertex: $(2, -6)$; Focus: $(-3, -6)$
SET 3: Given the Vertex & Directrix 	9. Vertex: $(0, 0)$; Directrix: $y = 8$	10. Vertex: $(0, 0)$; Directrix: $x = -2$
	11. Vertex: $(7, 4)$; Directrix: $x = \frac{3}{2}$	12. Vertex: $(-1, -1)$; Directrix: $y = -8$
SET 4: Given the Focus & Directrix 	13. Focus: $(3, 0)$; Directrix: $x = -3$	14. Focus: $(7, -2)$; Directrix: $y = -4$
	15. Focus: $(-8, 5)$; Directrix: $x = 4$	16. Focus: $\left(-4, -\frac{3}{2}\right)$; Directrix: $y = -\frac{1}{2}$

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

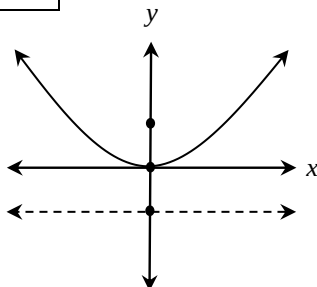
Homework 7: Writing Equations of Parabolas

**** This is a 2-page document! ****

Label p , h , and k on each diagram and give the equation of the parabola in standard form.

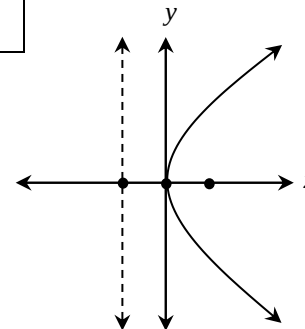
VERTICAL PARABOLA

Standard Form:



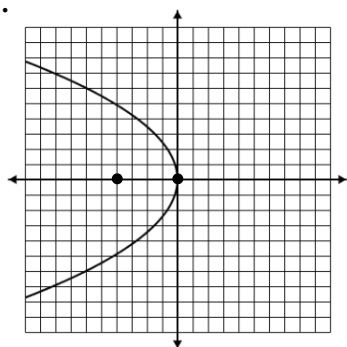
HORIZONTAL PARABOLA

Standard Form:

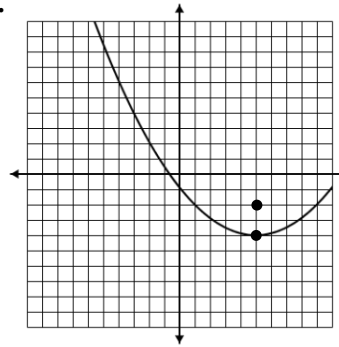


Write an equation for each parabola.

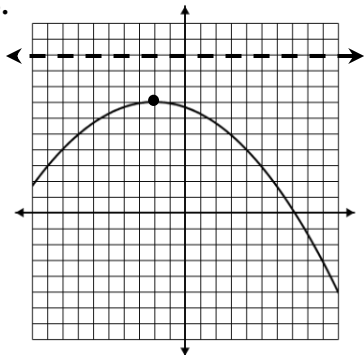
1.



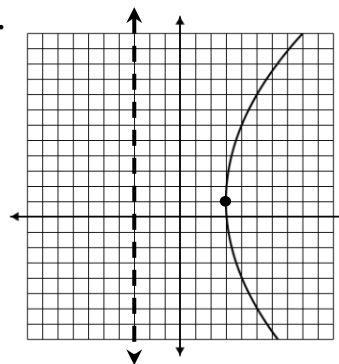
2.



3.



4.



Write an equation for each parabola with the given information.

5. Vertex: $(0, 0)$; Focus: $(2, 0)$

6. Vertex: $(0, 0)$; Focus: $\left(0, -\frac{7}{4}\right)$

7. Vertex: $(-6, -5)$; Focus: $(-6, -4)$	8. Vertex: $(3, -4)$; Focus: $\left(\frac{1}{2}, -4\right)$
9. Vertex: $(0, 0)$; Directrix: $y = -5$	10. Vertex: $(-4, -2)$; Directrix: $x = -7$
11. Vertex: $(5, 0)$; Directrix: $y = 6$	12. Vertex: $(-1, -1)$; Directrix: $x = -\frac{1}{2}$
13. Focus: $(0, -6)$; Directrix: $y = 6$	14. Focus: $(-1, 5)$; Directrix: $x = 7$
15. Focus: $(6, -4)$; Directrix: $x = 0$	16. Focus: $\left(-7, \frac{17}{4}\right)$; Directrix: $y = -\frac{1}{4}$

Name: _____

Algebra 2

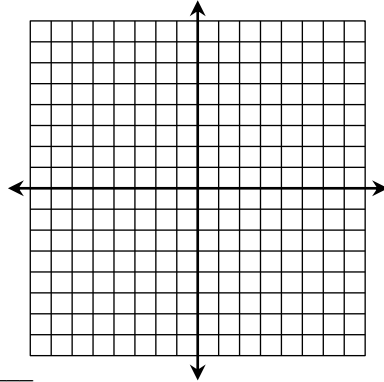
Date: _____ Per: _____

Unit 9: Conic Sections

Quiz 9-2: Hyperbolas and Parabolas

Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{16} - \frac{y^2}{49} = 1$



Center: _____

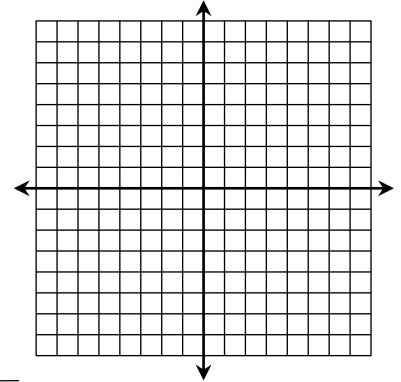
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

2. $\frac{(y+1)^2}{36} - \frac{(x-2)^2}{4} = 1$



Center: _____

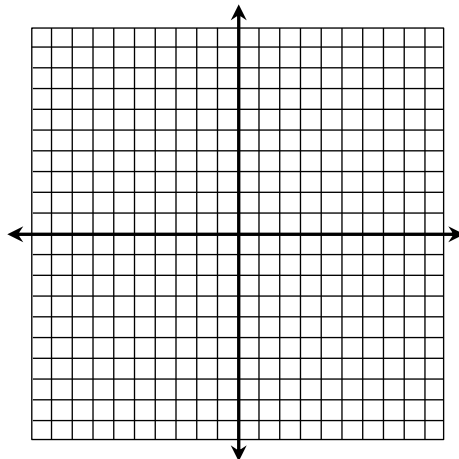
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

3. $x^2 - 25y^2 + 8x = 9$



Standard Form: _____

Center: _____

Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

Write the equation of the hyperbola in standard form with the given characteristics.

4. Vertices: (-5, 9) and (-5, -11)
Co-Vertices: (2, -1) and (-12, -1)

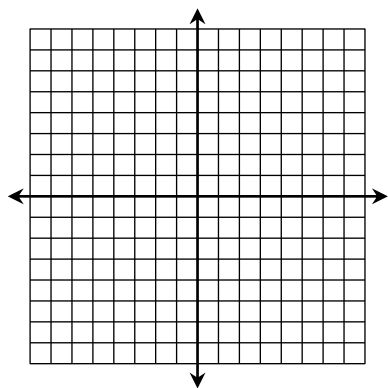
5. Vertices: (7, 2) and (1, 2)
Foci: (9, 2) and (-1, 2)

Standard Form: _____

Standard Form: _____

Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

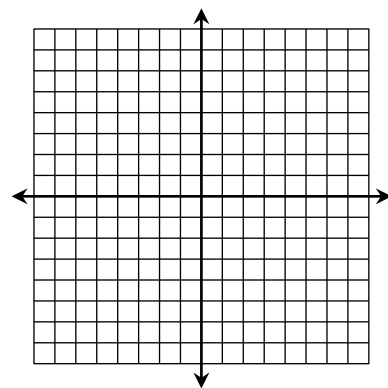
6. $x^2 = -4(y - 5)$



Vertex: _____ Axis of Sym: _____

Focus: _____ Directrix: _____

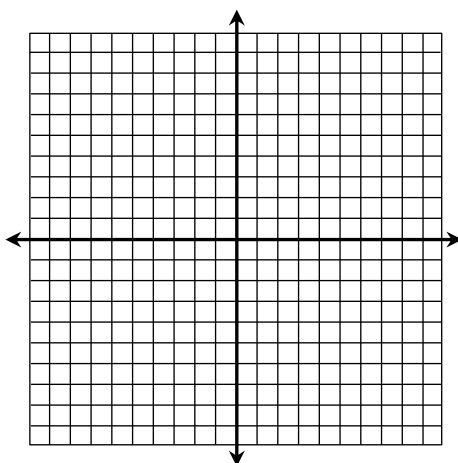
7. $(y + 3)^2 = 10(x - 2)$



Vertex: _____ Axis of Sym: _____

Focus: _____ Directrix: _____

8. $y^2 - 16x - 2y + 81 = 0$



Standard Form: _____

Vertex: _____

Axis of Symmetry: _____

Focus: _____

Directrix: _____

Write the equation of the parabola in standard form with the given characteristics.

9. Vertex: (0, 0); Focus: (0, -6)

10. Vertex: (3, -4); Focus: (5, -4)

Standard Form: _____

Standard Form: _____

11. Vertex: (-2, -6); Directrix: $x = -\frac{5}{2}$

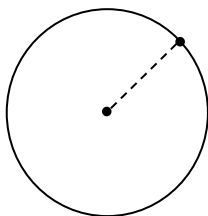
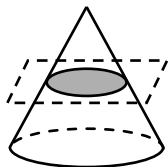
12. Focus: (-5, 2); Directrix: $y = 0$

Standard Form: _____

Standard Form: _____

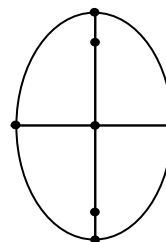
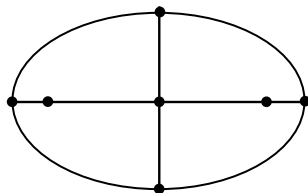
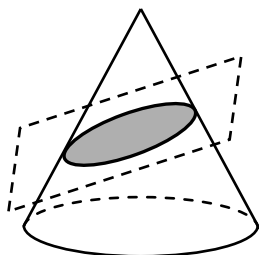
CONIC SECTIONS

CIRCLES



Standard Form:

ELLIPSES

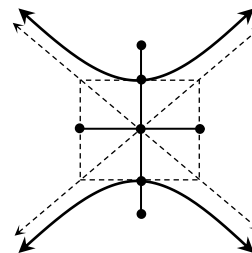
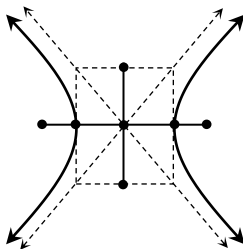
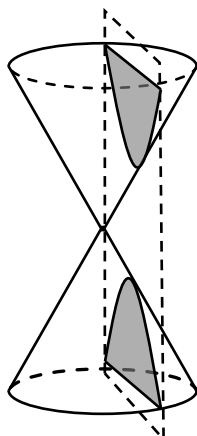


Standard Form:

Standard Form:

Formula for c:

HYPERBOLAS



Standard Form:

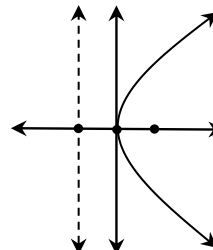
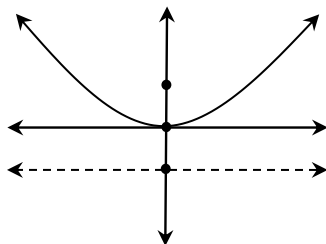
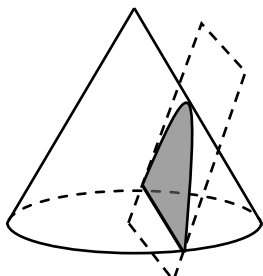
Standard Form:

Asymptotes:

Asymptotes:

Formula for c:

PARABOLAS



Standard Form:

Standard Form:

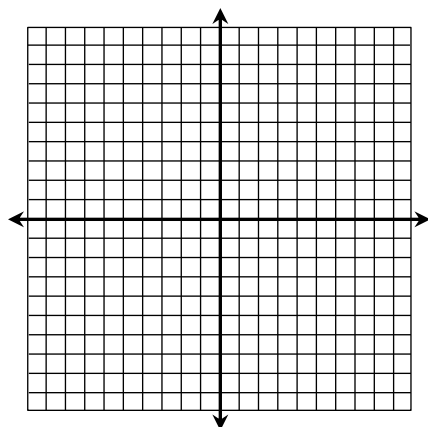
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
GENERAL FORM of a Conic Equation		
WHICH CONIC IS IT?	➤ Which variables are squared? ○ If just one is squared, it's a _____. ○ If both are squared, go to the next test. ➤ Look at the coefficients. ○ If they are exactly the same: _____ ○ If they are different but have the same sign: _____ ○ If they have opposite signs: _____	
EXAMPLES	State the conic represented by each equation below.	
	1. $4x^2 + y^2 - 48x - 14y + 189 = 0$	2. $x^2 - 18x + 6y + 99 = 0$
	3. $x^2 + y^2 - 10x - 6y + 18 = 0$	4. $-x^2 + 9y^2 + 8x - 54y + 56 = 0$

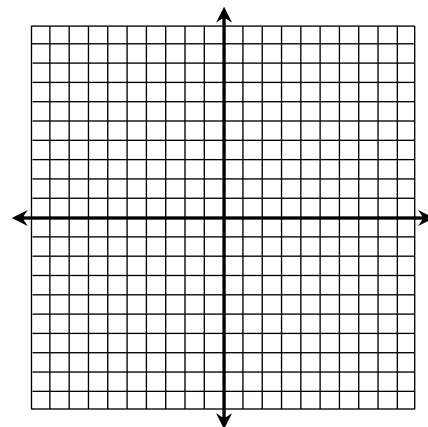
GRAPHING REVIEW

Determine the type of conic, write the equation in standard form, then graph.

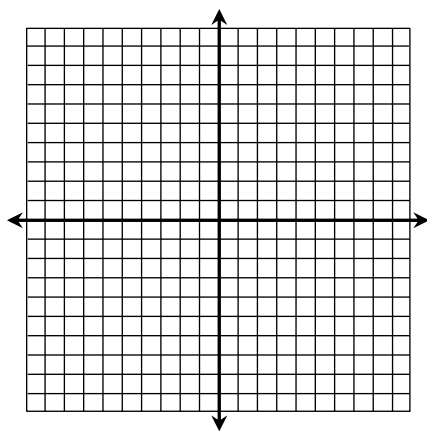
5. $x^2 - 8x + y + 14 = 0$



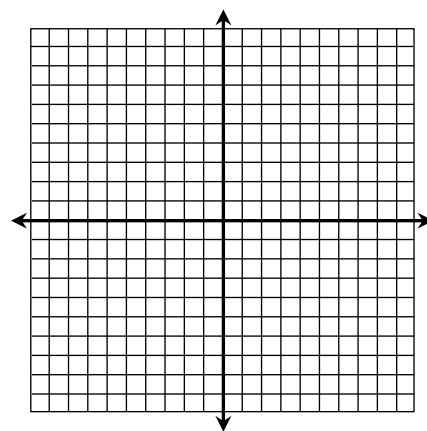
6. $x^2 - y^2 - 4x - 6y - 9 = 0$



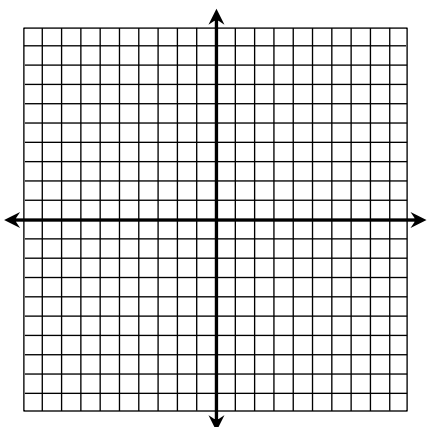
7. $x^2 + y^2 + 10x + 2y + 17 = 0$



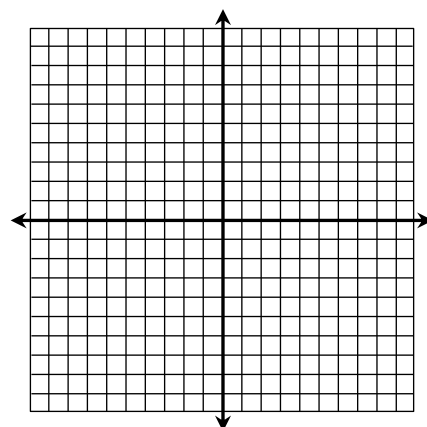
8. $4x^2 + y^2 + 2y - 35 = 0$



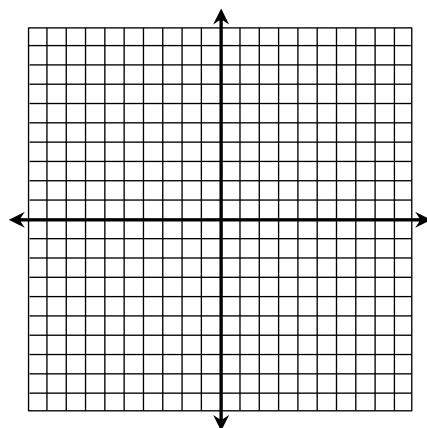
9. $y^2 - 4x + 8y + 4 = 0$



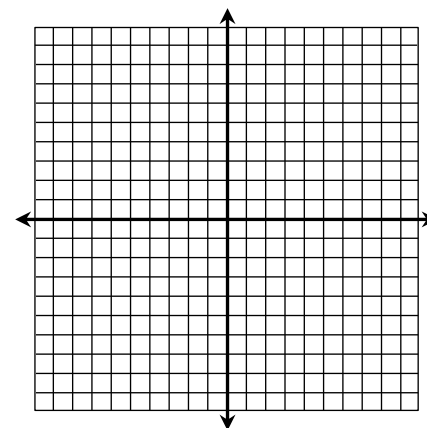
10. $x^2 + y^2 + 12x - 2y + 21 = 0$



11. $-16x^2 + 9y^2 - 64x - 18y - 199 = 0$



12. $x^2 + 9y^2 + 2x + 90y + 190 = 0$



Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 8: Identifying Conic Sections

**** This is a 2-page document! ****

State the conic represented by each equation below.

1. $9x^2 - 4y^2 - 36x + 16y - 16 = 0$

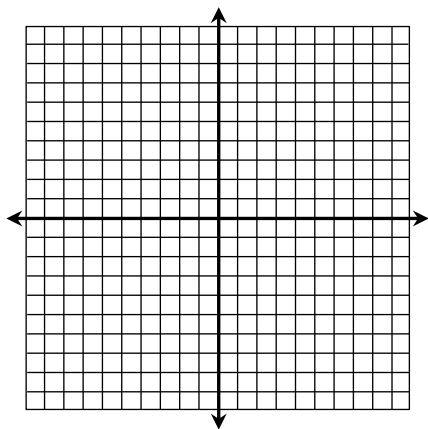
2. $x^2 + y^2 + 4x - 8y + 11 = 0$

3. $y^2 - x - 10y + 28 = 0$

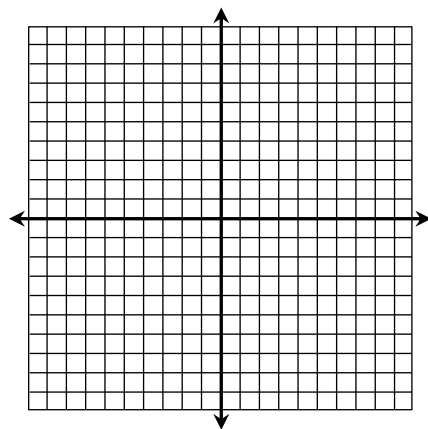
4. $49x^2 + 4y^2 - 98x - 147 = 0$

Determine the type of conic, write the equation in standard form, then graph.

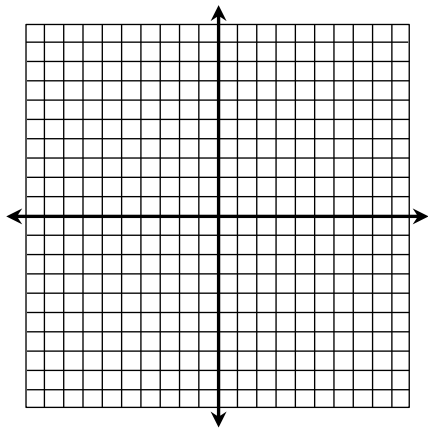
5. $x^2 + y^2 - 14x - 6y + 57 = 0$



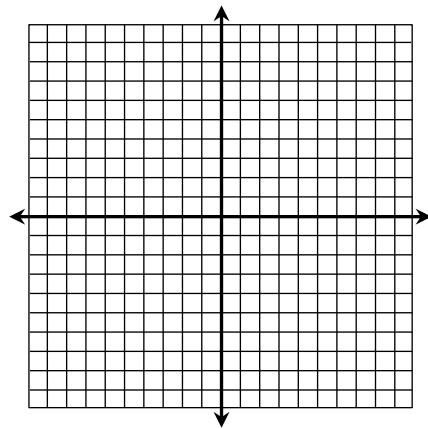
6. $16x^2 + 25y^2 + 64x - 50y - 311 = 0$



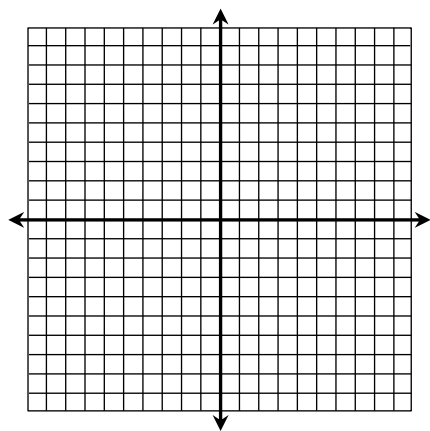
7. $y^2 + x + 2y = 0$



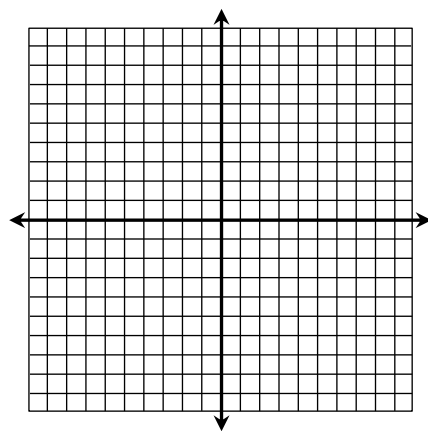
8. $-x^2 + y^2 + 6x + 6y - 4 = 0$



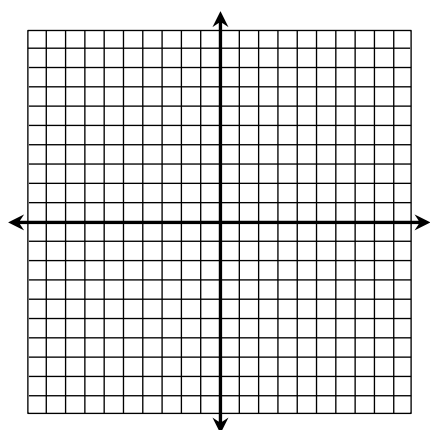
9. $2x^2 - 8x - 3y + 11 = 0$



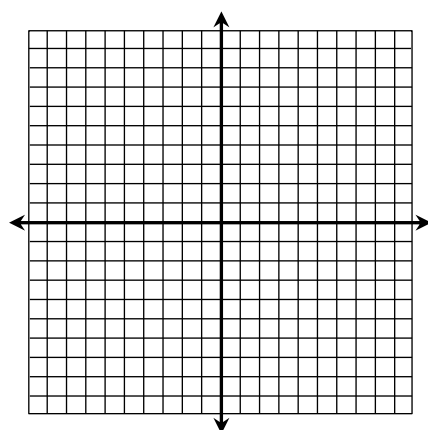
10. $x^2 + y^2 - 8x - 20 = 0$



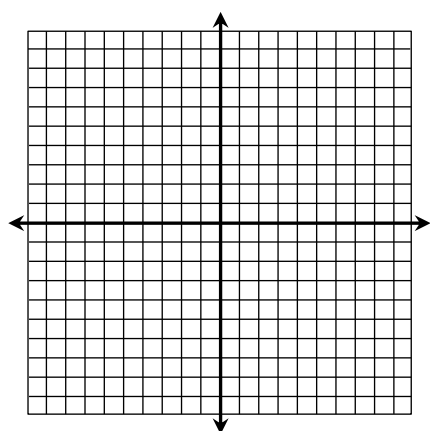
11. $4x^2 + y^2 - 56x + 6y + 189 = 0$



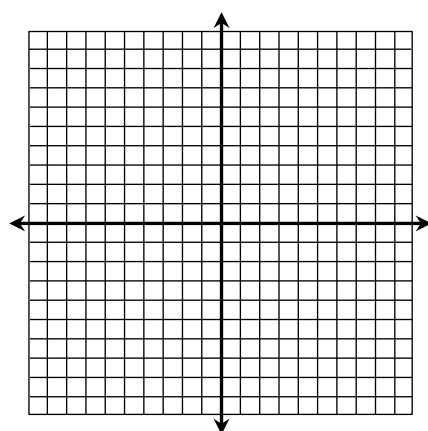
12. $x^2 - y^2 - 25 = 0$



13. $2x^2 + 2y^2 + 12x - 20y + 60 = 0$



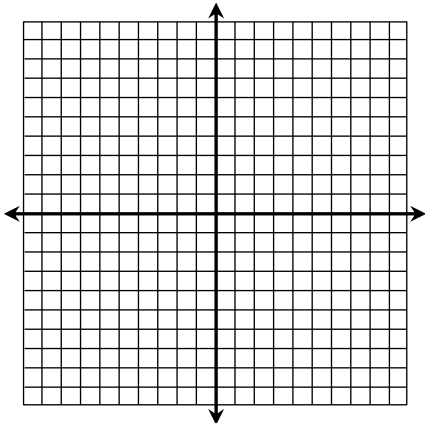
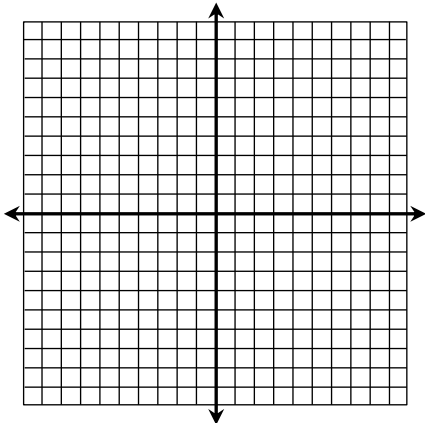
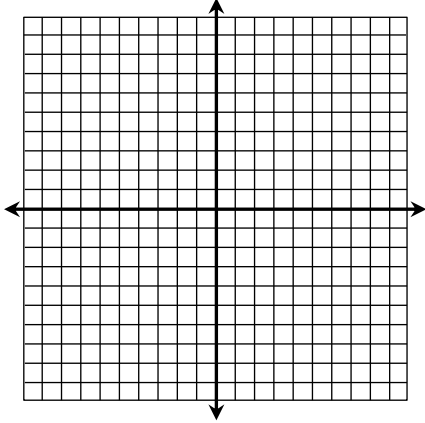
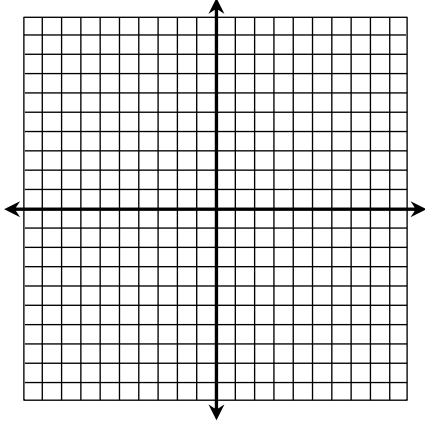
14. $y^2 - 12x - 2y - 71 = 0$



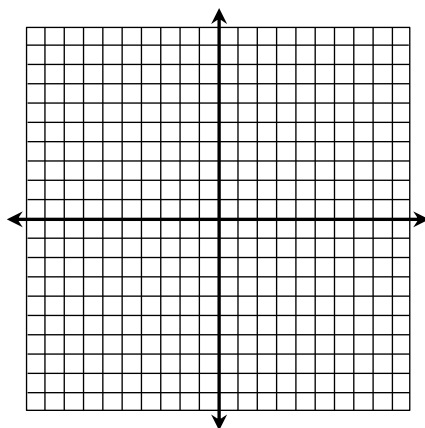
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
<i>Non-Linear</i> SYSTEM OF EQUATIONS	
SOLUTIONS <i>to Non-Linear Systems</i>	- The solution set to a system of equation is the set of points in which the equations _____. - There can be one solution, multiple solutions, or no real solutions.

SOLVING A NON-LINEAR SYSTEM *by* **GRAPHING**

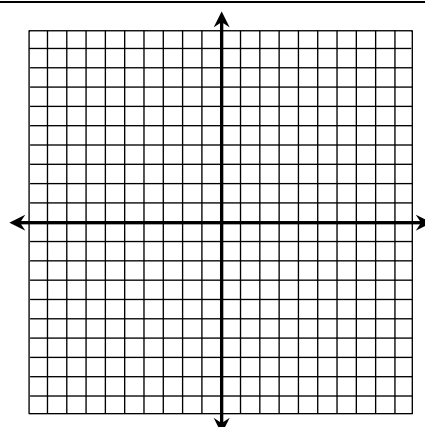
1. $y = -x^2 + 7$ $y = x + 5$  Solutions:	2. $y = 2x^2 - 8x + 3$ $y = -2x - 1$  Solutions:
3. $y = x^2 + 6x + 4$ $x + y = -2$  Solutions:	4. $y = x^2 + 2x - 8$ $4x - y = 5$  Solutions:

5. $y = x^2 - 9$
 $y = -x^2 - 6x - 9$



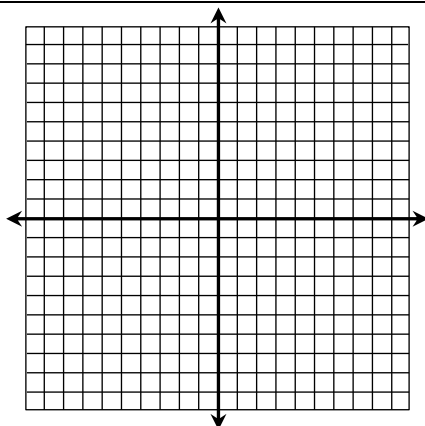
Solutions:

7. $y = -x^2 - 6x - 11$
 $y = x^2 - 4x$



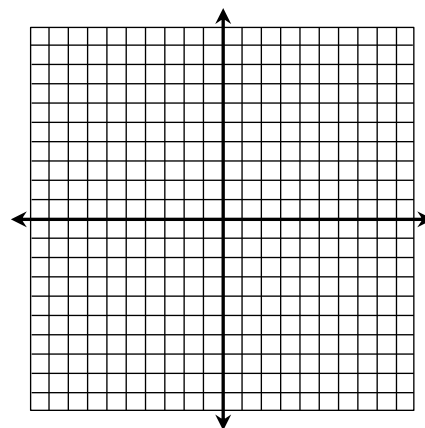
Solutions:

9. $x^2 + y^2 = 25$
 $3x - 4y = 0$



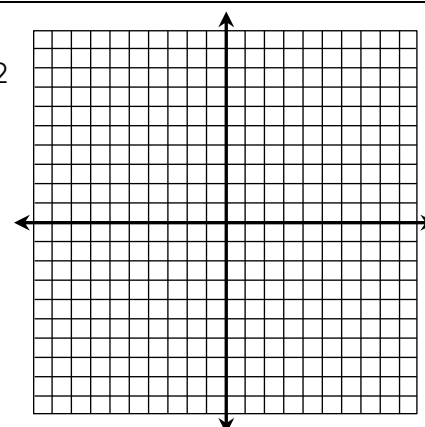
Solutions:

6. $y = 2x^2 - 8x + 7$
 $y = -x^2 + 4x - 2$



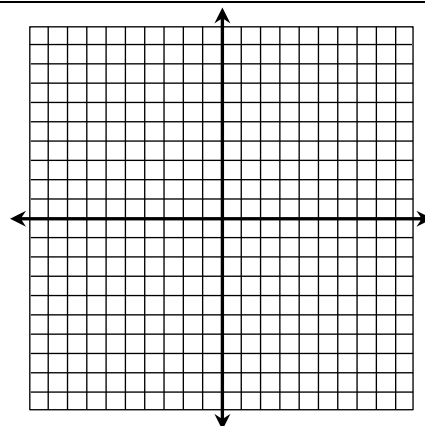
Solutions:

8. $y = x^2 - 3$
 $y = -2x^2 + 12x - 12$



Solutions:

10. $y^2 = -4x + 4$
 $4x^2 + 9y^2 = 36$



Solutions:

Name: _____

Unit 9: Conic Sections

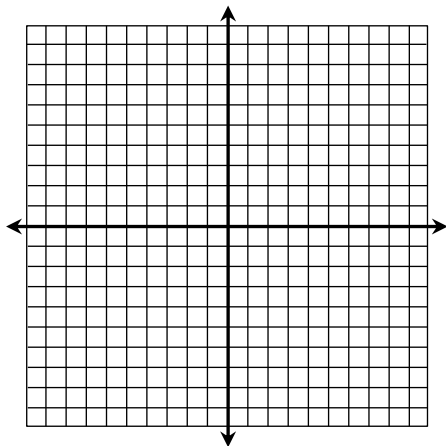
Date: _____ Per: _____

Homework 9: Non-Linear Systems (Day 1)

**** This is a 2-page document! ****

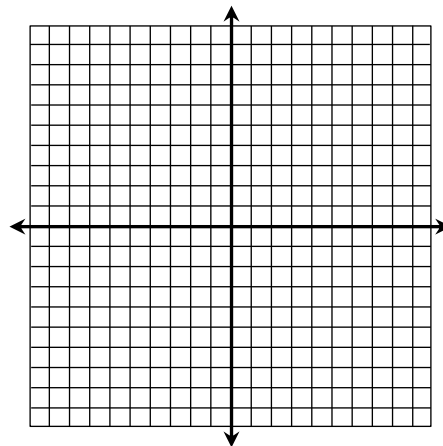
Solve each system of equations by graphing. Identify all possible solutions.

1. $y = x^2 + 6x$
 $y = x - 4$



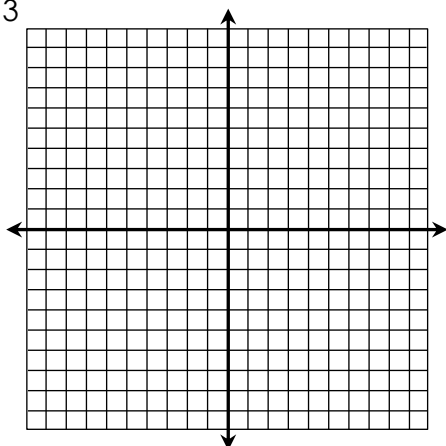
Solutions:

2. $y = x^2 + 6x + 2$
 $y = -x - 8$



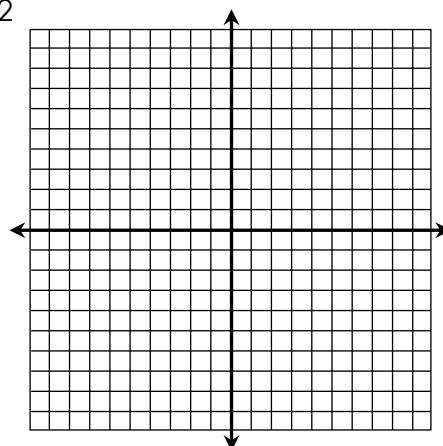
Solutions:

3. $x - 2y = -6$
 $y = -x^2 + 4x - 3$



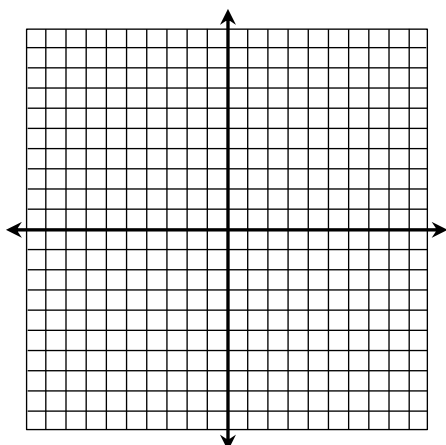
Solutions:

4. $4x + y = -4$
 $y = -3x^2 - 7x + 2$



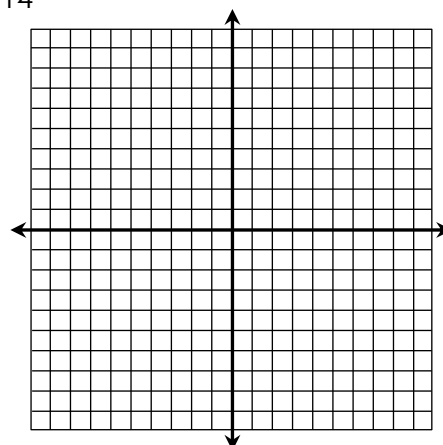
Solutions:

5. $y = -x^2 + 2x$
 $y = x^2 - 2x - 6$



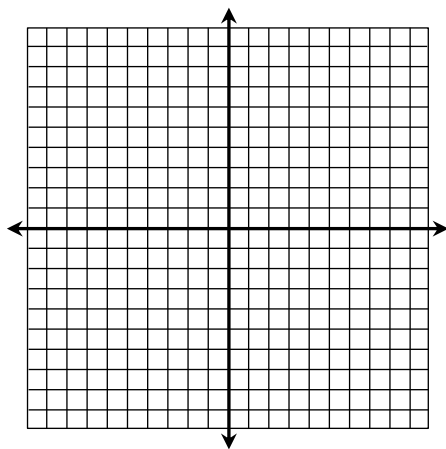
Solutions:

6. $y = x^2 - 8x + 10$
 $y = -2x^2 - 12x - 14$



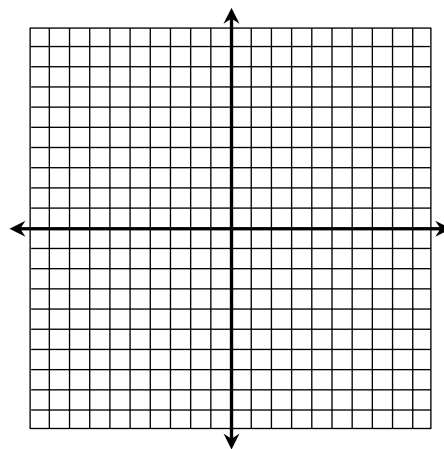
Solutions:

7. $y = 2x^2 + 20x + 47$
 $y = -x^2 - 10x - 28$



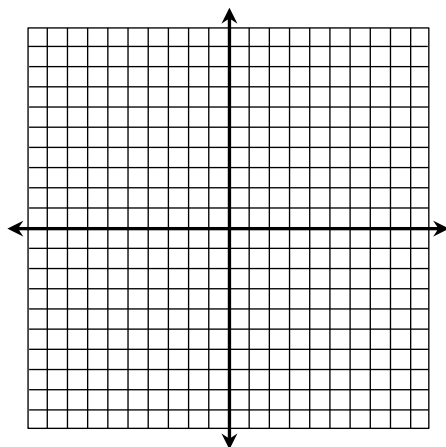
Solutions:

8. $y = x^2 - 4x + 3$
 $y = -x^2 + 12x - 27$



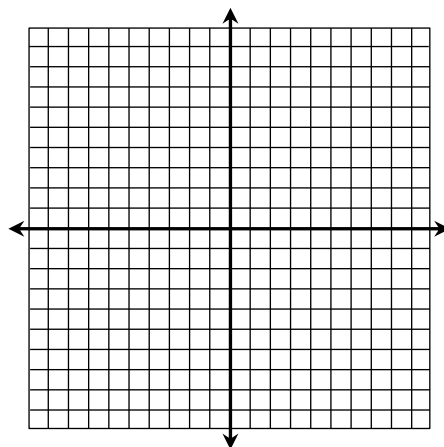
Solutions:

9. $x^2 + y^2 = 16$
 $x - y = 6$



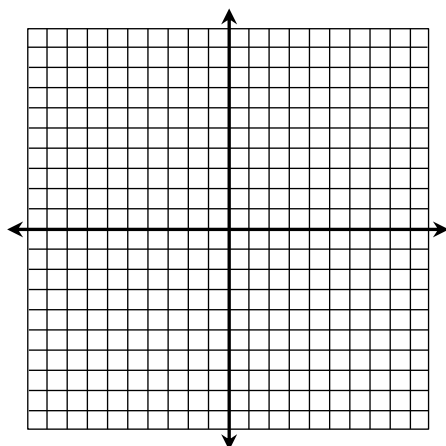
Solutions:

10. $5x + 3y = 0$
 $y^2 - x^2 = 16$



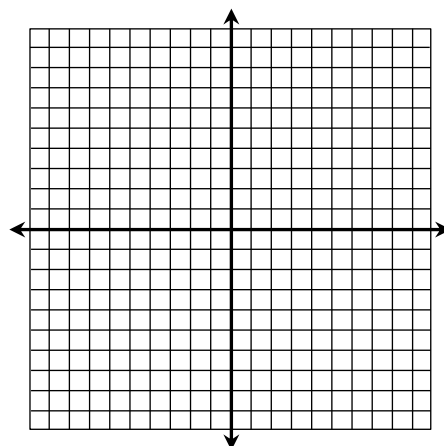
Solutions:

11. $(x+1)^2 + (y+4)^2 = 4$
 $y = -x^2 - 2x - 1$



Solutions:

12. $y^2 = 4x$
 $16(x-4)^2 + y^2 = 16$



Solutions:

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes/Examples
SUBSTITUTION <i>Method</i>	$\begin{cases} 4x + y = -20 \\ y = x^2 + 5x - 2 \end{cases}$
ELIMINATION <i>Method</i>	$\begin{cases} y = x^2 + 3x + 2 \\ y = x^2 + 5x - 4 \end{cases}$
EXAMPLES	Solve each system algebraically using your method of choice.
	1. $\begin{cases} y = x^2 - 3x - 2 \\ y = x + 3 \end{cases}$
	2. $\begin{cases} y = -x^2 - 8x - 6 \\ 2x + y = -1 \end{cases}$
	3. $\begin{cases} 3x - y = 7 \\ y = 2x^2 + 20x + 46 \end{cases}$

$$4. \begin{cases} x^2 + y^2 = 18 \\ y = x - 6 \end{cases}$$

$$5. \begin{cases} y^2 - x^2 + 75 = 0 \\ 2y - x = 0 \end{cases}$$

$$6. \begin{cases} y = 2x^2 + 3x - 19 \\ y = x^2 + 2x - 7 \end{cases}$$

$$7. \begin{cases} y = 4x^2 - 29 \\ y = x^2 - 17 \end{cases}$$

$$8. \begin{cases} x^2 + 2y^2 = 17 \\ 2x^2 - 3y^2 = 6 \end{cases}$$

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 10: Non-Linear Systems (Day 2)

**** This is a 2-page document! ****

Solve each system of equations algebraically using substitution or elimination.

1.
$$\begin{cases} y = x^2 - 9x + 12 \\ y = -5x + 9 \end{cases}$$

2.
$$\begin{cases} y = -x^2 - x - 5 \\ x - y = 5 \end{cases}$$

3.
$$\begin{cases} y = 2x^2 + 4x \\ x + y = -3 \end{cases}$$

4.
$$\begin{cases} y = x^2 - 2x - 3 \\ 2x - y = 7 \end{cases}$$

5.
$$\begin{cases} x^2 + y^2 = 40 \\ x - y = 4 \end{cases}$$

6.
$$\begin{cases} y^2 = 8x - 3 \\ 2x - y = -1 \end{cases}$$

$$7. \begin{cases} x^2 = 2y - 9 \\ 4x - y = 3 \end{cases}$$

$$8. \begin{cases} x^2 - 9y^2 = 28 \\ x = 4y \end{cases}$$

$$9. \begin{cases} y = x^2 + 5x - 13 \\ y = x^2 - 3x + 3 \end{cases}$$

$$10. \begin{cases} y = 3x^2 - 4x - 11 \\ y = 2x^2 - 7x - 13 \end{cases}$$

$$11. \begin{cases} y = x^2 + 9x + 7 \\ y = -x^2 + 3x + 7 \end{cases}$$

$$12. \begin{cases} x^2 + y^2 = 13 \\ x^2 + 4y^2 = 25 \end{cases}$$

Name: _____

Algebra 2

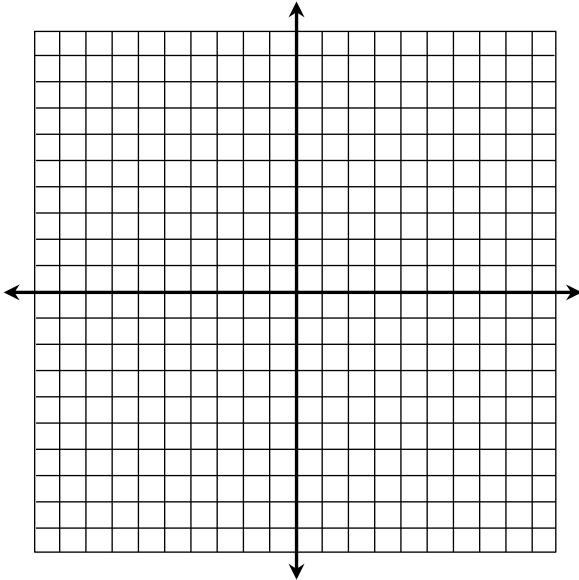
Date: _____ Per: _____

Unit 9: Conic Sections

Quiz 9-3: Non-Linear Systems of Equations

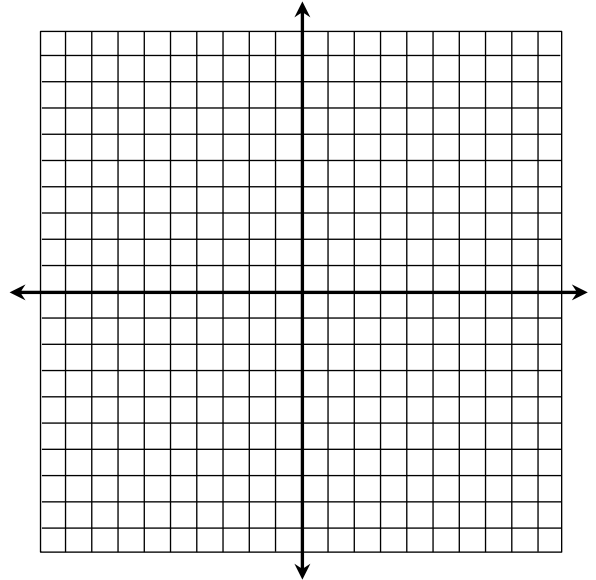
Solve each system by graphing.

1.
$$\begin{cases} y = x^2 - 10x + 17 \\ 4x + y = 9 \end{cases}$$



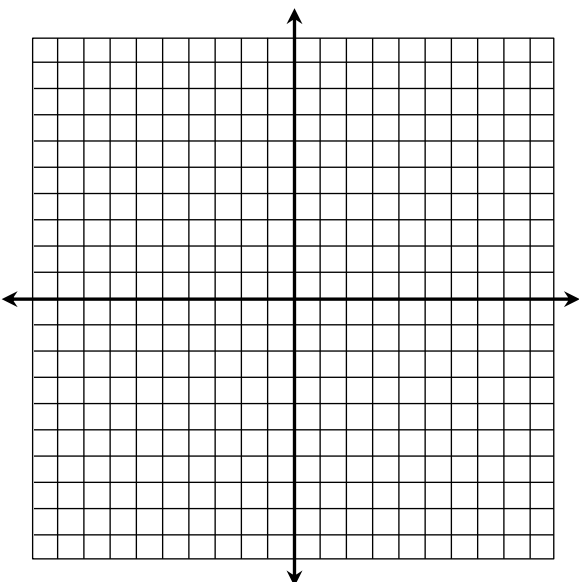
Solution(s): _____

2.
$$\begin{cases} x - y = -8 \\ y = -x^2 - 4x + 2 \end{cases}$$



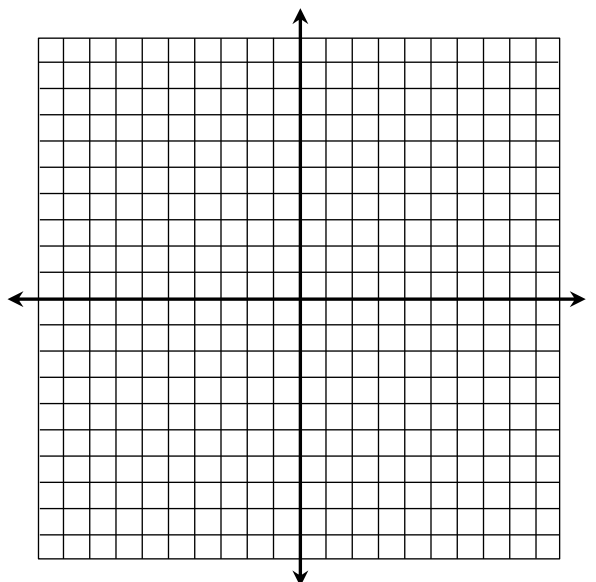
Solution(s): _____

3.
$$\begin{cases} y = x^2 - 6x + 5 \\ y = -x^2 + 6x - 11 \end{cases}$$



Solution(s): _____

4.
$$\begin{cases} y = 2x^2 + 20x + 47 \\ x^2 + y^2 = 4 \end{cases}$$



Solution(s): _____

Solve each system using substitution or elimination.

5.
$$\begin{cases} y = x^2 - 3x - 8 \\ y = x - 3 \end{cases}$$

6.
$$\begin{cases} y = -x^2 + 9x - 13 \\ 3x - y = 5 \end{cases}$$

Solution(s): _____

Solution(s): _____

7.
$$\begin{cases} x = y + 2 \\ x^2 + y^2 = 10 \end{cases}$$

8.
$$\begin{cases} y^2 - 2x - 10 = 0 \\ x + y = -1 \end{cases}$$

Solution(s): _____

Solution(s): _____

9.
$$\begin{cases} y = x^2 + 4x - 29 \\ y = x^2 + 2x - 45 \end{cases}$$

10.
$$\begin{cases} x^2 + y^2 = 41 \\ x^2 - y^2 = 9 \end{cases}$$

Solution(s): _____

Solution(s): _____

Unit 9 Test Study Guide

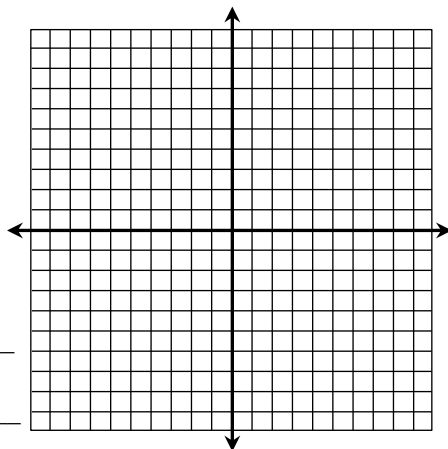
(Conic Sections)

Name: _____

Date: _____ Per: _____

Topic 1: Graphing Circles, Ellipses, Hyperbolas, and Parabolas

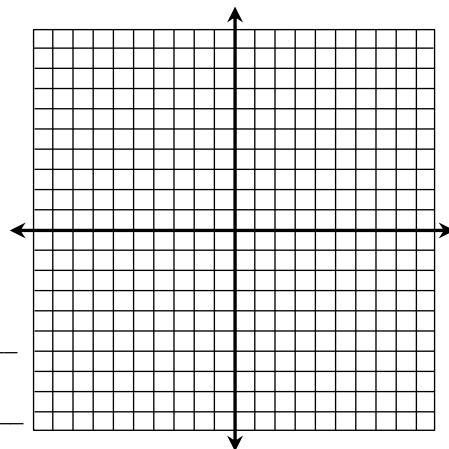
1. $8x^2 + 8y^2 = 72$



Center: _____

Radius: _____

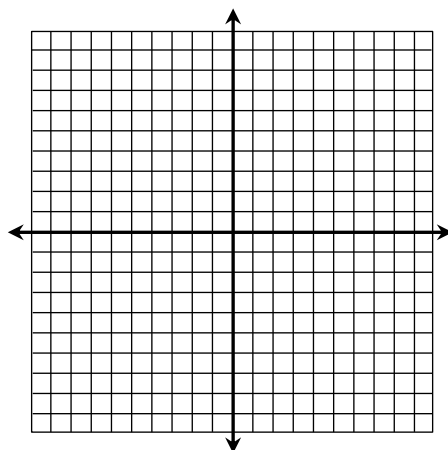
2. $(x + 3)^2 + (y - 1)^2 = 49$



Center: _____

Radius: _____

3. $x^2 + 36y^2 = 36$



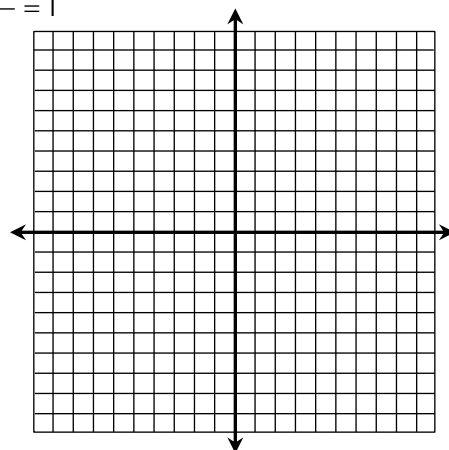
Center: _____

Vertices: _____

Co-Vertices: _____

Foci: _____

4. $\frac{(x + 5)^2}{25} + \frac{(y - 1)^2}{81} = 1$



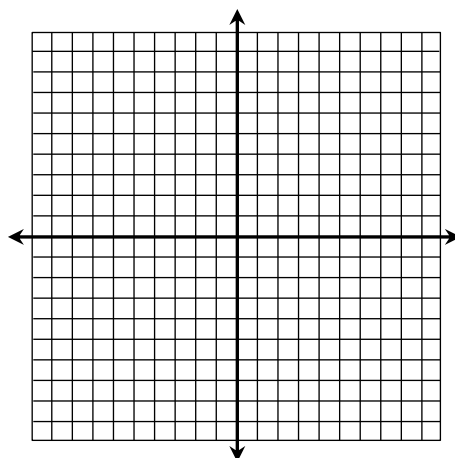
Center: _____

Vertices: _____

Co-Vertices: _____

Foci: _____

5. $16x^2 - 9y^2 = 144$



Center: _____

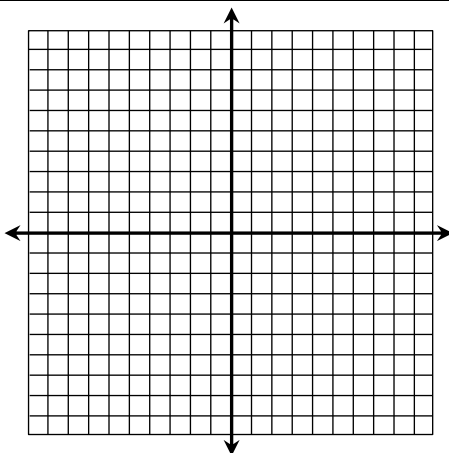
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

6. $\frac{(y-3)^2}{4} - \frac{(x-4)^2}{36} = 1$



Center: _____

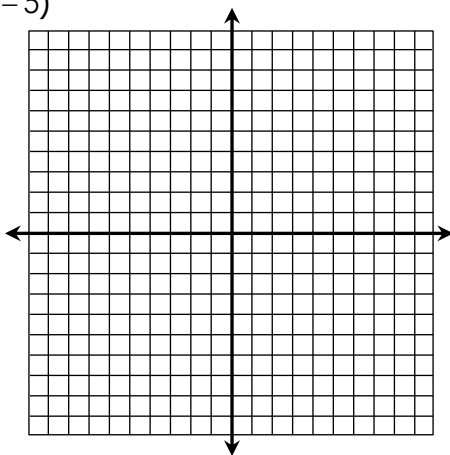
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

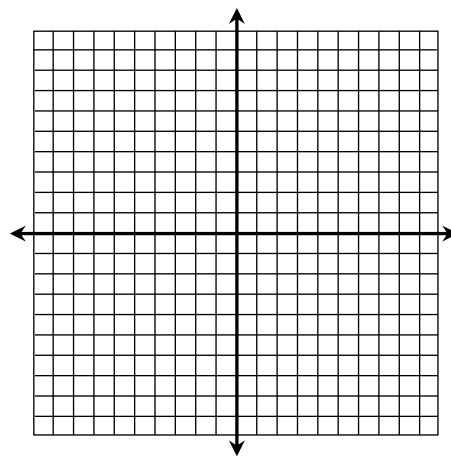
7. $(x+4)^2 = -8(y-5)$



Vertex: _____ Axis of Sym: _____

Focus: _____ Directrix: _____

8. $\frac{1}{20}y^2 = x+1$



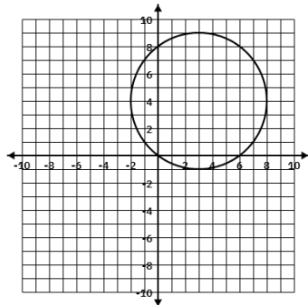
Vertex: _____ Axis of Sym: _____

Focus: _____ Directrix: _____

Topic 2: Writing Equations of Circles, Ellipses, Hyperbolas, and Parabolas

Write the equation of the circle in standard form with the information given.

9.



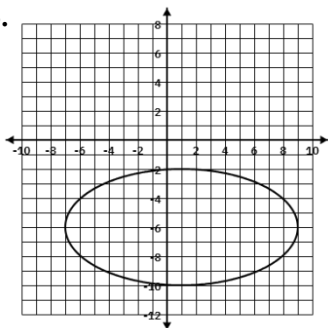
10. Center: $(-2, 7)$, Radius: $5\sqrt{3}$

11. Center: $(-4, -1)$, Endpoint: $(2, 3)$

12. Endpoints of Diameter: $(16, -2)$ and $(2, -8)$

Write the equation of the ellipse in standard form with the information given.

13.



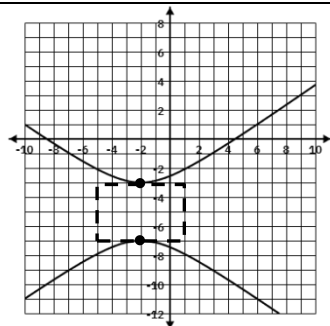
14. Vertices: $(-3, 12)$ and $(-3, -10)$
Co-Vertices: $(4, 1)$ and $(-10, 1)$

15. Vertices: $(8, 9)$ and $(-2, 9)$
Foci: $(7, 9)$ and $(-1, 9)$

16. Co-Vertices: $(4, -3)$ and $(-8, -3)$
Foci: $(-2, 5)$ and $(-2, -11)$

Write the equation of the hyperbola in standard form with the information given.

17.



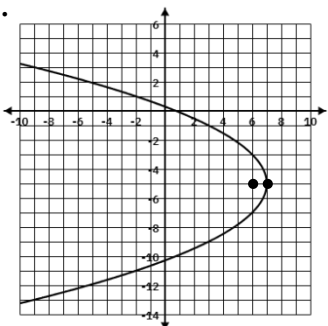
18. Vertices: $(9, 0)$ and $(-7, 0)$
Co-Vertices: $(1, 6)$ and $(1, -6)$

19. Vertices: $(3, -5)$ and $(3, -13)$
Foci: $(3, -4)$ and $(3, -14)$

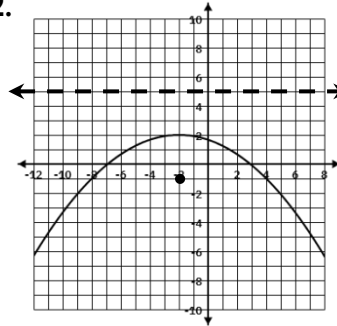
20. Co-Vertices: $(7, 16)$ and $(7, -8)$
Foci: $(20, 4)$ and $(-6, 4)$

Write the equation of the parabola in standard form with the information given.

21.



22.



23. Vertex: $(-4, 2)$; Focus: $(2, 2)$	24. Vertex: $(1, -3)$; Focus: $\left(1, -\frac{5}{2}\right)$
25. Vertex: $(6, 0)$; Directrix: $y = 10$	26. Focus: $\left(-\frac{11}{2}, -1\right)$; Directrix: $x = \frac{3}{2}$

Topic 3: Identifying Conic Sections

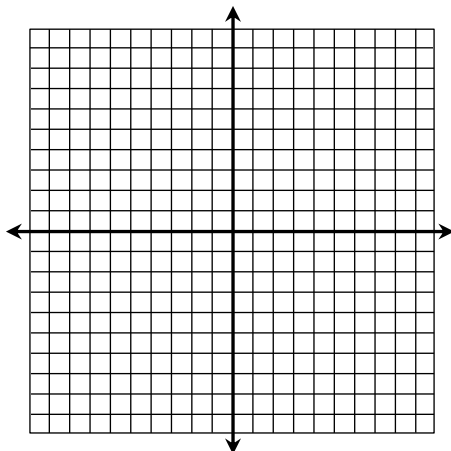
Classify the conic, then write the equation in standard form.

27. $9x^2 + y^2 + 54x - 2y + 46 = 0$	28. $x^2 + y^2 - 8x - 4y - 5 = 0$
29. $x^2 - 10x + 4y + 33 = 0$	30. $4x^2 - y^2 + 24x - 16y - 32 = 0$
31. $25x^2 + 4y^2 + 100x - 24y + 36 = 0$	32. $y^2 - 10x + 2y + 71 = 0$

Topic 4: Non-Linear Systems of Equations

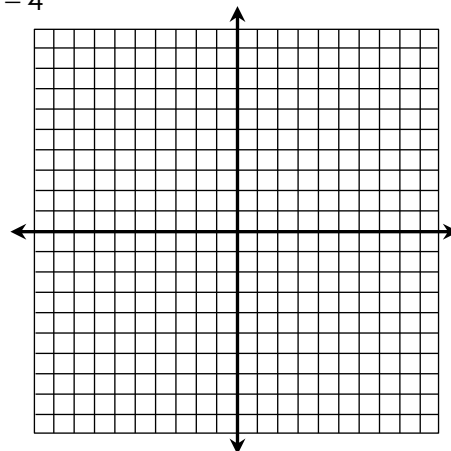
Solve by graphing.

33.
$$\begin{cases} y = -x^2 + 6x - 2 \\ x + 3y = 24 \end{cases}$$



Solution(s): _____

34.
$$\begin{cases} y = x^2 + 10x + 16 \\ y = -x^2 - 4x - 4 \end{cases}$$



Solution(s): _____

Solve algebraically.

35.
$$\begin{cases} 3x + y = 1 \\ y = x^2 - 3x \end{cases}$$

Solution(s): _____

36.
$$\begin{cases} y = -x^2 - 6x + 19 \\ y = -x^2 + 2x - 5 \end{cases}$$

Solution(s): _____

37.
$$\begin{cases} y = 2x^2 + 5x - 3 \\ y = x^2 + 5x - 2 \end{cases}$$

Solution(s): _____

38.
$$\begin{cases} x^2 + y^2 = 29 \\ x - y = -7 \end{cases}$$

Solution(s): _____

Name: _____

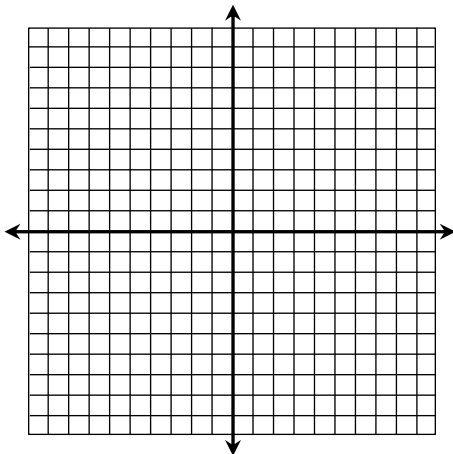
Date: _____ Per: _____

Unit 9 Test

Conic Sections

Graph each equation and identify its key characteristics.

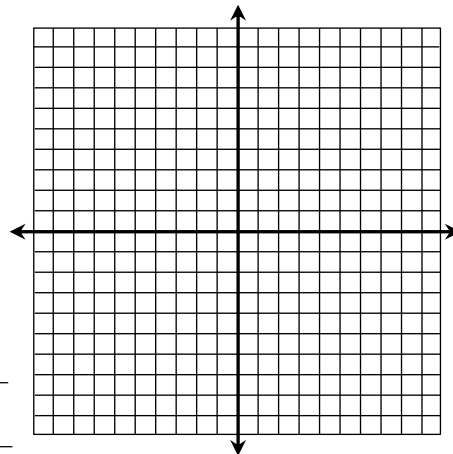
1. $5x^2 + 5y^2 = 80$



Center: _____

Radius: _____

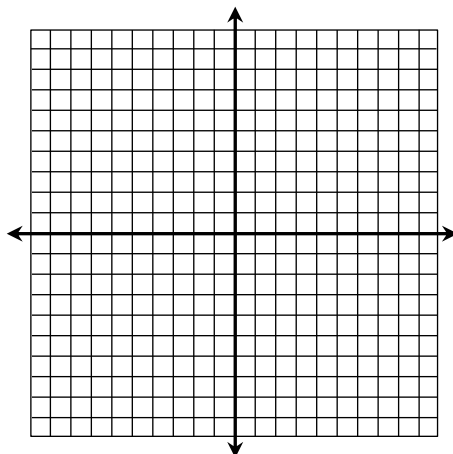
2. $(x + 2)^2 + (y - 1)^2 = 36$



Center: _____

Radius: _____

3. $16x^2 + 49y^2 = 784$



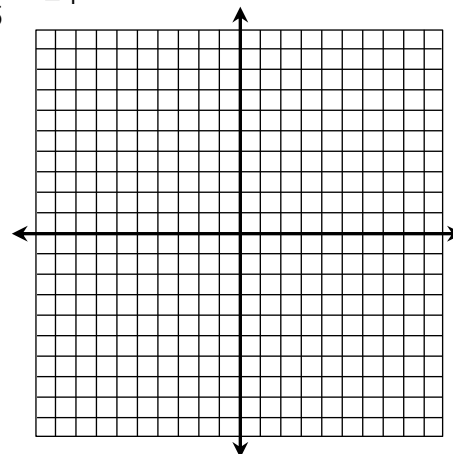
Center: _____

Vertices: _____

Co-Vertices: _____

Foci: _____

4. $\frac{(x - 5)^2}{9} + \frac{(y - 2)^2}{25} = 1$



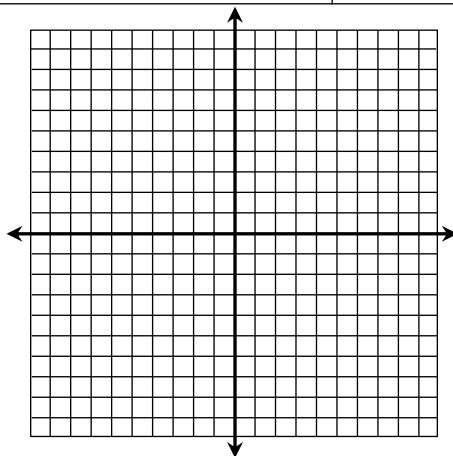
Center: _____

Vertices: _____

Co-Vertices: _____

Foci: _____

5. $4x^2 - y^2 = 4$



Center: _____

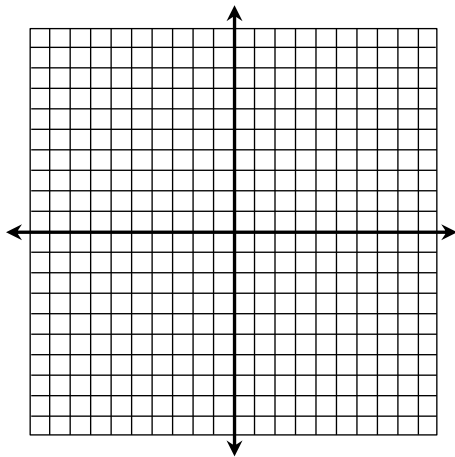
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

6. $\frac{(y+1)^2}{16} - \frac{(x+4)^2}{9} = 1$



Center: _____

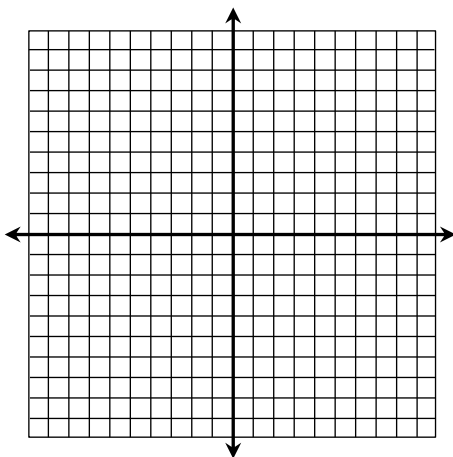
Vertices: _____

Co-Vertices: _____

Foci: _____

Asymptotes: _____

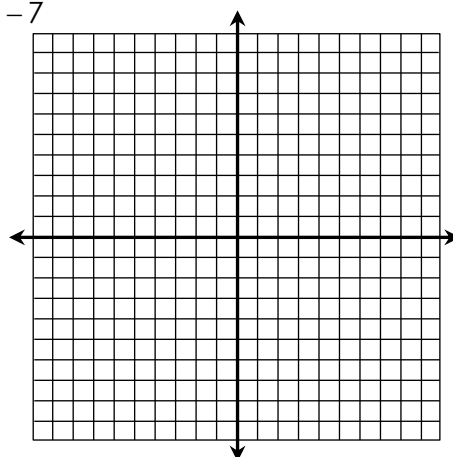
7. $x^2 = 12y$



Vertex: _____ Axis of Sym: _____

Focus: _____ Directrix: _____

8. $-\frac{1}{4}(y+2)^2 = x-7$

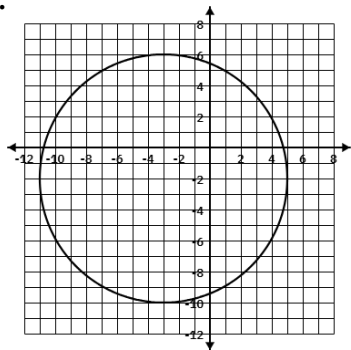


Vertex: _____ Axis of Sym: _____

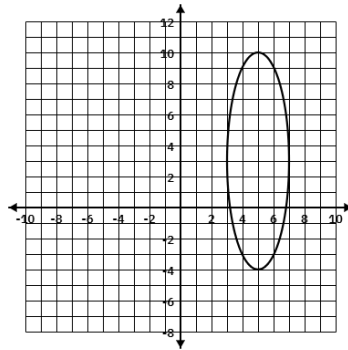
Focus: _____ Directrix: _____

Write the standard form of the equation given the graph.

9.



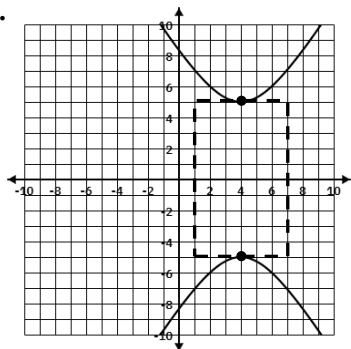
10.



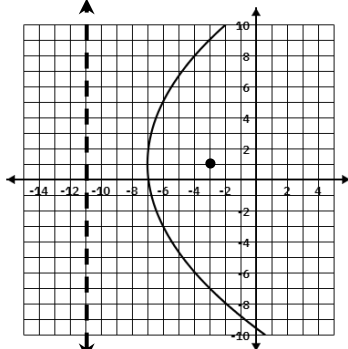
9. _____

10. _____

11.



12.



11. _____

12. _____

Write the equation of the circle in standard form with the given information.**13.** Center: $(1, -4)$, Point on Circle: $(-9, -2)$

Equation: _____

14. Endpoints of Diameter: $(-7, -4)$ and $(3, -8)$

Equation: _____

Write the equation of the ellipse in standard form with the given information.**15.** Vertices: $(-2, -3)$ and $(12, -3)$
Co-Vertices: $(5, -1)$ and $(5, -5)$

Equation: _____

16. Co-Vertices: $(-3, 2)$ and $(-11, 2)$
Foci: $(-7, 5)$ and $(-7, -1)$

Equation: _____

Write the equation of the hyperbola in standard form with the given information.**17.** Vertices: $(4, 6)$ and $(-4, 6)$
Co-Vertices: $(0, -3)$ and $(0, 15)$

Equation: _____

18. Vertices: $(2, -7)$ and $(2, 9)$
Foci: $(2, -9)$ and $(2, 11)$

Equation: _____

Write the equation of the parabola in standard form with the given information.**19.** Vertex: $(-4, 5)$; Focus: $(-7, 5)$

Equation: _____

20. Focus: $(-8, 2)$; Directrix: $y = 0$

Equation: _____

Classify each conic and write the equation in standard form.

21. $x^2 - 8x + 2y + 10 = 0$

- ☐ Circle
☐ Ellipse
☐ Hyperbola
☐ Parabola

22. $x^2 + y^2 + 4x - 10y - 52 = 0$

- ☐ Circle
☐ Ellipse
☐ Hyperbola
☐ Parabola

Equation: _____

Equation: _____

23. $-9x^2 + 16y^2 - 32y - 128 = 0$

- ☐ Circle
☐ Ellipse
☐ Hyperbola
☐ Parabola

24. $4x^2 + y^2 - 16x - 6y + 9 = 0$

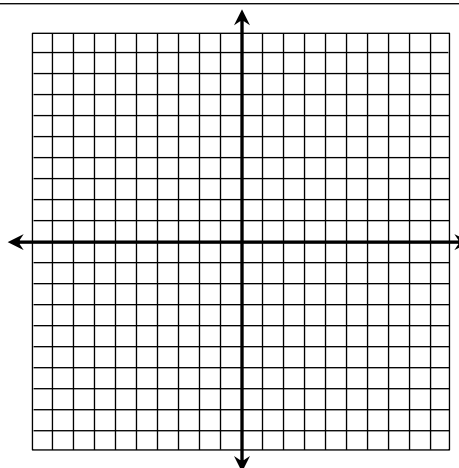
- ☐ Circle
☐ Ellipse
☐ Hyperbola
☐ Parabola

Equation: _____

Equation: _____

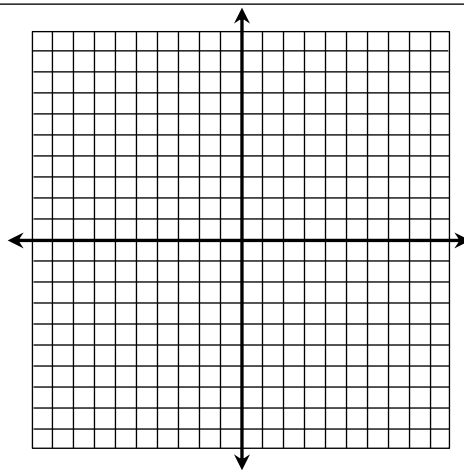
Solve the system by graphing.

25.
$$\begin{cases} y = x^2 - 8x + 11 \\ x + y = 5 \end{cases}$$



Solution(s):

26.
$$\begin{cases} y = 2x^2 + 12x + 17 \\ y = -2x^2 - 8x - 7 \end{cases}$$



Solution(s):

Solve the system algebraically.

27.
$$\begin{cases} y = -x^2 + 7x - 15 \\ 2x - y = 11 \end{cases}$$

Solution(s): _____

28.
$$\begin{cases} y = x^2 - 7x - 11 \\ y = x^2 - 4x - 35 \end{cases}$$

Solution(s): _____

29.
$$\begin{cases} y = 2x^2 - 23 \\ y = x^2 + x - 11 \end{cases}$$

Solution(s): _____

30.
$$\begin{cases} x^2 + y^2 = 53 \\ x - y = -5 \end{cases}$$

Solution(s): _____

Name:

Date:

Topic:

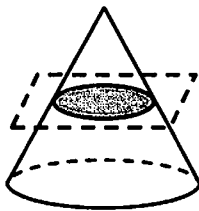
Class:

Main Ideas/Questions

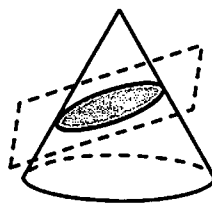
Notes/Examples

CONIC SECTIONS

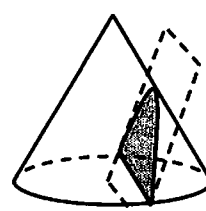
A **conic section** is formed by the intersection of a plane and a right circular cone. The angle of the plane creates the type of conic sections. Label each conic section below.



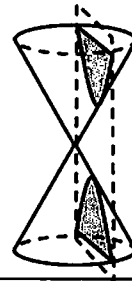
Circle



Ellipse



Parabola



Hyperbola

CIRCLES



Standard Form of the Equation of a Circle:

$$(x-h)^2 + (y-k)^2 = r^2$$

where (h,k) is the center and r is the radius

WRITING EQUATIONS of Circles

Set 1: Given the center and radius.

1. Center: $(-4, 6)$, Radius: 4

$$(x+4)^2 + (y-6)^2 = 16$$

2. Center: $(0, -2)$, Radius: 7

$$(x-0)^2 + (y+2)^2 = 49$$

$$x^2 + (y+2)^2 = 49$$

3. Center: $(1, -5)$, Radius: $\sqrt{10}$

$$(x-1)^2 + (y+5)^2 = 10$$

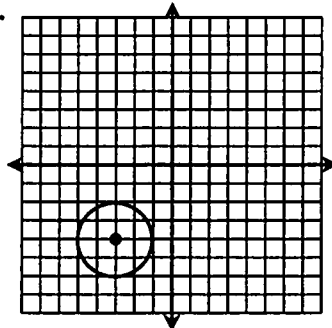
4. Center: $(9, 0)$, Radius: $5\sqrt{3}$

$$(x-9)^2 + (y-0)^2 = (5\sqrt{3})^2$$

$$(x-9)^2 + y^2 = 75$$

Set 2: Given a graph.

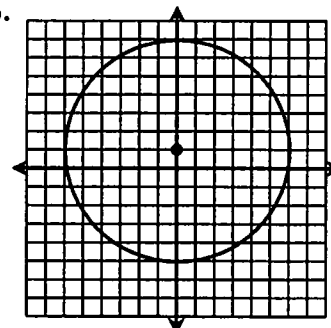
5.



$C = (-3, -4)$
 $r = 2$

$$(x+3)^2 + (y+4)^2 = 4$$

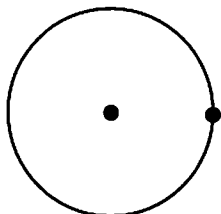
6.



$C = (0, 1)$
 $r = 6$

$$x^2 + (y-1)^2 = 36$$

RECALL →



Set 3: Given the center and an endpoint on the circle.

Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint Formula

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

7. Center: (-2, 6), Endpoint: (-4, 12)

$$r = \sqrt{(-2 - (-4))^2 + (6 - 12)^2}$$

$$r = \sqrt{4 + 36}$$

$$r = \sqrt{40}$$

$$(x + 2)^2 + (y - 6)^2 = 40$$

8. Center: (-4, -7), Endpoint: (-1, -5)

$$r = \sqrt{(-4 - (-1))^2 + (-7 - (-5))^2}$$

$$r = \sqrt{9 + 4}$$

$$r = \sqrt{13}$$

$$(x + 4)^2 + (y + 7)^2 = 13$$

9. Center: (0, 4), Endpoint: (7, -2)

$$r = \sqrt{(0 - 7)^2 + (4 - (-2))^2}$$

$$r = \sqrt{49 + 36}$$

$$r = \sqrt{85}$$

$$x^2 + (y - 4)^2 = 85$$

10. Center: (3, -3), Endpoint: (-4, 10)

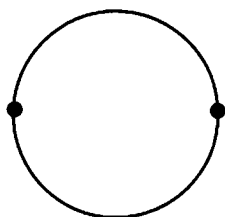
$$r = \sqrt{(3 - (-4))^2 + (-3 - 10)^2}$$

$$r = \sqrt{49 + 169}$$

$$r = \sqrt{218}$$

$$(x - 3)^2 + (y + 3)^2 = 218$$

Set 4: Given the endpoints of a diameter.



11. (-1, 8) and (5, 6)

$$d = \sqrt{(-1 - 5)^2 + (8 - 6)^2}$$

$$= \sqrt{36 + 4}$$

$$= \sqrt{40} = 2\sqrt{10}; r = \sqrt{10}$$

$$M = \left(\frac{-1 + 5}{2}, \frac{8 + 6}{2} \right) = (2, 7)$$

$$(x - 2)^2 + (y - 7)^2 = 10$$

12. (0, -3) and (-8, 3)

$$d = \sqrt{(0 - (-8))^2 + (-3 - 3)^2}$$

$$= \sqrt{64 + 36}$$

$$= \sqrt{100} = 10; r = 5$$

$$M = \left(\frac{0 - 8}{2}, \frac{-3 + 3}{2} \right) = (-4, 0)$$

$$(x + 4)^2 + y^2 = 25$$

13. (-4, 10) and (2, 4)

$$d = \sqrt{(-4 - 2)^2 + (10 - 4)^2}$$

$$= \sqrt{36 + 36}$$

$$= \sqrt{72} = 6\sqrt{2}; r = 3\sqrt{2}$$

$$M = \left(\frac{-4 + 2}{2}, \frac{10 + 4}{2} \right) = (-1, 7)$$

$$(x + 1)^2 + (y - 7)^2 = 18$$

14. (-16, -17) and (-14, -13)

$$d = \sqrt{(-16 - (-14))^2 + (-17 - (-13))^2}$$

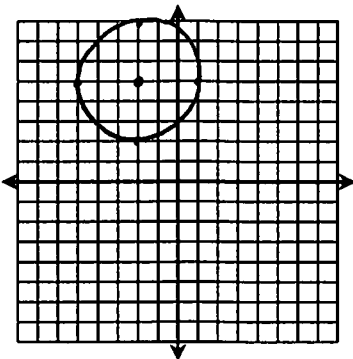
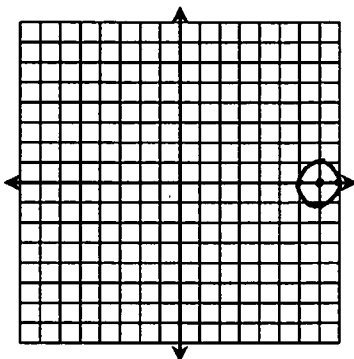
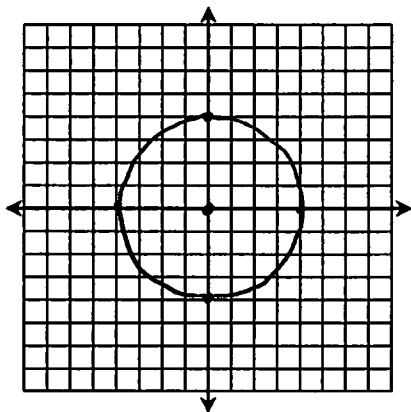
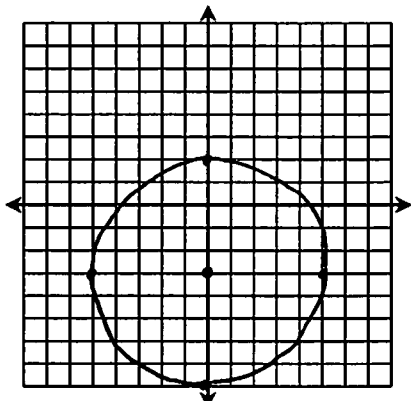
$$= \sqrt{4 + 16}$$

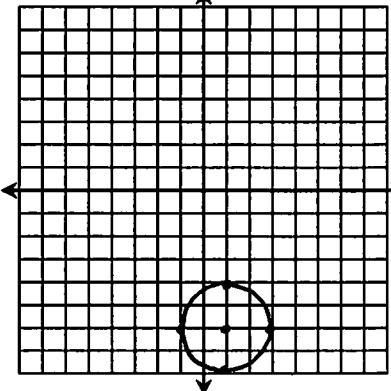
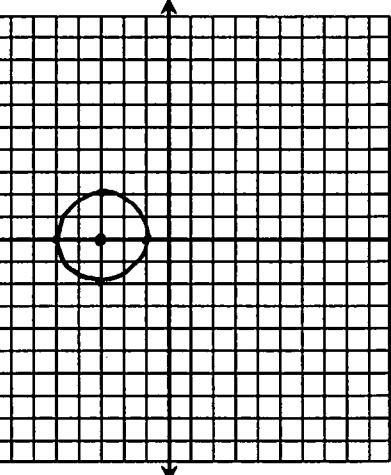
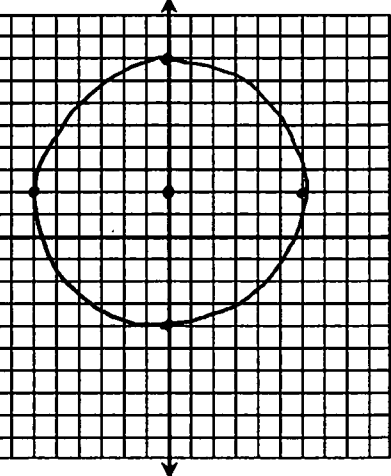
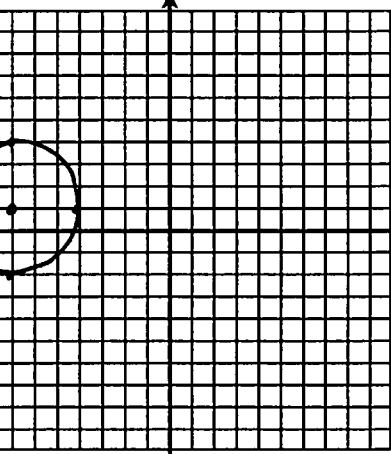
$$= \sqrt{20} = 2\sqrt{5}; r = \sqrt{5}$$

$$M = \left(\frac{-16 - 14}{2}, \frac{-17 - 13}{2} \right) = (-15, -15)$$

$$(x + 15)^2 + (y + 15)^2 = 5$$

Name: _____		Date: _____	
Topic: _____		Class: _____	

Main Ideas/Questions	Notes/Examples
Graphing CIRCLES	Standard Form of the Equation of a Circle: $(x-h)^2 + (y-k)^2 = r^2$ where (h, k) is the center and r is the radius Identify the center and radius of each circle, then graph. 1. $(x+2)^2 + (y-5)^2 = 9$ Center: $(-2, 5)$ Radius: $\sqrt{9} = 3$  2. $(x-7)^2 + y^2 = 1$ Center: $(7, 0)$ Radius: $\sqrt{1} = 1$ 
WHAT ABOUT: Equations NOT in Standard Form?	Write the equation in standard form, identify the center and radius, then graph. 3. $\frac{5x^2}{5} + \frac{5y^2}{5} = \frac{80}{5}$ $x^2 + y^2 = 16$ Center: $(0, 0)$ Radius: 4  4. $(y+3)^2 = 25 - x^2$ $x^2 + (y+3)^2 = 25$ Center: $(0, -3)$ Radius: 5 

	5. $2(x-1)^2 - 8 = -2(y+6)^2$ $\frac{2(x-1)^2}{2} + \frac{2(y+6)^2}{2} = \frac{8}{2}$ $(x-1)^2 + (y+6)^2 = 4$ Center: $(1, -6)$ Radius: 2	
Examples with COMPLETING THE SQUARE	6. $x^2 + y^2 + 6x = -5$ $x^2 + \underline{6x} + y^2 = -5$ $(3)^2 = 9$ $x^2 + 6x + \underline{9} + y^2 = -5 + \underline{9}$ $(x+3)^2 + y^2 = 4$ Center: $(-3, 0)$ Radius: 2	
	7. $x^2 + y^2 + 2y = 6y + 32$ $x^2 + y^2 - \underline{4y} = 32$ $(-2)^2 = 4$ $x^2 + y^2 - 4y + \underline{4} = 32 + \underline{4}$ $x^2 + (y-2)^2 = 36$ Center: $(0, 2)$ Radius: 6	
	8. $x^2 + y^2 + 14x = 2y - 41$ $x^2 + \underline{14x} + y^2 - \underline{2y} = -41$ $(7)^2 = 49 \quad (-1)^2 = 1$ $x^2 + 14x + \underline{49} + y^2 - 2y + \underline{1} = -41 + \underline{49} + \underline{1}$ $(x+7)^2 + (y-1)^2 = 9$ Center: $(-7, 1)$ Radius: 3	

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 1: Circles

**** This is a 2-page document! ******Write the equation of each circle with the given information.**

1. Center: $(-9, 0)$, Radius: 1
 $(x+9)^2 + (y-0)^2 = 1$

$$(x+9)^2 + y^2 = 1$$

2. Center: $(3, -3)$, Radius: $2\sqrt{6}$
 $(x-3)^2 + (y+3)^2 = (2\sqrt{6})^2$

$$(x-3)^2 + (y+3)^2 = 24$$

3. Center: $(4, 7)$, Endpoint: $(-1, -2)$

$$r = \sqrt{(4+1)^2 + (7+2)^2}$$

$$r = \sqrt{25+81}$$

$$r = \sqrt{106}$$

$$(x-4)^2 + (y-7)^2 = 106$$

4. Center: $(0, -8)$, Endpoint: $(2, -5)$

$$r = \sqrt{(0-2)^2 + (-8+5)^2}$$

$$r = \sqrt{4+9}$$

$$r = \sqrt{13}$$

$$x^2 + (y+8)^2 = 13$$

5. Endpoints of Diameter: $(-6, 2)$ and $(-8, 10)$

$$d = \sqrt{(-6+8)^2 + (2-10)^2}$$

$$d = \sqrt{4+64}$$

$$d = \sqrt{68} = 2\sqrt{17}; r = \sqrt{17}$$

$$M = \left(\frac{-6-8}{2}, \frac{2+10}{2} \right) = (-7, 6)$$

$$(x+7)^2 + (y-6)^2 = 17$$

6. Endpoints of Diameter: $(-13, -7)$ and $(11, 11)$

$$d = \sqrt{(-13-11)^2 + (-7-11)^2}$$

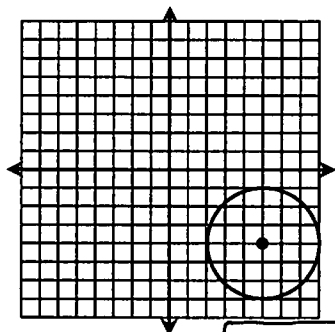
$$d = \sqrt{576+324}$$

$$d = \sqrt{900} = 30; r = 15$$

$$M = \left(\frac{-13+11}{2}, \frac{-7+11}{2} \right) = (-1, 2)$$

$$(x+1)^2 + (y-2)^2 = 225$$

7.

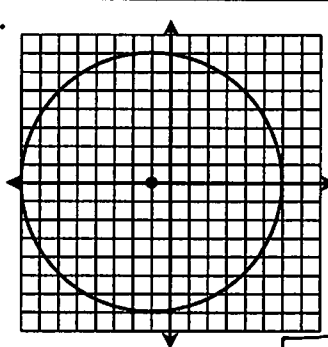


$$C = (5, -4)$$

$$r = 3$$

$$(x-5)^2 + (y+4)^2 = 9$$

8.



$$C = (-1, 0)$$

$$r = 7$$

$$(x+1)^2 + y^2 = 49$$

Identify the center and radius of each circle given the equations below.

9. $(x+1)^2 + (y+4)^2 = 81$

$$C = (-1, -4)$$

$$r = 9$$

10. $x^2 + (y-5)^2 = 7$

$$C = (0, 5)$$

$$r = \sqrt{7} \quad (2.65)$$

$$11. (x-7)^2 + (y+2)^2 = 40$$

$$C = (7, -2)$$

$$r = 2\sqrt{10} \quad (6.32)$$

$$12. \frac{2(x+6)^2}{2} + \frac{2y^2}{2} = \frac{144}{2}$$

$$(x+6)^2 + y^2 = 72$$

$$C = (-6, 0)$$

$$r = 6\sqrt{2} \quad (8.49)$$

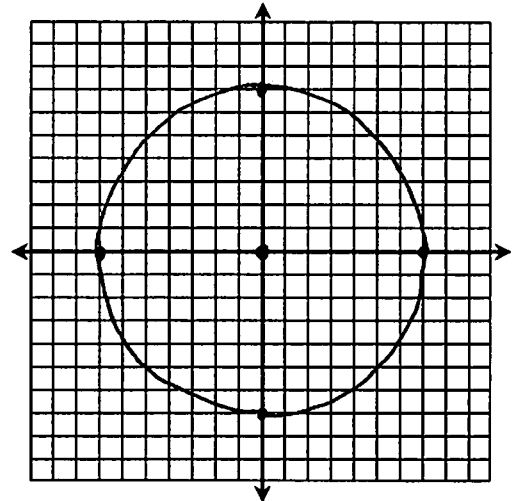
Write the equation in standard form, identify the center and radius, then graph the circle.

$$13. 3x^2 + 3y^2 - 5 = 142$$

$$\frac{3x^2}{3} + \frac{3y^2}{3} = \frac{147}{3}$$

$$x^2 + y^2 = 49$$

$$\text{Center: } (0, 0) \quad \text{Radius: } 7$$



$$14. x^2 + y^2 + 13y = y - 32$$

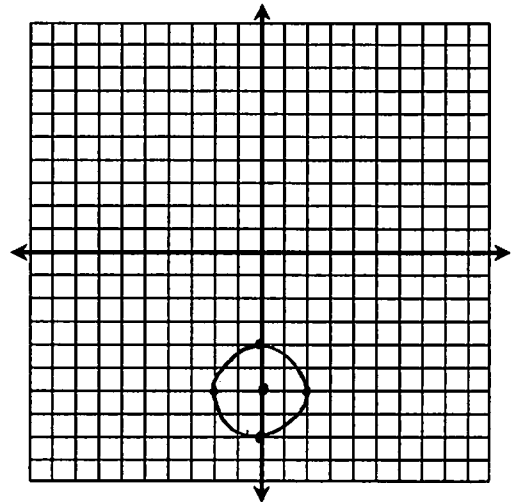
$$x^2 + y^2 + \underline{12y} = -32$$

$$6^2 = 36$$

$$x^2 + y^2 + 12y + \underline{36} = -32 + \underline{36}$$

$$x^2 + (y+6)^2 = 4$$

$$\text{Center: } (0, -6) \quad \text{Radius: } 2$$



$$15. x^2 + y^2 + 7y = 10x + 3y - 13$$

$$x^2 - \underline{10x} + y^2 + \underline{4y} = -13$$

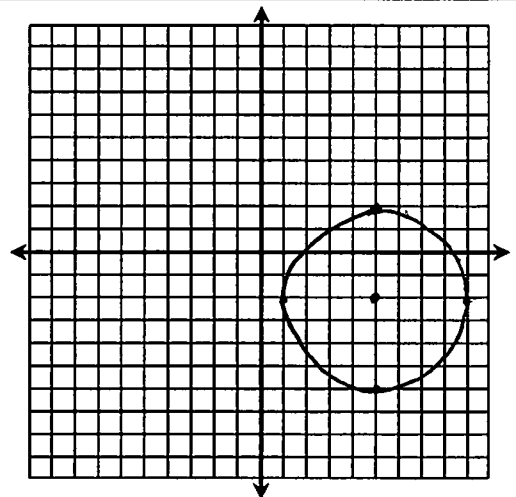
$$(-5)^2 = 25$$

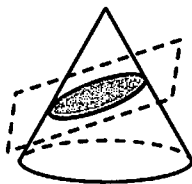
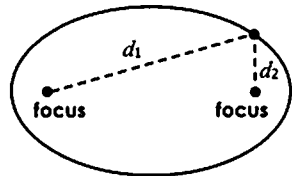
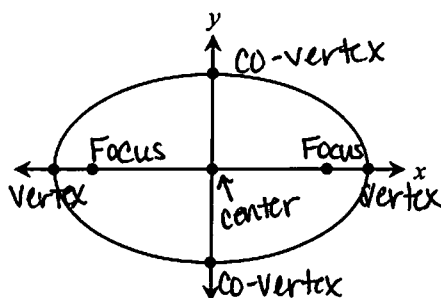
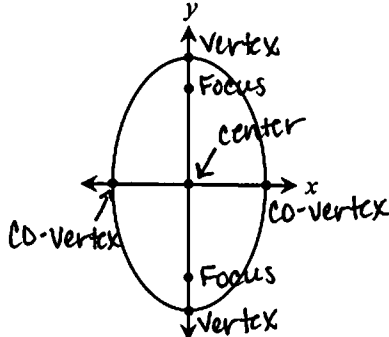
$$2^2 = 4$$

$$x^2 - 10x + \underline{25} + y^2 + 4y + \underline{4} = -13 + \underline{25} + \underline{4}$$

$$(x-5)^2 + (y+2)^2 = 16$$

$$\text{Center: } (5, -2) \quad \text{Radius: } 4$$



Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
ELLIPSES 		An ellipse is a set of points such that the sum of the distances from any point on the ellipse to two fixed points remains constant. These fixed points are called the <u>foci</u> of the ellipse. $d_1 + d_2$ remains constant for all points on the ellipse! 	
MORE PARTS		Major Axis: The <u>longer</u> axis of symmetry. Minor Axis: The <u>shorter</u> axis of symmetry. Center: The point at which the major and minor axis <u>intersect</u>. Vertices: The <u>endpoints</u> of the <u>major axis</u>. Co-Vertices: The <u>endpoints</u> of the <u>minor axis</u>.	
Types of ELLIPSES	HORIZONTAL ELLIPSE 		VERTICAL ELLIPSE 
	- The vertices are <u>$\pm a$</u> units away from the center. - The co-vertices are <u>$\pm b$</u> units away from the center. - Length of Major Axis: <u>$2a$</u>; Length of Minor Axis: <u>$2b$</u> - The foci are <u>$\pm c$</u> units away from the center. - Formula to find c: <u>$c^2 = a^2 - b^2$</u> * <u>Not</u> the Pythagorean Theorem!		
STANDARD FORM Centered at (0, 0)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$		$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
STANDARD FORM Centered at (h, k)	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$		$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$

Graphing ELLIPSES

- ✓ Write the equation in **standard form**.
- ✓ Graph the **center**, (h, k) .
- ✓ Graph the **vertices**, using the larger denominator, a .
- ✓ Graph the **co-vertices**, using the smaller denominator, b .
- ✓ Use the formula $c^2 = a^2 - b^2$ to find and graph the **foci**.

Directions: Graph each ellipse. Identify the center, vertices, co-vertices, and foci.

1. $\frac{x^2}{4} + \frac{y^2}{36} = 1$

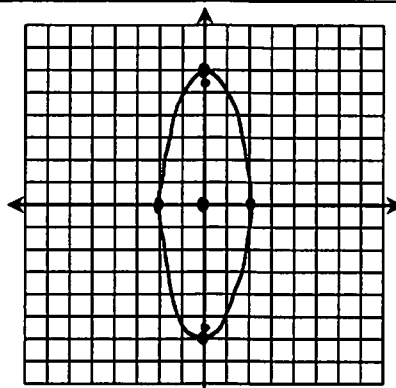
$a^2 = 36; a = 6$

$b^2 = 4; b = 2$

$c^2 = 36 - 4$

$c^2 = 32$

$c = 4\sqrt{2} \approx 5.66$



Center:

$(0, 0)$

Vertices:

$(0, \pm 6)$

Co-Vertices:

$(\pm 2, 0)$

Foci:

$(0, \pm 5.66)$

2. $\frac{x^2}{64} + \frac{y^2}{25} = 1$

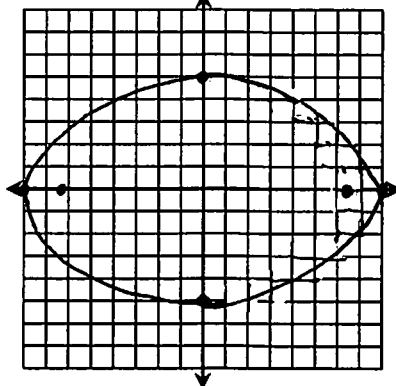
$a^2 = 64; a = 8$

$b^2 = 25; b = 5$

$c^2 = 64 - 25$

$c^2 = 39$

$c = \sqrt{39} \approx 6.24$



Center:

$(0, 0)$

Vertices:

$(\pm 8, 0)$

Co-Vertices:

$(0, \pm 5)$

Foci:

$(\pm 6.24, 0)$

3. $\frac{(x+1)^2}{49} + \frac{(y-2)^2}{9} = 1$

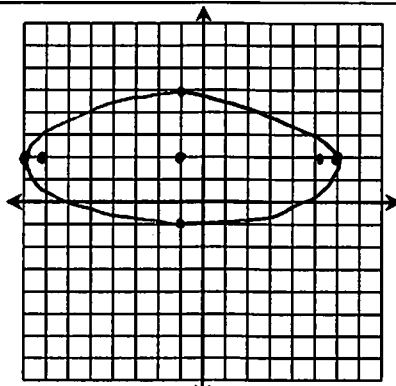
$a^2 = 49; a = 7$

$b^2 = 9; b = 3$

$c^2 = 49 - 9$

$c^2 = 40$

$c = 2\sqrt{10} \approx 6.32$



Center:

$(-1, 2)$

Vertices:

$(-8, 2) (6, 2)$

Co-Vertices:

$(-1, 5) (-1, -1)$

Foci:

$(5.32, 2) (-7.32, 2)$

4. $\frac{(x-3)^2}{25} + \frac{(y-1)^2}{4} = 1$

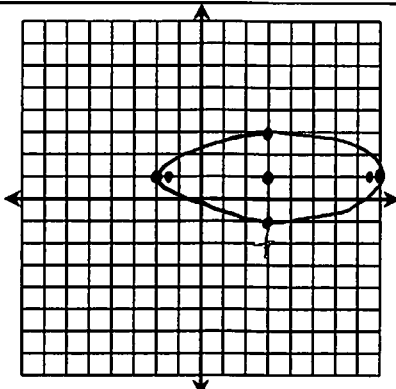
$a^2 = 25; a = 5$

$b^2 = 4; b = 2$

$c^2 = 25 - 4$

$c^2 = 21$

$c = \sqrt{21} \approx 4.58$



Center:

$(3, 1)$

Vertices:

$(-2, 1) (8, 1)$

Co-Vertices:

$(3, 3) (3, -1)$

Foci:

$(7.58, 1) (-1.58, 1)$

$$5. \frac{(x+4)^2}{20} + \frac{(y+2)^2}{36} = 1$$

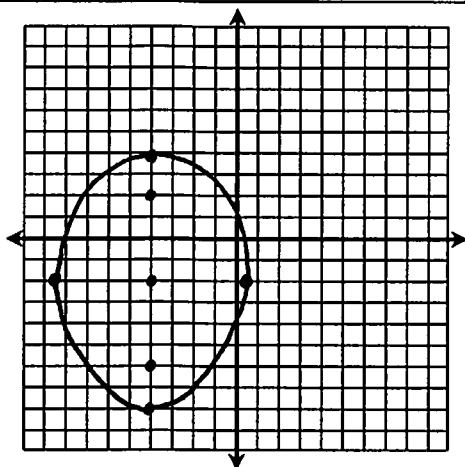
$$a^2 = 36; a = 6$$

$$b^2 = 20; b = 2\sqrt{5} \approx 4.47$$

$$c^2 = 36 - 20$$

$$c^2 = 16$$

$$c = 4$$



Center:

$$(-4, -2)$$

Vertices:

$$(-4, 4) \quad (-4, -8)$$

Co-Vertices:

$$(0.47, -2) \quad (-8.47, -2)$$

Foci:

$$(-4, 2) \quad (-4, -6)$$

$$6. \frac{9x^2}{144} + \frac{16y^2}{144} = \frac{144}{144} \rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$$

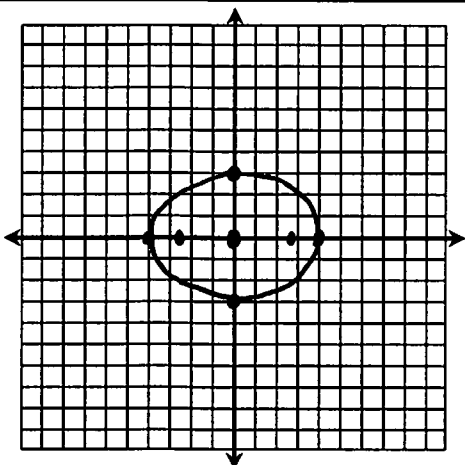
$$a^2 = 16; a = 4$$

$$b^2 = 9; b = 3$$

$$c^2 = 16 - 9$$

$$c^2 = 7$$

$$c = \sqrt{7} \approx 2.65$$



Center:

$$(0, 0)$$

Vertices:

$$(\pm 4, 0)$$

Co-Vertices:

$$(0, \pm 3)$$

Foci:

$$(\pm 2.65, 0)$$

$$7. x^2 + 4y^2 - 10x = -9$$

$$x^2 - 10x + 4y^2 = -9$$

$$x^2 - 10x + 25 + 4y^2 = -9 + 25$$

$$\frac{(x-5)^2}{16} + \frac{4y^2}{16} = \frac{16}{16}$$

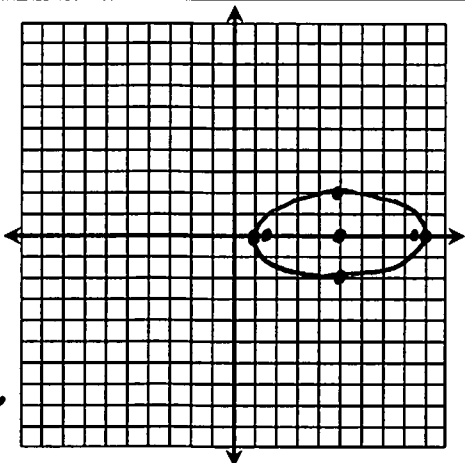
$$\frac{(x-5)^2}{16} + \frac{y^2}{4} = 1$$

$$a^2 = 16; a = 4$$

$$b^2 = 4; b = 2$$

$$c^2 = 16 - 4 = 12$$

$$c = 2\sqrt{3} \approx 3.46$$



Center:

$$(5, 0)$$

Vertices:

$$(1, 0) \quad (9, 0)$$

Co-Vertices:

$$(5, -2) \quad (5, 2)$$

Foci:

$$(8.46, 0) \quad (1.54, 0)$$

$$8. 4x^2 + y^2 - 32x + 8y + 76 = 0$$

$$4x^2 - 32x + y^2 + 8y = -76$$

$$4(x^2 - 8x) + y^2 + 8y = -76$$

$$4(x^2 - 8x + 16) + y^2 + 8y + 16 = -76 + 64 + 16$$

$$\frac{4(x-4)^2}{4} + \frac{(y+4)^2}{4} = \frac{4}{4}$$

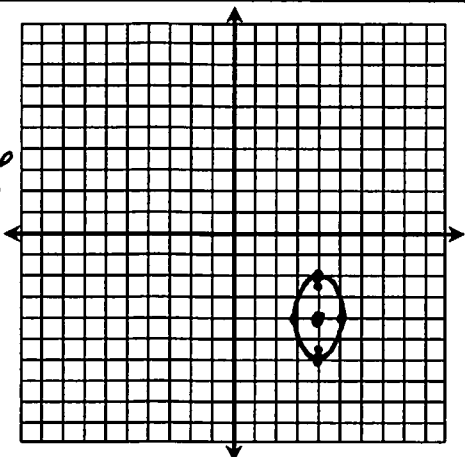
$$(x-4)^2 + \frac{(y+4)^2}{4} = 1$$

$$a^2 = 4; a = 2$$

$$b^2 = 1; b = 1$$

$$c^2 = 4 - 1 = 3$$

$$c = \sqrt{3} \approx 1.73$$



Center:

$$(4, -4)$$

Vertices:

$$(4, -2) \quad (4, -6)$$

Co-Vertices:

$$(5, -4) \quad (3, -4)$$

Foci:

$$(4, -2.27) \quad (4, -5.73)$$



Name: _____

Unit 9: Conic Sections

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Homework 2: Graphing Ellipses

**** This is a 2-page document! ****

Graph each ellipse. Identify the center, vertices, co-vertices, and foci.			
1. $\frac{x^2}{64} + \frac{y^2}{9} = 1$ $a^2 = 64 ; a = 4$ $b^2 = 9 ; b = 3$ $c^2 = 64 - 9$ $c^2 = 55$ $c = \sqrt{55} \approx 7.42$		Center: $(0, 0)$	
		Vertices: $(\pm 4, 0)$	
		Co-Vertices: $(0, \pm 3)$	
		Foci: $(\pm 7.42, 0)$	
2. $x^2 + \frac{y^2}{36} = 1$ $a^2 = 36 ; a = 6$ $b^2 = 1 ; b = 1$ $c^2 = 36 - 1$ $c^2 = 35$ $c = \sqrt{35} \approx 5.92$		Center: $(0, 0)$	
		Vertices: $(0, \pm 6)$	
		Co-Vertices: $(\pm 1, 0)$	
		Foci: $(0, \pm 5.92)$	
3. $\frac{x^2}{16} + \frac{y^2}{7} = 1$ $a^2 = 16 ; a = 4$ $b^2 = 7 ; b = \sqrt{7} \approx 2.65$ $c^2 = 16 - 7$ $c^2 = 9$ $c = 3$		Center: $(0, 0)$	
		Vertices: $(\pm 4, 0)$	
		Co-Vertices: $(0, \pm 2.65)$	
		Foci: $(\pm 3, 0)$	
4. $\frac{(x+1)^2}{25} + \frac{(y+1)^2}{49} = 1$ $a^2 = 49 ; a = 7$ $b^2 = 25 ; b = 5$ $c^2 = 49 - 25$ $c^2 = 24$ $c = 2\sqrt{6} \approx 4.9$		Center: $(-1, -1)$	
		Vertices: $(-1, 6) (-1, -8)$	
		Co-Vertices: $(4, -1) (-6, -1)$	
		Foci: $(-1, 3.9) (-1, -5.9)$	

$$5. \frac{(x+3)^2}{45} + \frac{(y-1)^2}{81} = 1$$

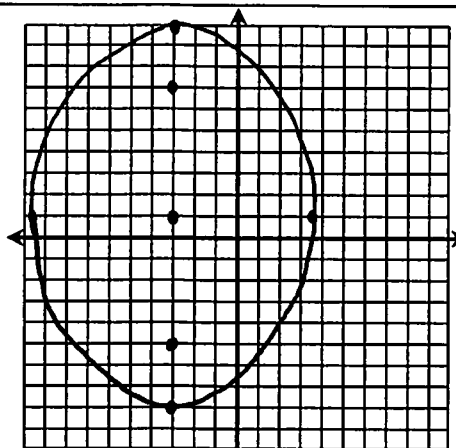
$$a^2 = 81; a = 9$$

$$b^2 = 45; b = 3\sqrt{5} \approx 6.71$$

$$c^2 = 81 - 45$$

$$c^2 = 36$$

$$c = 6$$



Center:

$$(-3, 1)$$

Vertices:

$$(-3, 10) (-3, -8)$$

Co-Vertices:

$$(-9.71, 1) (3.71, 1)$$

Foci:

$$(-3, 7) (-3, -5)$$

$$6. \frac{9x^2}{225} + \frac{25y^2}{225} = \frac{225}{225} \rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1$$

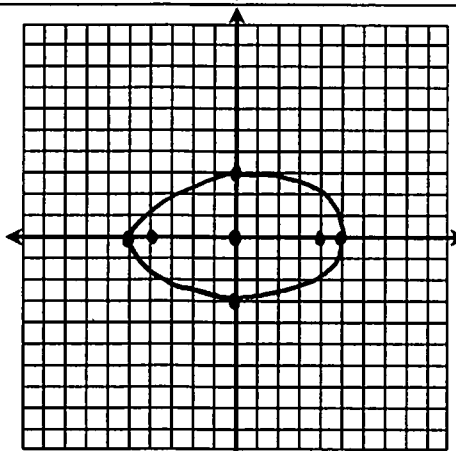
$$a^2 = 25; a = 5$$

$$b^2 = 9; b = 3$$

$$c^2 = 25 - 9$$

$$c^2 = 16$$

$$c = 4$$



Center:

$$(0, 0)$$

Vertices:

$$(\pm 5, 0)$$

Co-Vertices:

$$(0, \pm 3)$$

Foci:

$$(\pm 4, 0)$$

$$7. \frac{27(x-1)^2}{972} + \frac{36(y+2)^2}{972} = \frac{972}{972}$$

$$\frac{(x-1)^2}{36} + \frac{(y+2)^2}{27} = 1$$

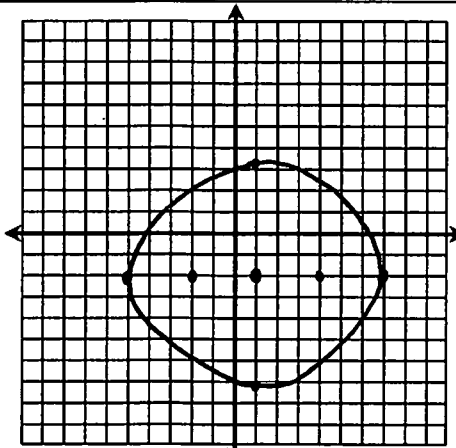
$$a^2 = 36; a = 6$$

$$b^2 = 27; b = 3\sqrt{3} \approx 5.2$$

$$c^2 = 36 - 27$$

$$c^2 = 9$$

$$c = 3$$



Center:

$$(1, -2)$$

Vertices:

$$(7, -2) (-5, -2)$$

Co-Vertices:

$$(1, 3.2) (1, -7.2)$$

Foci:

$$(4, -2) (-2, -2)$$

$$8. 9x^2 + y^2 + 144x = -567$$

$$9x^2 + 144x + y^2 = -567$$

$$9(x^2 + 16x) + y^2 = -567$$

$$9(x^2 + 16x + 64) + y^2 = -567 + 576$$

$$\frac{9(x+8)^2}{9} + \frac{y^2}{9} = \frac{9}{9}$$

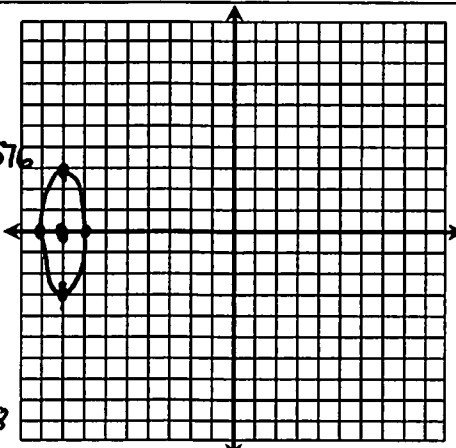
$$(x+8)^2 + \frac{y^2}{9} = 1$$

$$a^2 = 9; a = 3$$

$$b^2 = 1; b = 1$$

$$c^2 = 9 - 1 = 8$$

$$c = 2\sqrt{2} \approx 2.8$$



Center:

$$(-8, 0)$$

Vertices:

$$(-8, 3) (-8, -3)$$

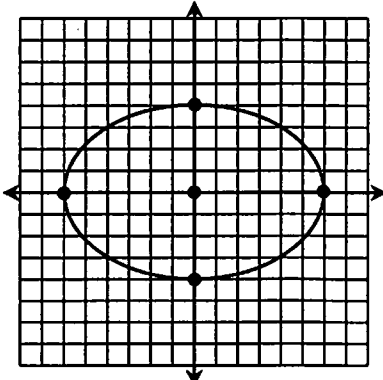
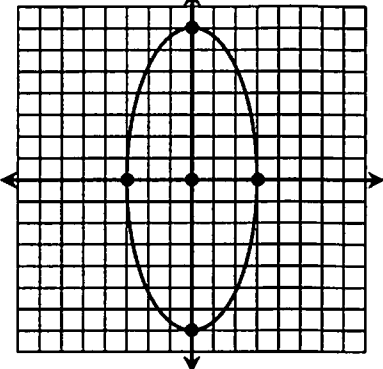
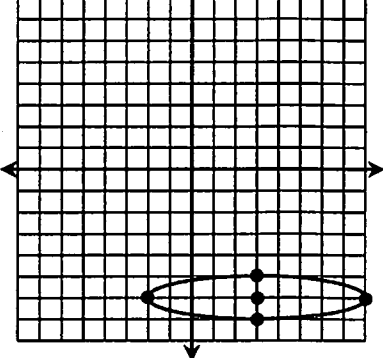
Co-Vertices:

$$(-9, 0) (-7, 0)$$

Foci:

$$(-8, 2.8) (-8, -2.8)$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ELLIPSE <i>Equations</i>	Standard Form for the Equation of an Ellipse at Center (h, k):	
	HORIZONTAL ELLIPSE $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	VERTICAL ELLIPSE: $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$
	a is the distance from the center to the vertices. b is the distance from the center to the co-vertices. - To find the foci use the formula, $c^2 = a^2 - b^2$ $\neq a > b$	
SET 1: Given a Graph $c^2 = 36 - 16$ $c^2 = 20$ $c = 2\sqrt{5} \approx 4.47$ $c^2 = 49 - 9$ $c^2 = 40$ $c = 2\sqrt{10} \approx 6.32$	Write an equation for each ellipse. Then identify all points.	
	1. 	Equation: $\frac{x^2}{36} + \frac{y^2}{16} = 1$
		Center: $(0, 0)$
		Vertices: $(\pm 6, 0)$
		Co-Vertices: $(0, \pm 4)$
		Foci: $(\pm 4.47, 0)$
	2. 	Equation: $\frac{x^2}{9} + \frac{y^2}{49} = 1$
		Center: $(0, 0)$
		Vertices: $(0, \pm 7)$
		Co-Vertices: $(\pm 3, 0)$
		Foci: $(0, \pm 6.32)$
	3. 	Equation: $\frac{(x-3)^2}{25} + (y+6)^2 = 1$
		Center: $(3, -6)$
		Vertices: $(8, -6)$ $(-2, -6)$
		Co-Vertices: $(3, -7)$ $(3, -5)$
		Foci: $(7.9, -6)$ $(-1.9, -6)$

SET 2:

Given Vertices,
Co-Vertices, & Foci

- ✓ Find the **center** (h, k) . (Use the midpoint formula with the vert. or co-vert.)
- ✓ Use the **vertices** to find a ; Use the **co-vertices** to find b .
*If given the foci, use the formula $c^2 = a^2 - b^2$ to find the missing part.
- ✓ Determine the direction of the ellipse to write the equation.

Write an equation for each ellipse with the given information.

4. Vertices: $(-5, 0)$ and $(5, 0)$ Horiz
Co-Vertices: $(0, -2)$ and $(0, 2)$
Center: $(0, 0)$
 $a = 5 \rightarrow a^2 = 25$
 $b = 2 \rightarrow b^2 = 4$

$$\frac{x^2}{25} + \frac{y^2}{4} = 1$$

5. Vertices: $(0, -8)$ and $(0, 8)$ Vert
Co-Vertices: $(-6, 0)$ and $(6, 0)$
Center: $(0, 0)$
 $a = 8 \rightarrow a^2 = 64$
 $b = 6 \rightarrow b^2 = 36$

$$\frac{x^2}{36} + \frac{y^2}{64} = 1$$

6. Vertices: $(-5, 1)$ and $(1, 1)$ Horiz
Co-Vertices: $(-2, 3)$ and $(-2, -1)$
Center: $\left(\frac{-5+1}{2}, \frac{3-1}{2}\right) = (-2, 1)$
 $a = 3 \rightarrow a^2 = 9$
 $b = 2 \rightarrow b^2 = 4$

$$\frac{(x+2)^2}{9} + \frac{(y-1)^2}{4} = 1$$

7. Vertices: $(5, 8)$ and $(5, -6)$ Vert
Co-Vertices: $(11, 1)$ and $(-1, 1)$
Center: $\left(\frac{5+5}{2}, \frac{8-6}{2}\right) = (5, 1)$
 $a = 7 \rightarrow a^2 = 49$
 $b = 6 \rightarrow b^2 = 36$

$$\frac{(x-5)^2}{36} + \frac{(y-1)^2}{49} = 1$$

8. Vertices: $(0, 6)$ and $(-10, 6)$ Horiz
Foci: $(-2, 6)$ and $(-8, 6)$
Center: $\left(\frac{0-10}{2}, \frac{6+6}{2}\right) = (-5, 6)$
 $a = 5 \rightarrow a^2 = 25$
 $c = 3 \rightarrow c^2 = 9$
 $9 = 25 - b^2$
 $-16 = -b^2$
 $16 = b^2$

$$\frac{(x+5)^2}{25} + \frac{(y-6)^2}{16} = 1$$

9. Vertices: $(-9, -2)$ and $(-9, -12)$ Vert
Foci: $(-9, -3)$ and $(-9, -11)$
Center: $\left(\frac{-9-9}{2}, \frac{-2-12}{2}\right) = (-9, -7)$
 $a = 5 \rightarrow a^2 = 25$
 $c = 4 \rightarrow c^2 = 16$
 $16 = 25 - b^2$
 $-9 = -b^2$
 $9 = b^2$

$$\frac{(x+9)^2}{9} + \frac{(y+7)^2}{25} = 1$$

10. Co-Vertices: $(5, 0)$ and $(-5, 0)$ Vert
Foci: $(0, 12)$ and $(0, -12)$
Center: $\left(\frac{5-5}{2}, \frac{0+0}{2}\right) = (0, 0)$
 $b = 5 \rightarrow b^2 = 25$
 $c = 12 \rightarrow c^2 = 144$
 $144 = a^2 - 25$
 $169 = a^2$

$$\frac{x^2}{25} + \frac{y^2}{169} = 1$$

11. Co-Vertices: $(2, 7)$ and $(2, -9)$ Horiz
Foci: $(-4, -1)$ and $(8, -1)$
Center: $\left(\frac{2+2}{2}, \frac{7-9}{2}\right) = (2, -1)$
 $b = 8 \rightarrow b^2 = 64$
 $c = 6 \rightarrow c^2 = 36$
 $36 = a^2 - 64$
 $100 = a^2$

$$\frac{(x-2)^2}{100} + \frac{(y+1)^2}{64} = 1$$

Name: _____

Unit 9: Conic Sections

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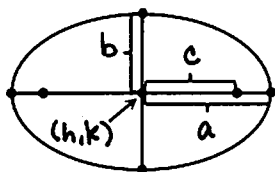
Homework 3: Writing Equations of Ellipses

** This is a 2-page document! **

Label a , b , c , h , and k on each diagram. Then give the standard form equation of each ellipse.**HORIZONTAL ELLIPSE**

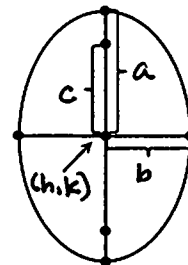
Standard Form:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

**VERTICAL ELLIPSE**

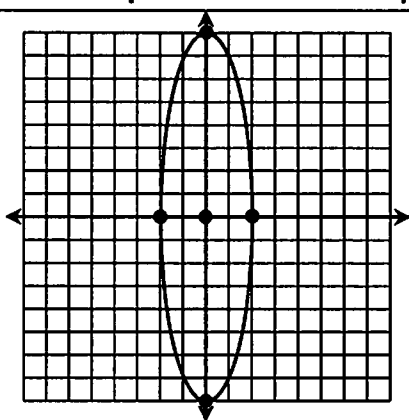
Standard Form:

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$



Write an equation for each ellipse. Identify the center, vertices, co-vertices, and foci.

1.



$a = 8$ - vert
 $b = 2$
 $c^2 = 64 - 4$
 $c^2 = 60$
 $c = 2\sqrt{15} \approx 7.75$

Equation:

$$\frac{x^2}{4} + \frac{y^2}{64} = 1$$

Center:

$$(0, 0)$$

Vertices:

$$(0, \pm 8)$$

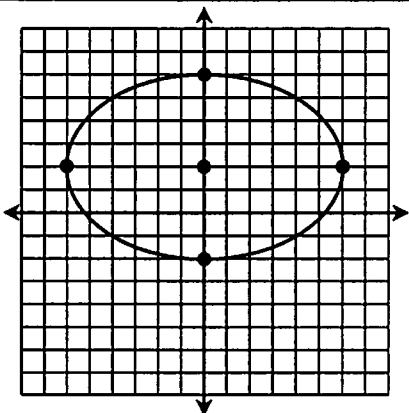
Co-Vertices:

$$(\pm 2, 0)$$

Foci:

$$(0, \pm 7.75)$$

2.



$a = 6$ - Horiz
 $b = 5$
 $c^2 = 36 - 25$
 $c^2 = 11$
 $c = \sqrt{11} \approx 3.32$

Equation:

$$\frac{x^2}{36} + \frac{(y-2)^2}{25} = 1$$

Center:

$$(0, 2)$$

Vertices:

$$(6, 2) \quad (-6, 2)$$

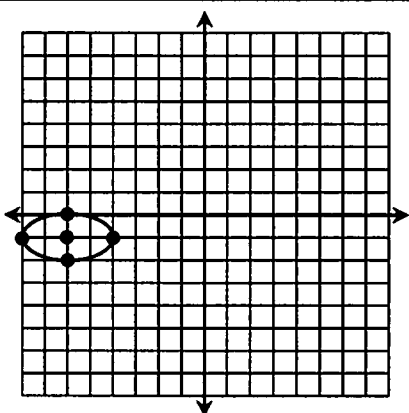
Co-Vertices:

$$(0, 7) \quad (0, -3)$$

Foci:

$$(3.32, 2) \quad (-3.32, 2)$$

3.



$a = 2$ - Horiz
 $b = 1$
 $c^2 = 4 - 1$
 $c^2 = 3$
 $c = \sqrt{3} \approx 1.73$

Equation:

$$\frac{(x+6)^2}{4} + (y+1)^2 = 1$$

Center:

$$(-6, -1)$$

Vertices:

$$(-8, -1) \quad (-4, -1)$$

Co-Vertices:

$$(-6, 0) \quad (-6, -2)$$

Foci:

$$(-4.27, -1) \quad (-7.73, -1)$$

Write an equation for each ellipse with the given information.

4. Vertices: (0, -9) and (0, 9) Vert
Co-Vertices: (-8, 0) and (8, 0)

$$\text{Center: } \left(\frac{-8+8}{2}, \frac{0+0}{2} \right) = (0, 0)$$

$$a = 9 \rightarrow a^2 = 81$$

$$b = 8 \rightarrow b^2 = 64$$

$$\boxed{\frac{x^2}{81} + \frac{y^2}{64} = 1}$$

5. Vertices: (-3, 0) and (3, 0) Horiz
Co-Vertices: (0, -1) and (0, 1)

$$\text{Center: } \left(\frac{-3+3}{2}, \frac{0+0}{2} \right) = (0, 0)$$

$$a = 3 \rightarrow a^2 = 9$$

$$b = 1 \rightarrow b^2 = 1$$

$$\boxed{\frac{x^2}{9} + y^2 = 1}$$

6. Vertices: (12, -4) and (-10, -4) Horiz.
Co-Vertices: (1, 2) and (1, -10)

$$\text{Center: } \left(\frac{12-10}{2}, \frac{-4-4}{2} \right) = (1, -4)$$

$$a = 11 \rightarrow a^2 = 121$$

$$b = 6 \rightarrow b^2 = 36$$

$$\boxed{\frac{(x-1)^2}{121} + \frac{(y+4)^2}{36} = 1}$$

7. Vertices: (-2, 17) and (-2, 3) Vert
Co-Vertices: (2, 10) and (-6, 10)

$$\text{Center: } \left(\frac{-2-2}{2}, \frac{17+3}{2} \right) = (-2, 10)$$

$$a = 7 \rightarrow a^2 = 49$$

$$b = 4 \rightarrow b^2 = 16$$

$$\boxed{\frac{(x+2)^2}{16} + \frac{(y-10)^2}{49} = 1}$$

8. Vertices: (12, 2) and (2, 2) Horiz
Foci: (10, 2) and (4, 2)

$$\text{Center: } \left(\frac{12+2}{2}, \frac{2+2}{2} \right) = (7, 2)$$

$$a = 5 \rightarrow a^2 = 25$$

$$c = 3 \rightarrow c^2 = 9$$

$$9 = 25 - b^2$$

$$-16 = -b^2$$

$$16 = b^2$$

$$\boxed{\frac{(x-7)^2}{25} + \frac{(y-2)^2}{16} = 1}$$

9. Vertices: (-4, -7) and (-4, 7) Vert
Foci: (-4, $-\sqrt{13}$) and (-4, $\sqrt{13}$)

$$\text{Center: } \left(\frac{-4-4}{2}, \frac{-7+7}{2} \right) = (-4, 0)$$

$$a = 7 \rightarrow a^2 = 49$$

$$c = \sqrt{13} \rightarrow c^2 = 13$$

$$13 = 49 - b^2$$

$$-36 = -b^2$$

$$36 = b^2$$

$$\boxed{\frac{(x+4)^2}{36} + \frac{y^2}{49} = 1}$$

10. Co-Vertices: (16, -3) and (-8, -3) Vert
Foci: (4, 2) and (4, -8)

$$\text{Center: } \left(\frac{16-8}{2}, \frac{-3-3}{2} \right) = (4, -3)$$

$$b = 12 \rightarrow b^2 = 144$$

$$c = 5 \rightarrow c^2 = 25$$

$$25 = a^2 - 144$$

$$169 = a^2$$

$$\boxed{\frac{(x-4)^2}{144} + \frac{(y+3)^2}{169} = 1}$$

11. Co-Vertices: (1, -13) and (1, 3) Horiz
Foci: (-5, -5) and (7, -5)

$$\text{Center: } \left(\frac{1+1}{2}, \frac{-13+3}{2} \right) = (1, -5)$$

$$b = 8 \rightarrow b^2 = 64$$

$$c = 6 \rightarrow c^2 = 36$$

$$36 = a^2 - 64$$

$$100 = a^2$$

$$\boxed{\frac{(x-1)^2}{100} + \frac{(y+5)^2}{64} = 1}$$

Name: _____

Algebra 2

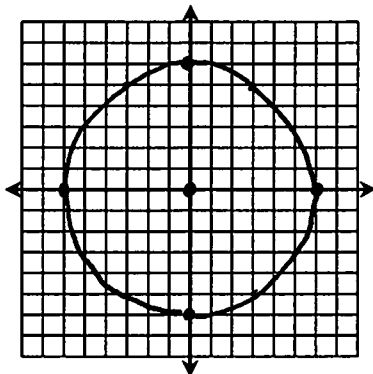
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Unit 9: Conic Sections

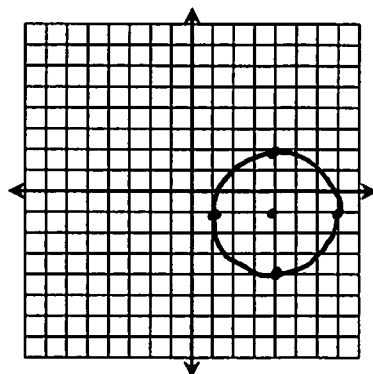
Quiz 9-1: Circles & Ellipses

Graph each circle. Identify the center and radius.

1. $x^2 + y^2 = 36$

Center: (0, 0)Radius: 6

2. $(x - 4)^2 + (y + 1)^2 = 9$

Center: (4, -1)Radius: 3

Write the equation of the circle in standard form. Identify the center and radius.

3. $7x^2 + 7y^2 + 3 = 31$

$$\frac{7x^2}{7} + \frac{7y^2}{7} = \frac{28}{7}$$

$$x^2 + y^2 = 4$$

Standard Form: $x^2 + y^2 = 4$ Center: (0, 0); Radius: 2

4. $x^2 + y^2 - 10x - 6y + 18 = 0$

$$x^2 - 10x + y^2 - 6y = -18$$

$$x^2 - 10x + 25 + y^2 - 6y + 9 = -18 + 25 + 9$$

$$(x - 5)^2 + (y - 3)^2 = 16$$

Standard Form: $(x - 5)^2 + (y - 3)^2 = 16$ Center: (5, 3); Radius: 4

Write the equation of the circle in standard form with the given characteristics.

5. Center: (-7, -2); Radius: 9

$$\hookrightarrow r^2 = 81$$

Standard Form: $(x + 7)^2 + (y + 2)^2 = 81$

6. Center: (4, 0); Radius: $2\sqrt{6}$

$$\hookrightarrow (2\sqrt{6})^2 = 24$$

Standard Form: $(x - 4)^2 + y^2 = 24$

7. Center: (5, 9); Point on Circle: (7, 8)

$$r = \sqrt{(7 - 5)^2 + (8 - 9)^2}$$

$$r = \sqrt{4 + 1}$$

$$r = \sqrt{5}$$

Standard Form: $(x - 5)^2 + (y - 9)^2 = 5$

8. Endpoints of Diameter: (-5, -8) and (7, 2)

$$\text{Center: } \left(\frac{-5 + 7}{2}, \frac{-8 + 2}{2} \right) = (1, -3)$$

$$\text{diameter} = \sqrt{(7 + 5)^2 + (2 + 8)^2}$$

$$= \sqrt{144 + 100}$$

$$= \sqrt{244} = 2\sqrt{61}$$

$$\hookrightarrow r = \sqrt{61}$$

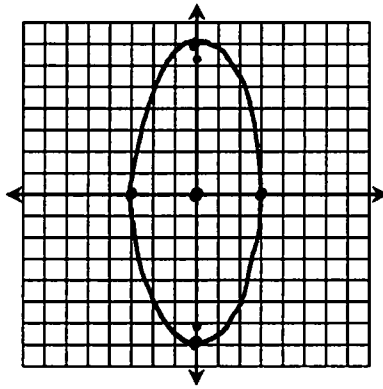
Standard Form: $(x - 1)^2 + (y + 3)^2 = 61$

Graph each ellipse. Identify the center, vertices, co-vertices, and foci.

9. $\frac{x^2}{9} + \frac{y^2}{49} = 1$

$$c^2 = 49 - 9 = 40$$

$$c = 2\sqrt{10} \approx 6.32$$



Center $(0, 0)$

Vertices: $(0, \pm 7)$

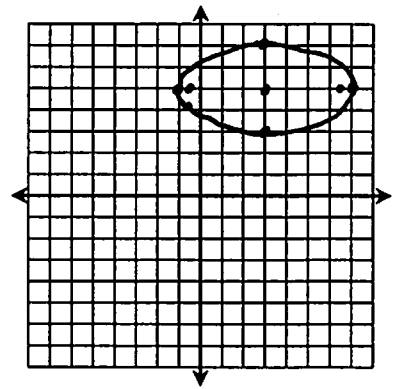
Co-Vertices: $(\pm 3, 0)$

Foci: $(0, \pm 6.32)$

10. $\frac{(x-3)^2}{16} + \frac{(y-5)^2}{4} = 1$

$$c^2 = 16 - 4 = 12$$

$$c = 2\sqrt{3} \approx 3.46$$



Center $(3, 5)$

Vertices: $(7, 5) (-1, 5)$

Co-Vertices: $(3, 7) (3, 3)$

Foci: $(6.46, 5) (-0.46, 5)$

Write the equation of the ellipse in standard form. Identify all characteristics.

11. $\frac{4x^2}{256} + \frac{64y^2}{256} = \frac{256}{256}$

$$c^2 = 64 - 4 = 60$$

$$c = 2\sqrt{15} \approx 7.75$$

$$\frac{x^2}{64} + \frac{y^2}{4} = 1$$

Standard Form: $\frac{x^2}{64} + \frac{y^2}{4} = 1$

Center $(0, 0)$

Vertices: $(\pm 8, 0)$

Co-Vertices: $(0, \pm 2)$

Foci: $(\pm 7.75, 0)$

12. $16x^2 + y^2 + 14y = -33$

$$16x^2 + y^2 + 14y + 49 = -33 + 49$$

$$\frac{16x^2}{16} + \frac{(y+7)^2}{16} = \frac{16}{16}$$

$$x^2 + \frac{(y+7)^2}{16} = 1$$

$$c^2 = 16 - 1 = 15$$

$$c = \sqrt{15} \approx 3.87$$

Standard Form: $x^2 + \frac{(y+7)^2}{16} = 1$

Center $(0, -7)$

Vertices: $(0, -3) (0, -11)$

Co-Vertices: $(1, -7) (-1, -7)$

Foci: $(0, -3.13) (0, -10.87)$

Write the equation of the ellipse in standard form with the given characteristics.

13. Vertices: $(2, 1)$ and $(2, 7)$ Vert.
Co-Vertices: $(0, 4)$ and $(4, 4)$

$$\text{Center: } \left(\frac{2+2}{2}, \frac{1+7}{2}\right) = (2, 4)$$

$$a = 3 \rightarrow a^2 = 9$$

$$b = 2 \rightarrow b^2 = 4$$

Standard Form: $\frac{(x-2)^2}{4} + \frac{(y-4)^2}{9} = 1$

14. Vertices: $(-3, 4)$ and $(5, 4)$ Horiz.
Foci: $(-1, 4)$ and $(3, 4)$

$$\text{Center: } \left(\frac{-3+5}{2}, \frac{4+4}{2}\right) = (1, 4)$$

$$a = 4 \rightarrow a^2 = 16$$

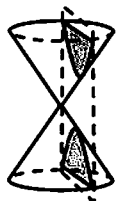
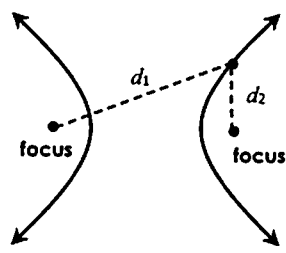
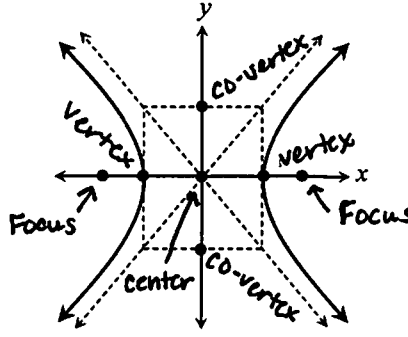
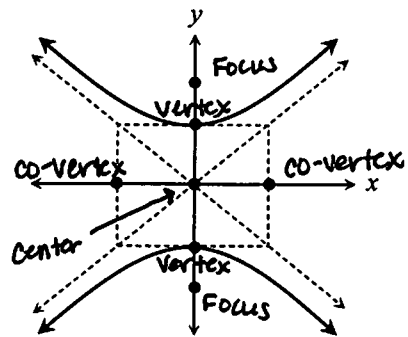
$$c = 2 \rightarrow c^2 = 4$$

$$4 = 16 - b^2$$

$$-12 = -b^2$$

$$12 = b^2$$

Standard Form: $\frac{(x-1)^2}{16} + \frac{(y-4)^2}{12} = 1$

Name:		Date:
Topic:		Class:
Main Ideas/Questions	Notes/Examples	
HYPERBOLAS 	A hyperbola is a set of points such that the difference of the distances from any point on the hyperbola to two fixed points remains constant. These fixed points are called the foci. $d_1 - d_2$ remains constant for all points on the hyperbola! 	
Types of HYPERBOLAS	HORIZONTAL (opens left and right) 	VERTICAL (opens up and down) 
	• The center (h, k) is the <u>midpoint</u> of the foci. • The asymptotes are the diagonal lines that "guide" the curves. • The vertices (turning points) are <u>$\pm a$</u> units away from the center. • The co-vertices are <u>$\pm b$</u> units away from the center. • The foci are <u>$\pm c$</u> units away from the center. • Formula to find c: <u>$c^2 = a^2 + b^2$</u>	
STANDARD FORM Centered at (h, k)	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$
ASYMPTOTE Equations	$y = \pm \frac{b}{a} (x-h) + k$	$y = \pm \frac{a}{b} (x-h) + k$
Graphing HYPERBOLAS	✓ Write the equation in standard form. ✓ Graph the center, (h, k). ✓ Graph the vertices. Use the first denominator, a. ✓ Graph the co-vertices. Use the second denominator, b. ✓ Use the formula $c^2 = a^2 + b^2$ to find and graph the foci. ✓ Write equations for the asymptotes and graph them.	

Directions: Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

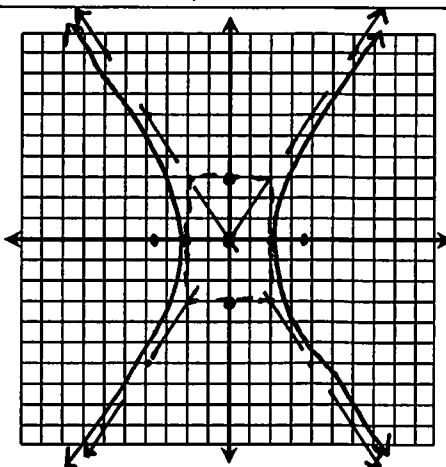
$a^2 = 4 \rightarrow a = 2$

$b^2 = 9 \rightarrow b = 3$

$c^2 = 9 + 4$

$c^2 = 13$

$c = \sqrt{13} \approx 3.61$



Center:

$(0, 0)$

Vertices:

$(\pm 2, 0)$

Co-Vertices:

$(0, \pm 3)$

Foci:

$(\pm 3.61, 0)$

Asymptotes:

$y = \pm \frac{3}{2}x$

2. $\frac{y^2}{36} - x^2 = 1$

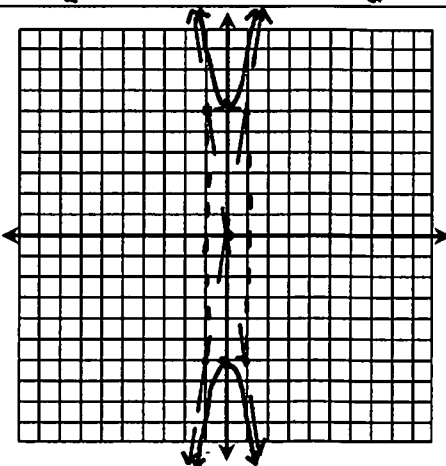
$a^2 = 36 \rightarrow a = 6$

$b^2 = 1 \rightarrow b = 1$

$c^2 = 36 + 1$

$c^2 = 37$

$c = \sqrt{37} \approx 6.08$



Center:

$(0, 0)$

Vertices:

$(0, \pm 6)$

Co-Vertices:

$(\pm 1, 0)$

Foci:

$(0, \pm 6.08)$

Asymptotes:

$y = \pm 6x$

3. $\frac{(x-3)^2}{25} - \frac{(y-2)^2}{16} = 1$

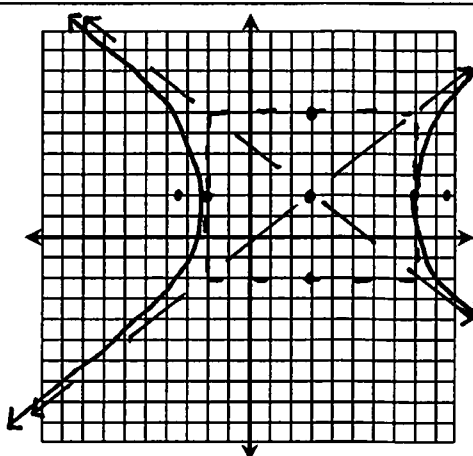
$a^2 = 25 \rightarrow a = 5$

$b^2 = 16 \rightarrow b = 4$

$c^2 = 25 + 16$

$c^2 = 41$

$c = \sqrt{41} \approx 6.40$



Center:

$(3, 2)$

Vertices:

$(8, 2) \quad (-2, 2)$

Co-Vertices:

$(3, 6) \quad (3, -2)$

Foci:

$(9.4, 2) \quad (-3.4, 2)$

Asymptotes:

$y = \pm \frac{4}{5}(x-3) + 2 \rightarrow$

$y = \frac{4}{5}x - \frac{2}{5} \quad \& \quad y = -\frac{4}{5}x + \frac{22}{5}$

4. $\frac{(y-3)^2}{36} - \frac{(x+1)^2}{9} = 1$

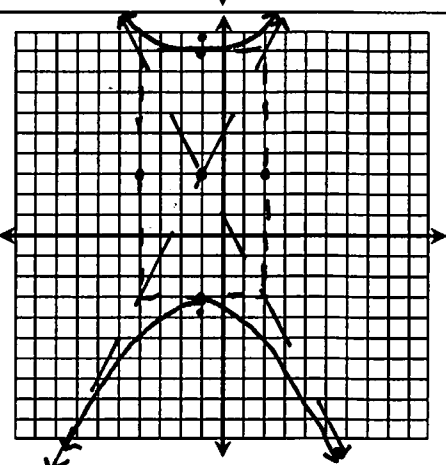
$a^2 = 36 \rightarrow a = 6$

$b^2 = 9 \rightarrow b = 3$

$c^2 = 36 + 9$

$c^2 = 45$

$c = 3\sqrt{5} \approx 6.71$



Center:

$(-1, 3)$

Vertices:

$(-1, 9) \quad (-1, -3)$

Co-Vertices:

$(-4, 3) \quad (2, 3)$

Foci:

$(-1, 9.71) \quad (-1, -3.71)$

Asymptotes:

$y = \pm 2(x+1) + 3 \rightarrow$

$y = 2x + 5 \quad \& \quad y = -2x - 1$

$$5. \frac{4y^2}{100} - \frac{25x^2}{100} = \frac{100}{100}$$

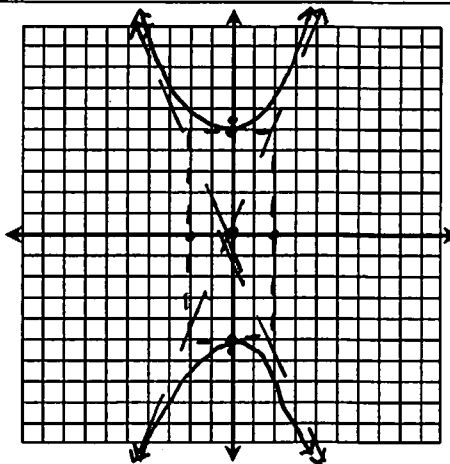
$$\frac{y^2}{25} - \frac{x^2}{4} = 1$$

$$a^2 = 25 \rightarrow a = 5$$

$$b^2 = 4 \rightarrow b = 2$$

$$c^2 = 25 + 4 = 29$$

$$c = \sqrt{29} \approx 5.39$$



Center:

$$(0, 0)$$

Vertices:

$$(0, \pm 5)$$

Co-Vertices:

$$(\pm 2, 0)$$

Foci:

$$(0, \pm 5.39)$$

Asymptotes:

$$y = \pm \frac{5}{2} x$$

$$6. x^2 - 9y^2 + 10x = -16$$

$$x^2 + 10x + 25 - 9y^2 = -16 + 25$$

$$\frac{(x+5)^2}{9} - \frac{9y^2}{9} = \frac{9}{9}$$

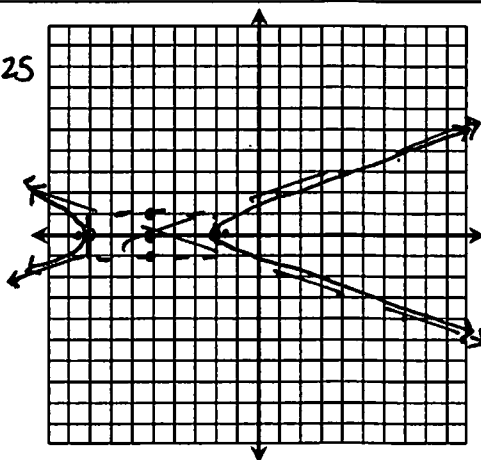
$$\frac{(x+5)^2}{9} - y^2 = 1$$

$$a^2 = 9 \rightarrow a = 3$$

$$b^2 = 1 \rightarrow b = 1$$

$$c^2 = 9 + 1 = 10$$

$$c = \sqrt{10} \approx 3.16$$



Center:

$$(-5, 0)$$

Vertices:

$$(-8, 0) (-2, 0)$$

Co-Vertices:

$$(-5, 1) (-5, -1)$$

Foci:

$$(-1.84, 0) (-8.16, 0)$$

Asymptotes: $y = \pm \frac{1}{3}(x+5) + 0 \rightarrow$

$$y = \frac{1}{3}x + \frac{5}{3} \quad \& \quad y = -\frac{1}{3}x - \frac{5}{3}$$

$$7. -36x^2 + y^2 - 6y = -45$$

$$-36x^2 + y^2 - 6y + 9 = -45 + 9$$

$$\frac{-36x^2}{-36} + \frac{(y-3)^2}{-36} = \frac{-36}{-36}$$

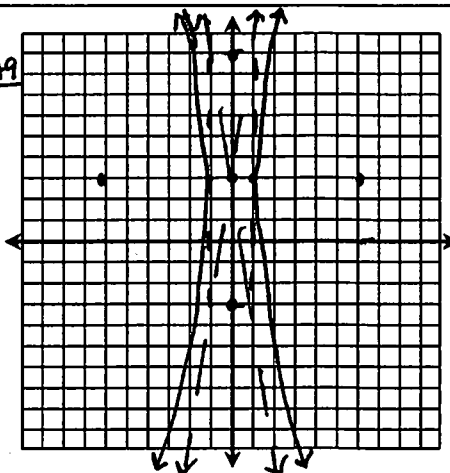
$$x^2 - \frac{(y-3)^2}{36} = 1$$

$$a^2 = 1 \rightarrow a = 1$$

$$b^2 = 36 \rightarrow b = 6$$

$$c^2 = 1 + 36 = 37$$

$$c = \sqrt{37} \approx 6.08$$



Center:

$$(0, 3)$$

Vertices:

$$(-1, 3) (1, 3)$$

Co-Vertices:

$$(0, 9) (0, -3)$$

Foci:

$$(6.08, 3) (-6.08, 3)$$

Asymptotes:

$$y = \pm 6(x-0) + 3 \rightarrow$$

$$y = 6x + 3 \quad \& \quad y = -6x + 3$$

$$8. -4x^2 + y^2 + 40x + 12y = 80$$

$$-4(x^2 - 10x) + y^2 + 12y = 80$$

$$-4(x^2 - 10x + 25) + y^2 + 12y + 36 = 80 - 100 + 36$$

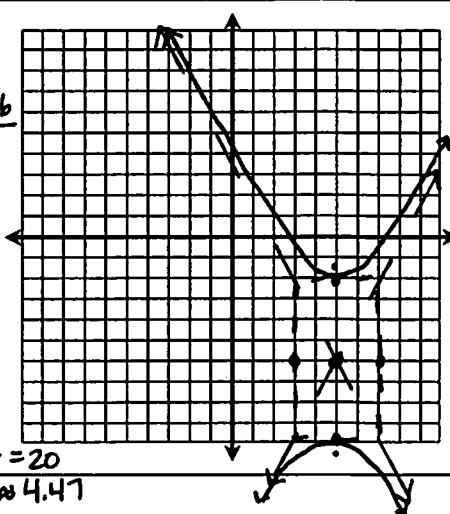
$$\frac{-4(x-5)^2}{16} + \frac{(y+6)^2}{16} = \frac{16}{16}$$

$$\frac{(x-5)^2}{-4} + \frac{(y+6)^2}{16} = 1$$

$$\frac{(y+6)^2}{16} - \frac{(x-5)^2}{4} = 1$$

$$c^2 = 16 + 4 = 20$$

$$c = 2\sqrt{5} \approx 4.47$$



Center:

$$(5, -6)$$

Vertices:

$$(5, -2) (5, -10)$$

Co-Vertices:

$$(3, -6) (7, -6)$$

Foci:

$$(5, -1.53) (5, -10.47)$$

Asymptotes:

$$y = \pm 2(x-5) - 6 \rightarrow$$

$$y = 2x - 16 \quad \& \quad y = -2x + 4$$

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 4: Graphing Hyperbolas

** This is a 2-page document! **

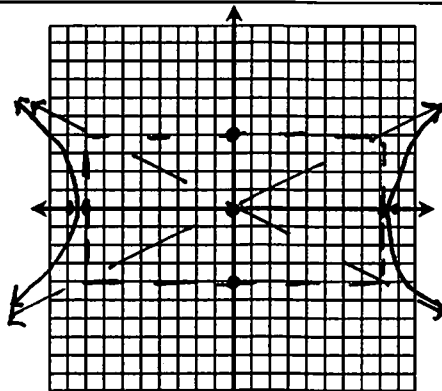
Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{64} - \frac{y^2}{16} = 1$

$C^2 = 64 + 16$

$C^2 = 80$

$C = 4\sqrt{5} \approx 8.94$

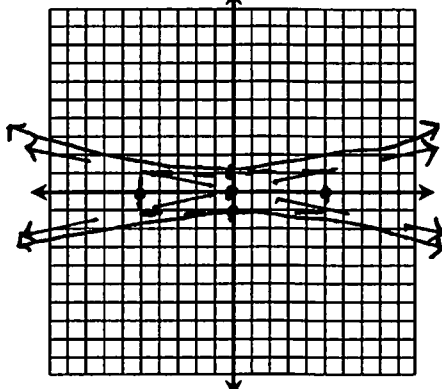
Center: $(0, 0)$ Vertices: $(\pm 8, 0)$ Co-Vertices: $(0, \pm 4)$ Foci: $(\pm 8.94, 0)$ Asymptotes: $y = \pm \frac{1}{2}x$

2. $y^2 - \frac{x^2}{25} = 1$

$C^2 = 1 + 25$

$C^2 = 26$

$C = \sqrt{26} \approx 5.1$

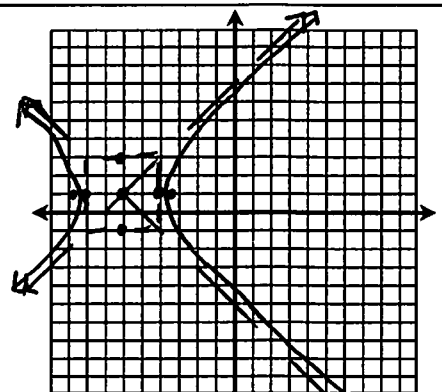
Center: $(0, 0)$ Vertices: $(0, \pm 1)$ Co-Vertices: $(\pm 5, 0)$ Foci: $(0, \pm 5.1)$ Asymptotes: $y = \pm \frac{1}{5}x$

3. $\frac{(x+6)^2}{4} - \frac{(y-1)^2}{4} = 1$

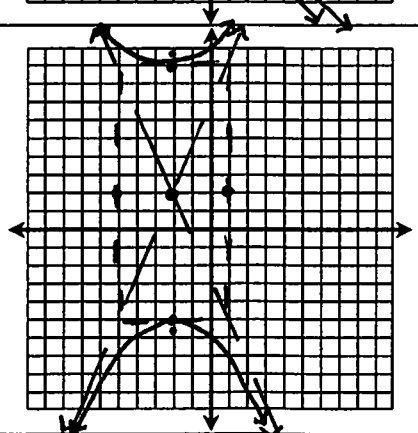
$C^2 = 4 + 4$

$C^2 = 8$

$C = 2\sqrt{2} \approx 2.8$

Center: $(-6, 1)$ Vertices: $(-8, 1)$ $(-4, 1)$ Co-Vertices: $(-6, 3)$ $(-6, -1)$ Foci: $(-8.8, 1)$ $(-3.2, 1)$ Asymptotes: $y = \pm 1(x+6)+1 \rightarrow$
 $y = x+7$ & $y = -x-5$

4. $\frac{(y-2)^2}{49} - \frac{(x+2)^2}{9} = 1$

Center: $(-2, 2)$ Vertices: $(-2, 9)$ $(-2, -5)$ Co-Vertices: $(-5, 2)$ $(1, 2)$ Foci: $(-2, 9.62)$ $(-2, -5.62)$ Asymptotes: $y = \pm \frac{7}{3}(x+2)+2 \rightarrow$
 $y = \frac{7}{3}x + \frac{20}{3}$ & $y = -\frac{7}{3}x - \frac{8}{3}$

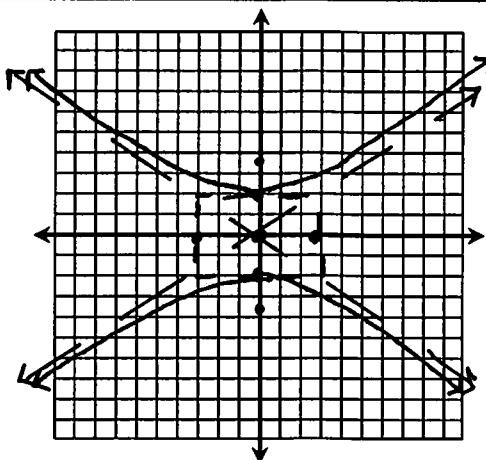
5. $9y^2 - 3x^2 + 1 = x^2 + 37$

$$\frac{9y^2}{36} - \frac{4x^2}{36} = \frac{36}{36}$$

$$\frac{y^2}{4} - \frac{x^2}{9} = 1$$

$$c^2 = 4 + 9 = 13$$

$$c = \sqrt{13} \approx 3.61$$



Center:

$$(0, 6)$$

Vertices:

$$(0, \pm 2)$$

Co-Vertices:

$$(\pm 3, 0)$$

Foci:

$$(0, \pm 3.61)$$

Asymptotes:

$$y = \pm \frac{2}{3}x$$

6. $4x^2 - y^2 - 6y = 25$

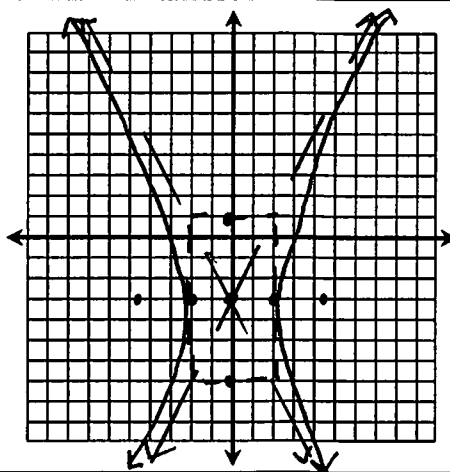
$$4x^2 - (y^2 + 6y + 9) = 25 - 9$$

$$\frac{4x^2}{16} - \frac{(y+3)^2}{16} = \frac{16}{16}$$

$$\frac{x^2}{4} - \frac{(y+3)^2}{16} = 1$$

$$c^2 = 4 + 16 = 20$$

$$c = \sqrt{20} = 2\sqrt{5} \approx 4.47$$



Center:

$$(0, -3)$$

Vertices:

$$(-2, -3) (2, -3)$$

Co-Vertices:

$$(0, 1) (0, -7)$$

Foci:

$$(-4.47, -3) (4.47, -3)$$

Asymptotes:

$$y = \pm 2(x - 0) - 3 \rightarrow y = 2x - 3 \text{ \& } y = -2x - 3$$

7. $x^2 - 16y^2 - 2x = 63$

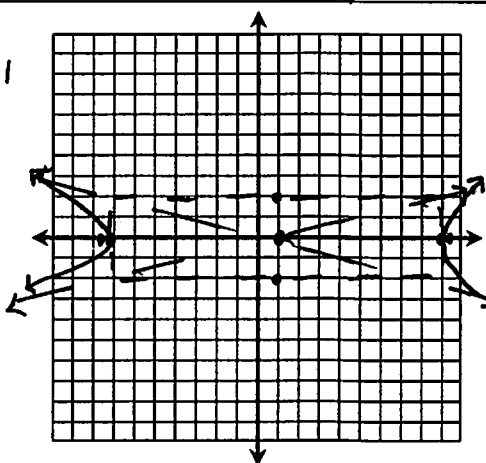
$$x^2 - 2x + 1 - 16y^2 = 63 + 1$$

$$\frac{(x-1)^2}{64} - \frac{16y^2}{64} = \frac{64}{64}$$

$$\frac{(x-1)^2}{64} - \frac{y^2}{4} = 1$$

$$c^2 = 64 + 4 = 68$$

$$c = \sqrt{68} = 2\sqrt{17} \approx 8.25$$



Center:

$$(1, 0)$$

Vertices:

$$(9, 0) (-7, 0)$$

Co-Vertices:

$$(1, 2) (1, -2)$$

Foci:

$$(9.25, 0) (-7.25, 0)$$

Asymptotes:

$$y = \pm \frac{1}{4}(x-1) + 0 \rightarrow y = \frac{1}{4}x - \frac{1}{4} \text{ \& } y = -\frac{1}{4}x + \frac{1}{4}$$

8. $-x^2 + 9y^2 - 4x + 72y = -131$

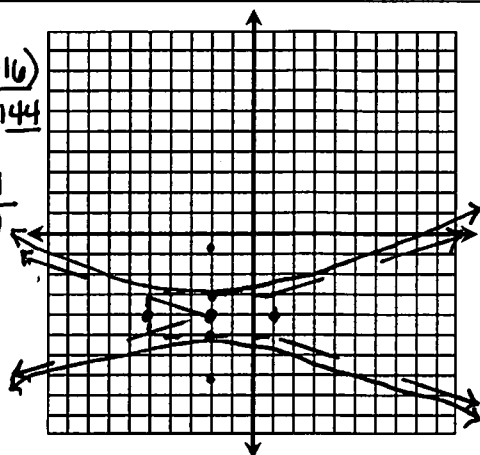
$$-(x^2 + 4x + 4) + 9(y^2 + 8y + 16) = -131 - 4 + 144$$

$$-\frac{(x+2)^2}{9} + \frac{9(y+4)^2}{9} = \frac{9}{9}$$

$$(y+4)^2 - \frac{(x+2)^2}{9} = 1$$

$$c^2 = 1 + 9 = 10$$

$$c = \sqrt{10} \approx 3.16$$



Center:

$$(-2, -4)$$

Vertices:

$$(-2, -3) (-2, -5)$$

Co-Vertices:

$$(1, -4) (-5, -4)$$

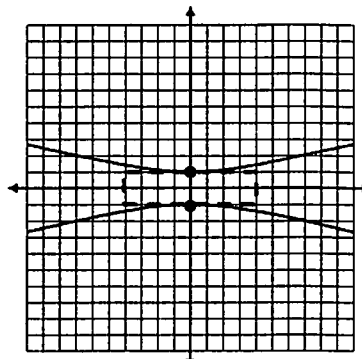
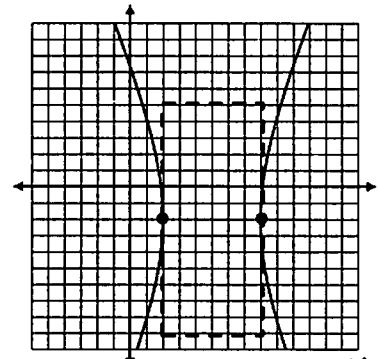
Foci:

$$(-2, -0.84) (-2, -7.16)$$

Asymptotes:

$$y = \pm \frac{1}{3}(x+2) - 4 \rightarrow y = \frac{1}{3}x - \frac{10}{3} \text{ \& } y = -\frac{1}{3}x - \frac{14}{3}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
HYPERBOLA <i>Equations</i>	Standard Form for the Equation of a Hyperbola at Center (h, k): <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px;"> HORIZONTAL HYPERBOLA $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ </td> <td style="width: 50%; padding: 5px;"> VERTICAL HYPERBOLA $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$ </td> </tr> </table> • <u>(h, k)</u> is the <u>Center</u> • <u>a</u> is the distance from the <u>center</u> to the <u>vertices</u> • <u>b</u> is the distance from the <u>center</u> to the <u>co-vertices</u>	HORIZONTAL HYPERBOLA $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	VERTICAL HYPERBOLA $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$
HORIZONTAL HYPERBOLA $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	VERTICAL HYPERBOLA $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$		
SET 1: Given a Graph	1.  Center $(0,0)$ $a=1, b=4$ $y^2 - \frac{x^2}{16} = 1$ Vert! 2.  center $(5,-2)$ $a=3, b=7$ $\frac{(x-5)^2}{9} - \frac{(y+2)^2}{49} = 1$ Horiz!		
SET 2: Given Vertices, Co-Vertices, & Foci	✓ Find the center (h, k). (Use the midpoint formula with the vert. or co-vert.) ✓ Use the vertices to find a; Use the co-vertices to find b. *If given the foci, use the formula $c^2 = a^2 + b^2$ to find the missing part. ✓ Determine the direction of the hyperbola to write the equation.		
3. Vertices: $(\pm 5, 0)$ Co-Vertices: $(0, \pm 2)$ Center: $(0, 0)$ $a = 5$ $b = 2$ $\frac{x^2}{25} - \frac{y^2}{4} = 1$ Horiz!	4. Vertices: $(4, 8)$ and $(4, -8)$ Co-Vertices: $(1, 0)$ and $(7, 0)$ Center: $(4, 0)$ $a = 8$ $b = 3$ $\frac{y^2}{64} - \frac{(x-4)^2}{9} = 1$ Vert!		

5. Vertices: (12, -6) and (-2, -6) Horiz
 Co-Vertices: (5, -12) and (5, 0)
 Center: $\left(\frac{12-2}{2}, \frac{-6-6}{2}\right) = (5, -6)$

$$a=7$$

$$b=6$$

$$\frac{(x-5)^2}{49} - \frac{(y+6)^2}{36} = 1$$

6. Vertices: (-9, 3) and (-9, -5) Vert
 Co-Vertices: (1, -1) and (-19, 1)
 Center: $\left(\frac{-9-9}{2}, \frac{3-5}{2}\right) = (-9, -1)$

$$a=4$$

$$b=10$$

$$\frac{(y+1)^2}{16} - \frac{(x+9)^2}{100} = 1$$

7. Vertices: (0, ± 3) Vert
 Foci: (0, $\pm \sqrt{73}$)

Center: (0, 0)

$$a=3$$

$$c=\sqrt{73}$$

$$73 = a^2 + b^2$$

$$64 = b^2$$

$$8 = b$$

$$\frac{y^2}{9} - \frac{x^2}{64} = 1$$

8. Vertices: (2, 5) and (6, 5) Horiz
 Foci: (1, 5) and (7, 5)

Center: $\left(\frac{2+6}{2}, \frac{5+5}{2}\right) = (4, 5)$

$$a=2$$

$$c=3$$

$$9 = 4 + b^2$$

$$5 = b^2$$

$$\sqrt{5} = b$$

$$\frac{(x-4)^2}{4} - \frac{(y-5)^2}{5} = 1$$

9. Vertices: (1, -9) and (-7, -9) Horiz
 Foci: (2, -9) and (-8, -9)

Center: $\left(\frac{1-7}{2}, \frac{-9-9}{2}\right) = (-3, -9)$

$$a=4$$

$$c=5$$

$$25 = 16 + b^2$$

$$9 = b^2$$

$$3 = b$$

$$\frac{(x+3)^2}{16} - \frac{(y+9)^2}{9} = 1$$

10. Vertices: (3, -2) and (3, 10) Vert
 Foci: (3, $4 \pm 2\sqrt{10}$)

Center: $\left(\frac{3+3}{2}, \frac{-2+10}{2}\right) = (3, 4)$

$$a=6$$

$$c=2\sqrt{10}$$

$$40 = 36 + b^2$$

$$4 = b^2$$

$$2 = b$$

$$\frac{(y-4)^2}{36} - \frac{(x-3)^2}{4} = 1$$

11. Co-Vertices: (5, -3) and (-7, -3) Vert
 Foci: (-1, 7) and (-1, -13)

Center: $\left(\frac{5-7}{2}, \frac{-3-3}{2}\right) = (-1, -3)$

$$b=6$$

$$c=10$$

$$100 = a^2 + 36$$

$$64 = a^2$$

$$8 = a$$

$$\frac{(y+3)^2}{64} - \frac{(x+1)^2}{36} = 1$$

12. Co-Vertices: (-2, 0) and (-2, 10) Horiz
 Foci: $(-2 \pm \sqrt{51}, 5)$

Center: $\left(\frac{-2-2}{2}, \frac{0+10}{2}\right) = (-2, 5)$

$$b=5$$

$$c=\sqrt{51}$$

$$51 = a^2 + 25$$

$$26 = a^2$$

$$\sqrt{26} = a$$

$$\frac{(x+2)^2}{26} - \frac{(y-5)^2}{25} = 1$$

Name: _____

Unit 9: Conic Sections

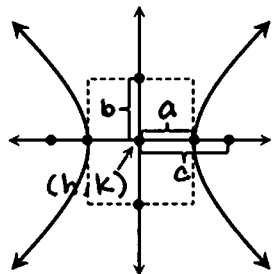
Date: _____ Per: _____

Homework 5: Writing Equations of Hyperbolas

**** This is a 2-page document! ******Directions:** Label a , b , c , h , and k on each diagram. Then give the standard form for the equation of a hyperbola.**HORIZONTAL HYPERBOLA**

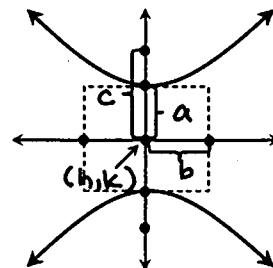
Standard Form:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

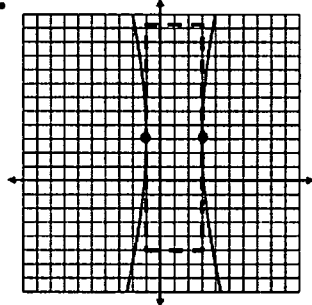
**VERTICAL HYPERBOLA**

Standard Form:

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

**Directions:** Write an equation for each hyperbola.

1.



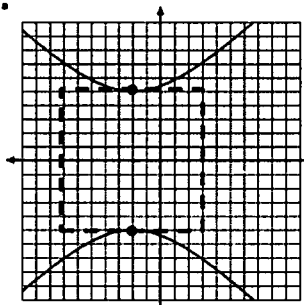
center = $(1, 3)$

$a = 2$

$b = 8$

$$\frac{(x-1)^2}{4} - \frac{(y-3)^2}{64} = 1$$

2.



center = $(-2, 0)$

$a = 5$

$b = 5$

$$\frac{y^2}{25} - \frac{(x+2)^2}{25} = 1$$

Directions: Write an equation for each hyperbola with the given information.3. Vertices: $(0, \pm 3)$ Co-Vertices: $(\pm 1, 0)$

center = $(0, 0)$

$a = 3$

$b = 1$

$$\frac{y^2}{9} - x^2 = 1$$

Vert.

4. Vertices: $(14, -4)$ and $(-4, -4)$ Co-Vertices: $(5, 1)$ and $(5, -9)$

center = $\left(\frac{14-4}{2}, \frac{-4-4}{2}\right) = (5, -4)$

$a = 9$

$b = 5$

$$\frac{(x-5)^2}{81} - \frac{(y+4)^2}{25} = 1$$

Horiz.

5. Vertices: $(4, -6)$ and $(-6, -6)$ Co-Vertices: $(-1, 3)$ and $(-1, -15)$

center = $\left(\frac{4-6}{2}, \frac{-6-6}{2}\right) = (-1, -6)$

$a = 5$

$b = 9$

$$\frac{(x+1)^2}{25} - \frac{(y+6)^2}{81} = 1$$

Horiz.

6. Vertices: $(-2, -8)$ and $(-2, 8)$ Co-Vertices: $(-6, 0)$ and $(2, 0)$

center = $\left(\frac{-2-2}{2}, \frac{-8+8}{2}\right) = (-2, 0)$

$a = 8$

$b = 4$

$$\frac{y^2}{64} - \frac{(x+2)^2}{16} = 1$$

Vert.

7. Vertices: $(\pm 1, 0)$

Horiz.

Foci: $(\pm\sqrt{37}, 0)$

$$\text{Center} = (0, 0)$$

$$a = 1$$

$$c = \sqrt{37}$$

$$37 = 1 + b^2$$

$$36 = b^2$$

$$b = 6$$

$$\boxed{x^2 - \frac{y^2}{36} = 1}$$

8. Vertices: $(-4, -9)$ and $(-10, -9)$

Horiz.

Foci: $(-2, -9)$ and $(-12, -9)$

$$\text{Center} = \left(\frac{-4-10}{2}, \frac{-9-9}{2} \right) = (-7, -9)$$

$$a = 3$$

$$c = 5$$

$$25 = 9 + b^2$$

$$16 = b^2$$

$$4 = b$$

$$\boxed{\frac{(x+7)^2}{9} - \frac{(y+9)^2}{16} = 1}$$

9. Vertices: $(4, 11)$ and $(4, -13)$

Vert.

Foci: $(4, 12)$ and $(4, -14)$

$$\text{Center} = \left(\frac{4+4}{2}, \frac{11-13}{2} \right) = (4, -1)$$

$$a = 12$$

$$c = 13$$

$$169 = 144 + b^2$$

$$25 = b^2$$

$$5 = b$$

$$\boxed{\frac{(y+1)^2}{144} - \frac{(x-4)^2}{25} = 1}$$

10. Vertices: $(0, 9)$ and $(0, 5)$

Vert.

Foci: $(0, 7 \pm \sqrt{29})$

$$\text{Center} = \left(\frac{0+0}{2}, \frac{9+5}{2} \right) = (0, 7)$$

$$a = 2$$

$$c = \sqrt{29}$$

$$29 = 4 + b^2$$

$$25 = b^2$$

$$5 = b$$

$$\boxed{\frac{(y-7)^2}{4} - \frac{x^2}{25} = 1}$$

11. Co-Vertices: $(-2, 3)$ and $(-8, 3)$

Vert.

Foci: $(-5, 8)$ and $(-5, -2)$

$$\text{Center} = \left(\frac{-2-8}{2}, \frac{3+3}{2} \right) = (-5, 3)$$

$$b = 3$$

$$c = 5$$

$$25 = a^2 + 9$$

$$16 = a^2$$

$$a = 4$$

$$\boxed{\frac{(y-3)^2}{16} + \frac{(x+5)^2}{9} = 1}$$

12. Co-Vertices: $(8, -1)$ and $(8, -5)$

Horiz.

Foci: $(8 \pm \sqrt{5}, -3)$

$$\text{Center} = \left(\frac{8+8}{2}, \frac{-1-5}{2} \right) = (8, -3)$$

$$b = 2$$

$$c = \sqrt{5}$$

$$5 = a^2 + 4$$

$$1 = a^2$$

$$1 = a$$

$$\boxed{(x-8)^2 - \frac{(y+3)^2}{4} = 1}$$

Challenge! Hint: A graph may be helpful to write the equations of the hyperbolas below.

13. Vertices: $(\pm 6, 0)$

Horiz.

Asymptotes: $y = \pm \frac{1}{3}x$

$$\left(\pm \frac{b}{a} \right)$$

$$\text{Center} = (0, 0)$$

$$a = 6$$

$$b = 2$$

$$\boxed{\frac{x^2}{36} - \frac{y^2}{4} = 1}$$

14. Co-Vertices: $(1, -1)$ and $(5, -1)$

Vert.

Asymptotes: $y = 2x - 7$ and $y = -2x + 5$

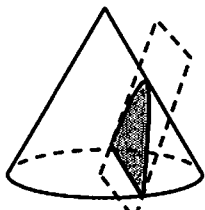
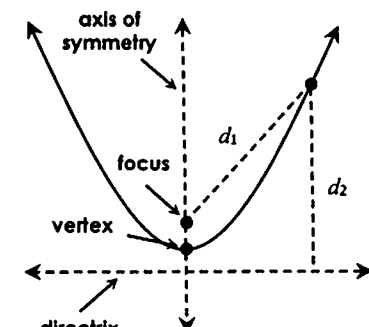
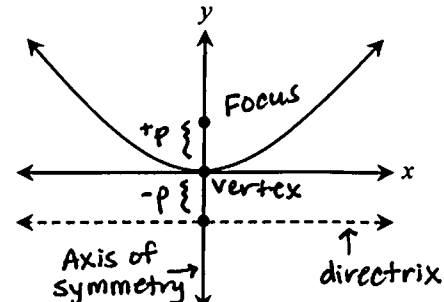
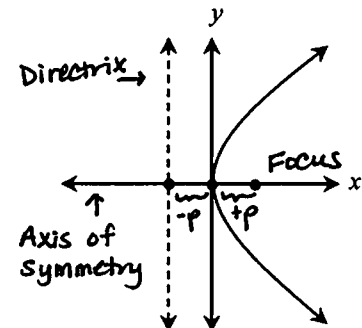
$$\left(\pm \frac{a}{b} \right)$$

$$\text{Center} = (3, -1)$$

$$b = 2$$

$$a = 4$$

$$\boxed{\frac{(y+1)^2}{16} - \frac{(x-3)^2}{4} = 1}$$

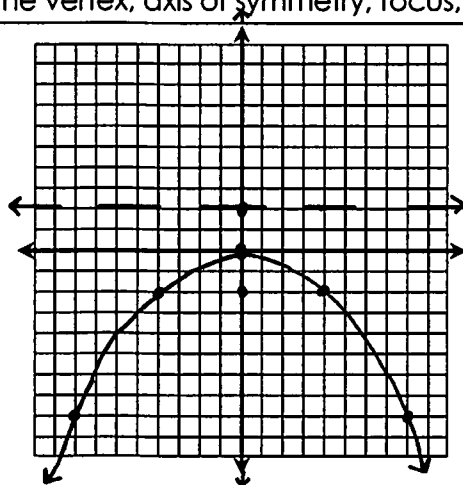
Name:		Date:
Topic:		Class:
Main Ideas/Questions	Notes/Examples	
PARABOLAS 	A parabola is a set of points such that each point is equidistant to a fixed point called the focus and a line called the directrix. <u>$d_1 = d_2$</u> for all points on the parabola! 	
Types of PARABOLAS	VERTICAL (opens up/down) 	HORIZONTAL (opens left/right) 
	- The <u>focus</u> always lies inside the parabola. - The <u>directrix</u> always lies outside the parabola. - The <u>directrix</u> and <u>axis of symmetry</u> are perpendicular. - The vertex is <u>p</u> units away from the focus and directrix.	
STANDARD FORM Centered at (h, k)	$(x-h)^2 = 4p(y-k)$	$(y-k)^2 = 4p(x-h)$
DIRECTION OF OPENING	Opens UP if <u>$p > 0$</u> Opens DOWN if <u>$p < 0$</u>	Opens RIGHT if <u>$p > 0$</u> Opens LEFT if <u>$p < 0$</u>
Graphing PARABOLAS	✓ Write the equation in standard form. Determine the type of parabola. ✓ Graph the vertex, (h, k). ✓ Solve for p to determine the direction of the opening. ✓ Use p to graph the focus and directrix. ✓ Sketch the parabola.	

Directions: Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

1. $x^2 = -8y$

Vertical

$$\begin{aligned} -8 &= 4p \\ -2 &= p \\ &\text{(down)} \end{aligned}$$



Vertex:

$$(0, 0)$$

Axis of Symmetry:

$$x = 0$$

Focus:

$$(0, -2)$$

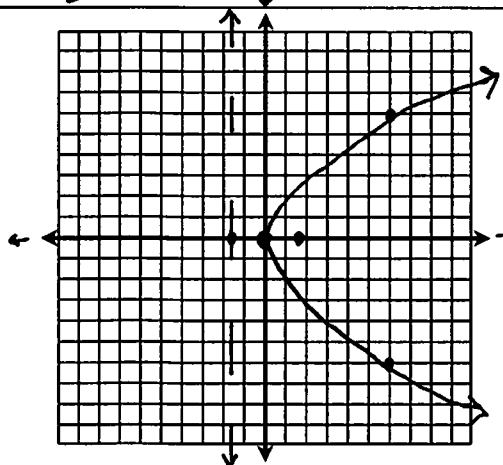
Directrix:

$$y = 2$$

2. $y^2 = 6x$

horizontal

$$\begin{aligned} 6 &= 4p \\ \frac{3}{2} &= p \\ &\text{(right)} \end{aligned}$$



Vertex:

$$(0, 0)$$

Axis of Symmetry:

$$y = 0$$

Focus:

$$\left(\frac{3}{2}, 0\right)$$

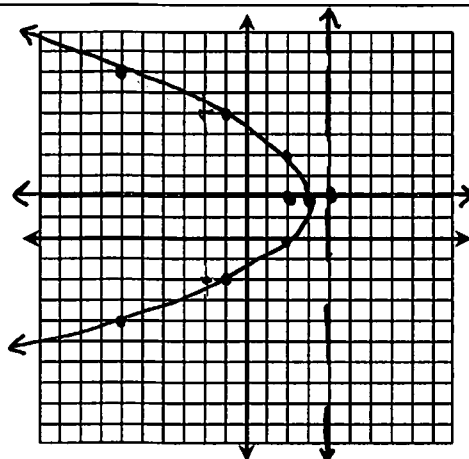
Directrix:

$$x = -\frac{3}{2}$$

3. $(y - 2)^2 = -4(x - 3)$

horizontal

$$\begin{aligned} -4 &= 4p \\ -1 &= p \\ &\text{(left)} \end{aligned}$$



Vertex:

$$(3, 2)$$

Axis of Symmetry:

$$y = 2$$

Focus:

$$(2, 2)$$

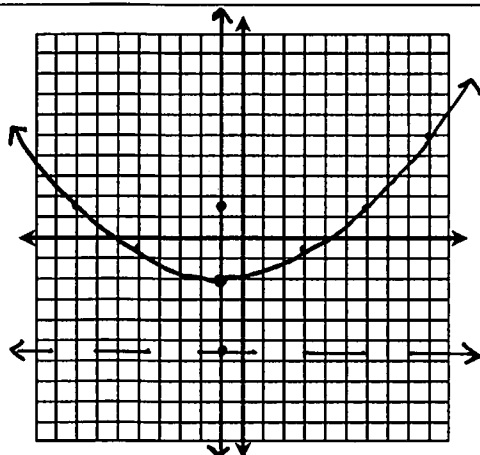
Directrix:

$$x = 4$$

4. $(x + 1)^2 = 14(y + 2)$

Vertical

$$\begin{aligned} 14 &= 4p \\ \frac{7}{2} &= p \\ &\text{(up)} \end{aligned}$$



Vertex:

$$(-1, -2)$$

Axis of Symmetry:

$$x = -1$$

Focus:

$$\left(-1, \frac{3}{2}\right)$$

Directrix:

$$y = -\frac{11}{2}$$

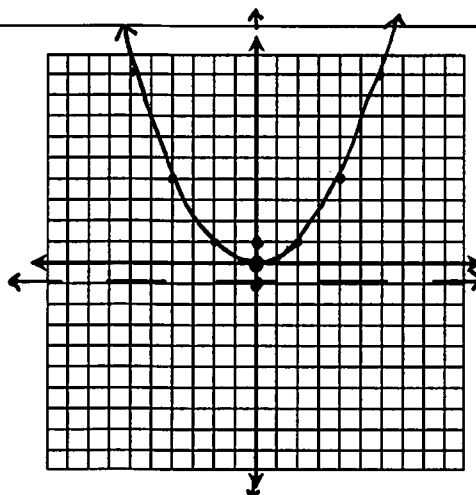
$$5. \frac{1}{4}x^2 = y$$

$$\underline{x^2 = 4y}$$

 vertical

$$4 = 4p$$

 $1 = p$
 (up)


 Vertex:
 $(0, 0)$

 Axis of Symmetry:
 $x = 0$

 Focus:
 $(0, 1)$

 Directrix:
 $y = -1$

$$6. -\frac{1}{10}(y+2)^2 - 3 = x$$

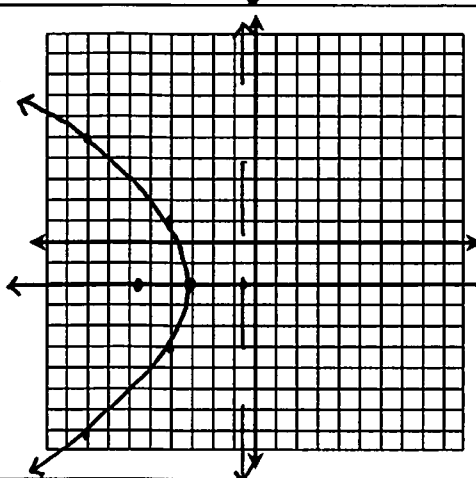
$$-\frac{1}{10}(y+2)^2 = x+3$$

$$\underline{(y+2)^2 = -10(x+3)}$$

horizontal

$$-10 = 4p$$

 $-\frac{5}{2} = p$
 (left)


 Vertex:
 $(-3, -2)$

 Axis of Symmetry:
 $y = -2$

 Focus:
 $(-\frac{11}{2}, -2)$

 Directrix:
 $x = -\frac{1}{2}$

$$7. y^2 - 16x - 2y = -33$$

$$y^2 - 2y = 16x - 33$$

$$y^2 - 2y + 1 = 16x - 33 + 1$$

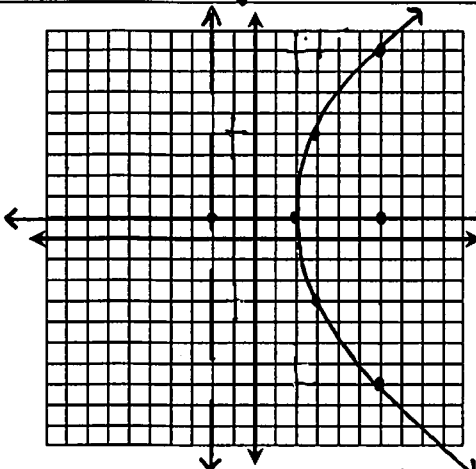
$$(y-1)^2 = 16x - 32$$

$$\underline{(y-1)^2 = 16(x-2)}$$

horizontal

$$16 = 4p$$

 $4 = p$
 (right)


 Vertex:
 $(2, 1)$

 Axis of Symmetry:
 $y = 1$

 Focus:
 $(6, 1)$

 Directrix:
 $x = -2$

$$8. x^2 + 10x + 4y + 1 = 0$$

$$x^2 + 10x = -4y - 1$$

$$x^2 + 10x + 25 = -4y - 1 + 25$$

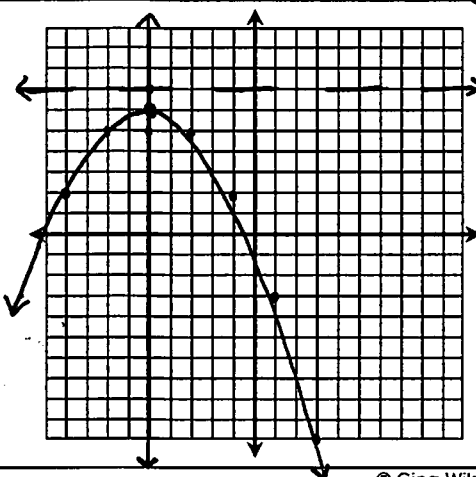
$$(x+5)^2 = -4y + 24$$

$$\underline{(x+5)^2 = -4(y-6)}$$

vertical

$$-4 = 4p$$

 $-1 = p$
 (down)


 Vertex:
 $(-5, 6)$

 Axis of Symmetry:
 $x = -5$

 Focus:
 $(-5, 5)$

 Directrix:
 $y = 7$

Name: _____

Unit 9: Conic Sections

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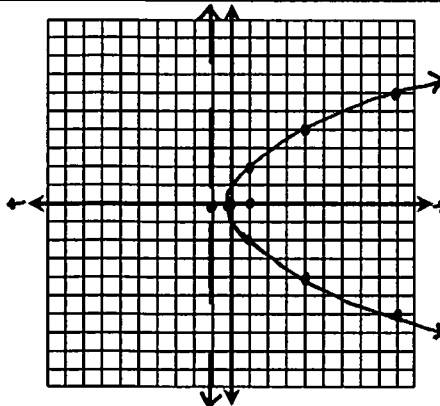
Homework 6: Graphing Parabolas

**** This is a 2-page document! ******Directions:** Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

1. $y^2 = 4x$

Horiz,
Right

$$4 = 4p$$
$$1 = p$$



Vertex:

$(0,0)$

Axis of Symmetry:

$y = 0$

Focus:

$(1,0)$

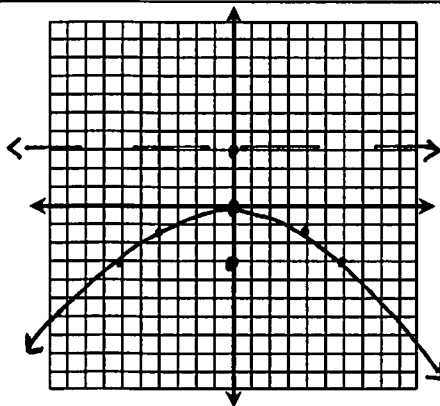
Directrix:

$x = -1$

2. $x^2 = -12y$

vert.,
down

$$-12 = 4p$$
$$-3 = p$$



Vertex:

$(0,0)$

Axis of Symmetry:

$x = 0$

Focus:

$(0,-3)$

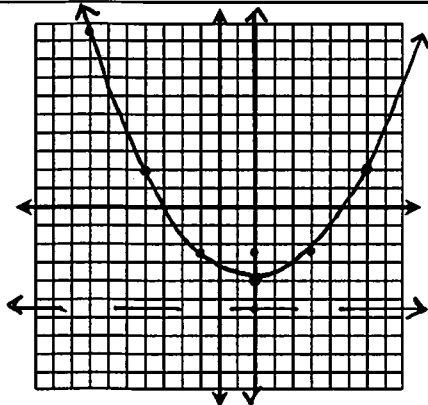
Directrix:

$y = 3$

3. $(x-2)^2 = 6(y+4)$

vert.,
up

$$6 = 4p$$
$$\frac{3}{2} = p$$



Vertex:

$(2,-4)$

Axis of Symmetry:

$x = 2$

Focus:

$(2, -\frac{5}{2})$

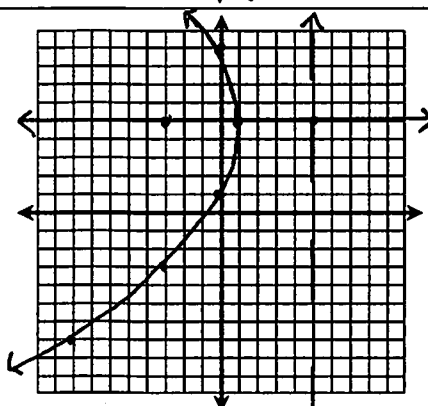
Directrix:

$y = -\frac{11}{2}$

4. $(y-5)^2 = -16(x-1)$

Horiz.,
left

$$-16 = 4p$$
$$-4 = p$$



Vertex:

$(1,5)$

Axis of Symmetry:

$y = 5$

Focus:

$(-3,5)$

Directrix:

$x = 5$

5. $\frac{1}{20}x^2 + 6 = y$

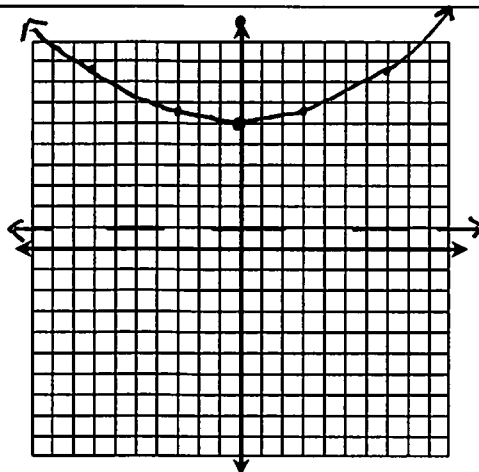
$$\frac{1}{20}x^2 = y - 6$$

$$x^2 = 20(y - 6)$$

$$20 = 4p$$

$$5 = p$$

vert., up



Vertex:

$$(0, 6)$$

Axis of Symmetry:

$$x = 0$$

Focus:

$$(0, 11)$$

Directrix:

$$y = 1$$

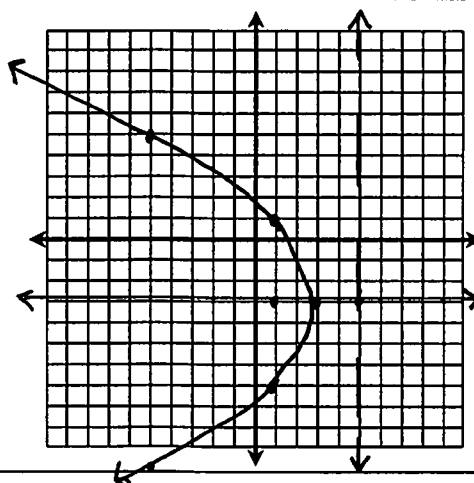
6. $-\frac{1}{8}(y + 3)^2 = x - 3$

$$(y + 3)^2 = -8(x - 3)$$

$$-8 = 4p$$

$$-2 = p$$

Horiz., left



Vertex:

$$(3, -3)$$

Axis of Symmetry:

$$y = -3$$

Focus:

$$(1, -3)$$

Directrix:

$$x = 5$$

7. $y^2 + 4x + 10y + 33 = 0$

$$y^2 + 10y + 25 = -4x - 33 + 25$$

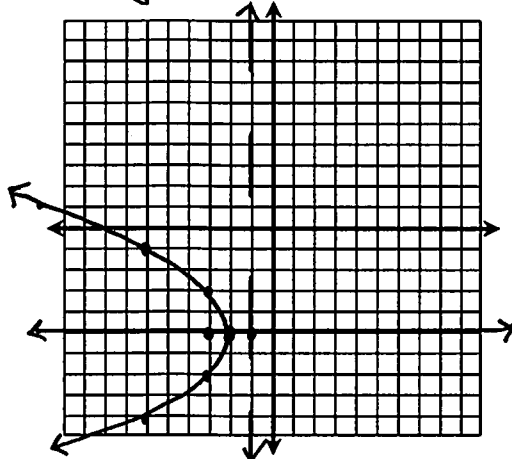
$$(y + 5)^2 = -4x - 8$$

$$(y + 5)^2 = -4(x + 2)$$

$$-4 = 4p$$

$$-1 = p$$

Horiz., left



Vertex:

$$(-2, -5)$$

Axis of Symmetry:

$$y = -5$$

Focus:

$$(-3, -5)$$

Directrix:

$$x = -1$$

8. $x^2 - 6x - 18y - 27 = 0$

$$x^2 - 6x + 9 = 18y + 27 + 9$$

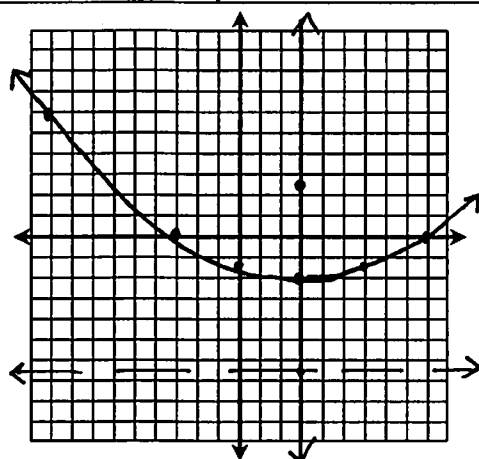
$$(x - 3)^2 = 18y + 36$$

$$(x - 3)^2 = 18(y + 2)$$

$$18 = 4p$$

$$\frac{9}{2} = p$$

vert., up



Vertex:

$$(3, -2)$$

Axis of Symmetry:

$$x = 3$$

Focus:

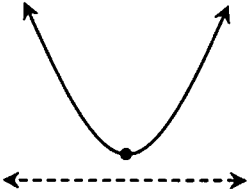
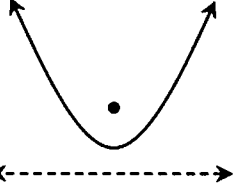
$$(3, \frac{5}{2})$$

Directrix:

$$y = -\frac{13}{2}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
PARABOLA Equations	Standard Form for the Equation of a Parabola at Vertex (h, k):		
	VERTICAL PARABOLA $(x-h)^2 = 4p(y-k)$	HORIZONTAL PARABOLA $(y-k)^2 = 4p(x-h)$	
	_____ is the distance from the vertex to the focus and the directrix.		
SET 1: Given a Graph	1. Vertex = (0, 5) p = 2 $(x-0)^2 = 4(2)(y-5)$ $x^2 = 8(y-5)$	2. Vertex = (3, -2) p = 5 $(y+2)^2 = 4(5)(x-3)$ $(y+2)^2 = 20(x-3)$	
	3. Vertex = (-6, 1) p = 1 $(y-1)^2 = 4(1)(x+6)$ $(y-1)^2 = 4(x+6)$	4. Vertex = (-4, -5) p = -10 $(x+4)^2 = 4(-10)(y+5)$ $(x+4)^2 = -40(y+5)$	
	5. Vertex: (0, 0); Focus: (0, 4) vertical, up 4 = p 16 = 4p $x^2 = 16y$		6. Vertex: (0, 0); Focus: (5/2, 0) Horiz., right p = 5/2 4p = 10 $y^2 = 10x$
	SET 2: Given the Vertex & Focus 		

	7. Vertex: $(-7, \underline{1})$; Focus: $(-7, \underline{\frac{3}{4}})$ vertical, down $p = -\frac{1}{4}$ $4p = -1$ $(x+7)^2 = -1(y-1)$	8. Vertex: $(\underline{2}, -6)$; Focus: $(\underline{-3}, -6)$ Horiz, left $p = -5$ $4p = -20$ $(y+6)^2 = -20(x-2)$
SET 3: Given the Vertex & Directrix 	9. Vertex: $(0, \underline{0})$; Directrix: $y = \underline{8}$ vert, down $p = -8$ $4p = -32$ $x^2 = -32y$	10. Vertex: $(0, \underline{0})$; Directrix: $x = \underline{-2}$ Horiz, right $p = 2$ $4p = 8$ $y^2 = 8x$
	11. Vertex: $(\underline{7}, 4)$; Directrix: $x = \underline{\frac{3}{2}}$ Horiz, right $p = \frac{11}{2}$ $4p = 22$ $(y-4)^2 = 22(x-7)$	12. Vertex: $(-1, \underline{-1})$; Directrix: $y = \underline{-8}$ vert, up $p = 7$ $4p = 28$ $(x+1)^2 = 28(y+1)$
SET 4: Given the Focus & Directrix 	13. Focus: $(\underline{3}, 0)$; Directrix: $x = \underline{-3}$ $\frac{3-3}{2} = 0 \rightarrow$ vertex $(0,0)$ $p = 3$ $4p = 12$ Horiz, right $y^2 = 12x$	14. Focus: $(\underline{7}, -2)$; Directrix: $y = \underline{-4}$ $\frac{-2-4}{2} = -3 \rightarrow$ vertex $(7,-3)$ $p = 1$ $4p = 4$ vert, up $(x-7)^2 = 4(y+3)$
	15. Focus: $(\underline{-8}, 5)$; Directrix: $x = \underline{4}$ $\frac{-8+4}{2} = -2 \rightarrow$ vertex $(-2,5)$ $p = -6$ $4p = -24$ Horiz, left $(y-5)^2 = -24(x+2)$	16. Focus: $(\underline{-4}, \underline{-\frac{3}{2}})$; Directrix: $y = \underline{-\frac{1}{2}}$ $\frac{-\frac{3}{2}-\frac{1}{2}}{2} = -1 \rightarrow$ vertex $(-4,-1)$ $p = -\frac{1}{2}$ $4p = -2$ vert, down $(x+4)^2 = -2(y+1)$

Name: _____

Unit 9: Conic Sections

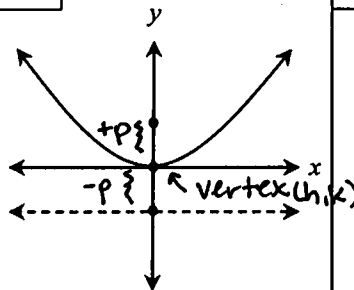
Date: _____ Per: _____

Homework 7: Writing Equations of Parabolas

**** This is a 2-page document! ****Label p , h , and k on each diagram and give the equation of the parabola in standard form.**VERTICAL PARABOLA**

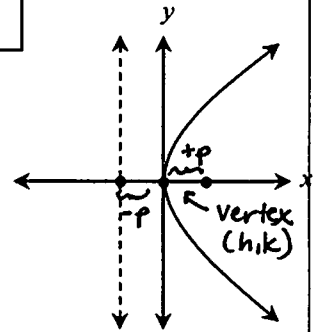
Standard Form:

$$(x-h)^2 = 4p(y-k)$$

**HORIZONTAL PARABOLA**

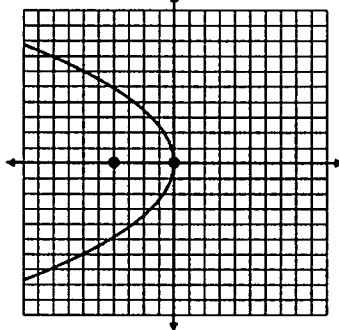
Standard Form:

$$(y-k)^2 = 4p(x-h)$$



Write an equation for each parabola.

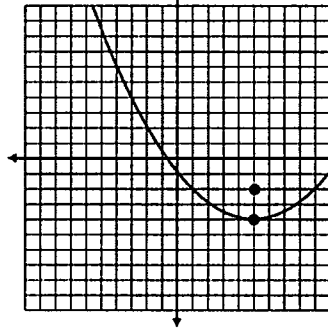
1.



vertex $(0, 0)$
 $p = -4$
 $4p = -16$

$$y^2 = -16x$$

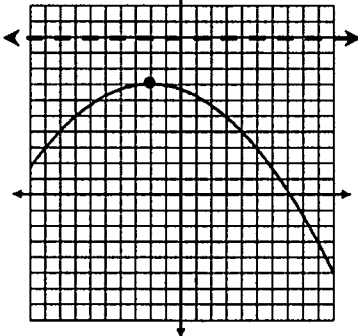
2.



vertex $(5, -4)$
 $p = 2$
 $4p = 8$

$$(x-5)^2 = 8(y+4)$$

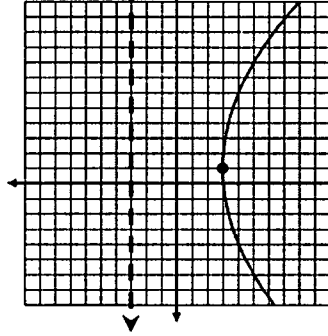
3.



vertex $(-2, 7)$
 $p = -3$
 $4p = -12$

$$(x+2)^2 = -12(y-7)$$

4.



vertex $(3, 1)$
 $p = 6$
 $4p = 24$

$$(y-1)^2 = 24(x-3)$$

Write an equation for each parabola with the given information.

5. Vertex: $(0, 0)$; Focus: $(2, 0)$

Horiz, right

$p = 2$
 $4p = 8$

$$y^2 = 8x$$

6. Vertex: $(0, 0)$; Focus: $(0, -\frac{7}{4})$

vert, down

$p = -\frac{7}{4}$
 $4p = -7$

$$x^2 = -7y$$

7. Vertex: $(-6, -5)$; Focus: $(-6, -4)$

vert, up

$$p=1$$
$$4p=4$$

$$(x+6)^2 = 4(y+5)$$

8. Vertex: $(3, -4)$; Focus: $(\frac{1}{2}, -4)$

horiz, left

$$p = -\frac{5}{2}$$
$$4p = -10$$

$$(y+4)^2 = -10(x-3)$$

9. Vertex: $(0, 0)$; Directrix: $y = -5$

vert, up

$$p=5$$
$$4p=20$$

$$x^2 = 20y$$

10. Vertex: $(-4, -2)$; Directrix: $x = -7$

horiz, right

$$p=3$$
$$4p=12$$

$$(y+2)^2 = 12(x+4)$$

11. Vertex: $(5, 0)$; Directrix: $y = 6$

vert, down

$$p=-6$$
$$4p=-24$$

$$(x-5)^2 = -24y$$

12. Vertex: $(-1, -1)$; Directrix: $x = -\frac{1}{2}$

horiz, left

$$p = -\frac{1}{2}$$
$$4p = -2$$

$$(y+1)^2 = -2(x+1)$$

13. Focus: $(0, -6)$; Directrix: $y = 6$

$$\frac{-6+6}{2} = 0 \rightarrow \text{vertex } (0, 0)$$

vert,
down

$$p=-6$$
$$4p=-24$$

$$x^2 = -24y$$

14. Focus: $(-1, 5)$; Directrix: $x = 7$

$$\frac{-1+7}{2} = 3 \rightarrow \text{vertex } (3, 5)$$

horiz,
left

$$p=-4$$
$$4p=-16$$

$$(y-5)^2 = -16(x-3)$$

15. Focus: $(6, -4)$; Directrix: $x = 0$

$$\frac{6+0}{2} = 3 \rightarrow \text{vertex } (3, -4)$$

horiz,
right

$$p=3$$
$$4p=12$$

$$(y+4)^2 = 12(x-3)$$

16. Focus: $(-7, \frac{17}{4})$; Directrix: $y = -\frac{1}{4}$

$$\frac{\frac{17}{4} - \frac{1}{4}}{2} = 2 \rightarrow \text{vertex } (-7, 2)$$

vert, up

$$p = \frac{9}{4}$$
$$4p = 9$$

$$(x+7)^2 = 9(y-2)$$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 9: Conic Sections

Quiz 9-2: Hyperbolas and Parabolas

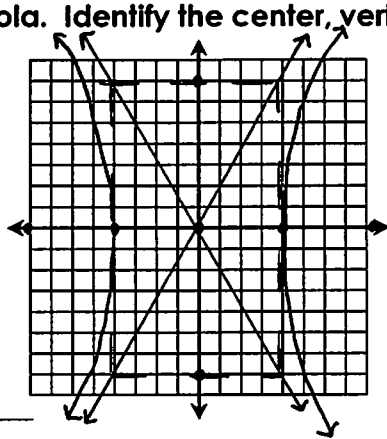
Graph each hyperbola. Identify the center, vertices, co-vertices, foci, and asymptotes.

1. $\frac{x^2}{16} - \frac{y^2}{49} = 1$

$$c^2 = 16 + 49$$

$$c^2 = 65$$

$$c = \sqrt{65} \approx 8.06$$

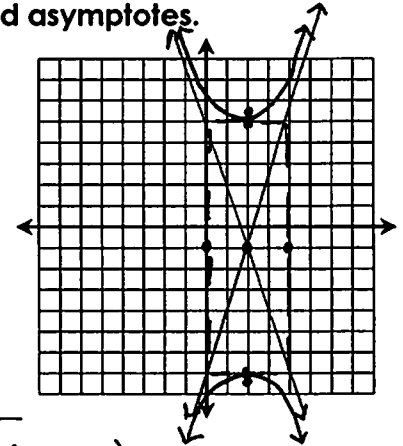
Center: $(0, 0)$ Vertices: $(\pm 4, 0)$ Co-Vertices: $(0, \pm 7)$ Foci: $(\pm 8.06, 0)$ Asymptotes: $y = \pm \frac{7}{4}x$

2. $\frac{(y+1)^2}{36} - \frac{(x-2)^2}{4} = 1$

$$c^2 = 36 + 4$$

$$c^2 = 40$$

$$c = 2\sqrt{10} \approx 6.32$$

Center: $(2, -1)$ Vertices: $(2, 5)$ $(2, -7)$ Co-Vertices: $(0, -1)$ $(4, -1)$ Foci: $(2, 5.32)$ $(2, -7.32)$ Asymptotes: $y = \pm 3(x-2) - 1 \rightarrow$

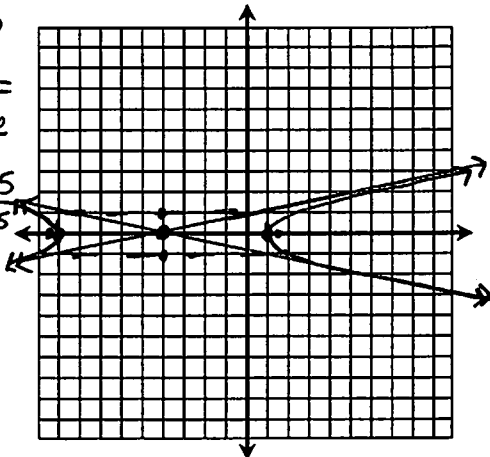
$$y = 3x - 7 \quad \& \quad y = -3x + 5$$

3. $x^2 - 25y^2 + 8x = 9$

$$x^2 + 8x + 16 - 25y^2 = 9 + 16$$

$$\frac{(x+4)^2}{25} - \frac{25y^2}{25} = \frac{25}{25}$$

$$\frac{(x+4)^2}{25} - y^2 = 1$$



Standard Form: $\frac{(x+4)^2}{25} - y^2 = 1$

Center: $(-4, 0)$ Vertices: $(-9, 0)$ $(-1, 0)$ Co-Vertices: $(-4, 1)$ $(-4, -1)$ Foci: $(-9.1, 0)$ $(-0.9, 0)$ Asymptotes: $y = \pm \frac{1}{5}(x+4) \rightarrow$

$$y = \frac{1}{5}x + \frac{4}{5} \quad \& \quad y = -\frac{1}{5}x - \frac{4}{5}$$

Write the equation of the hyperbola in standard form with the given characteristics.

4. Vertices: $(-5, 9)$ and $(-5, -11)$ Vert.

Co-Vertices: $(2, -1)$ and $(-12, -1)$

Center = $\left(\frac{-5-5}{2}, \frac{9-11}{2}\right) = (-5, -1)$

$$a = 10$$

$$b = 7$$

Standard Form: $\frac{(y+1)^2}{100} - \frac{(x+5)^2}{49} = 1$

5. Vertices: $(7, 2)$ and $(1, 2)$ Horiz.

Foci: $(9, 2)$ and $(-1, 2)$

Center = $\left(\frac{7+1}{2}, \frac{2+2}{2}\right) = (4, 2)$

$$a = 3$$

$$c = 5$$

$$25 = a^2 + b^2$$

$$16 = b^2$$

$$4 = b$$

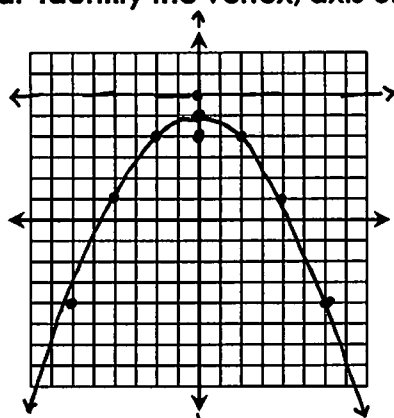
$$\frac{(x-4)^2}{9} - \frac{(y-2)^2}{16} = 1$$

Standard Form: _____

Graph each parabola. Identify the vertex, axis of symmetry, focus, and directrix.

6. $x^2 = -4(y - 5)$

Vert, down
 $-4 = 4p$
 $-1 = p$



Vertex: $(0, 5)$

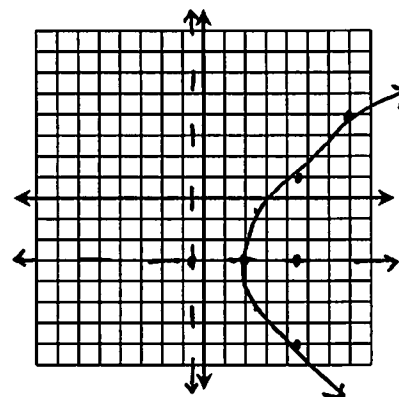
Axis of Sym: $x = 0$

Focus: $(0, 4)$

Directrix: $y = 6$

7. $(y + 3)^2 = 10(x - 2)$

horiz, right
 $10 = 4p$
 $\frac{5}{2} = p$



Vertex: $(2, -3)$

Axis of Sym: $y = -3$

Focus: $(\frac{9}{2}, -3)$

Directrix: $x = -\frac{1}{2}$

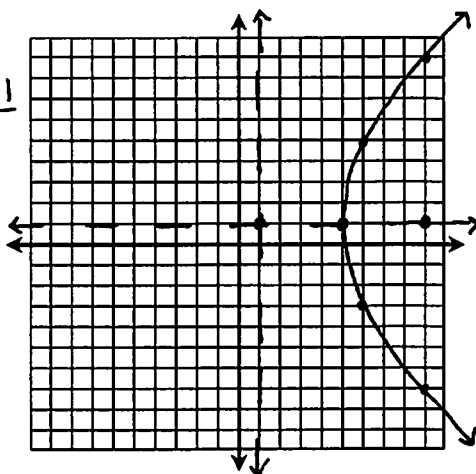
8. $y^2 - 16x - 2y + 81 = 0$

$y^2 - 2y + 1 - 16x = -81 + 1$

$(y - 1)^2 = 16x - 80$

$(y - 1)^2 = 16(x - 5)$

horiz, right
 $16 = 4p$
 $4 = p$



Standard Form: $(y - 1)^2 = 16(x - 5)$

Vertex: $(5, 1)$

Axis of Symmetry: $y = 1$

Focus: $(9, 1)$

Directrix: $x = 1$

Write the equation of the parabola in standard form with the given characteristics.

9. Vertex: $(0, 0)$; Focus: $(0, -6)$

Vert, down
 $p = -6$
 $4p = -24$

Standard Form: $x^2 = -24y$

10. Vertex: $(3, -4)$; Focus: $(5, -4)$

horiz, right
 $p = 2$
 $4p = 8$

Standard Form: $(y + 4)^2 = 8(x - 3)$

11. Vertex: $(-2, -6)$; Directrix: $x = -\frac{5}{2}$

horiz, left
 $p = -\frac{1}{2}$
 $4p = -2$

Standard Form: $(y + 6)^2 = -2(x + 2)$

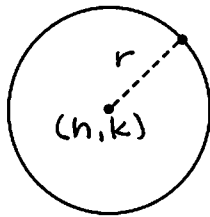
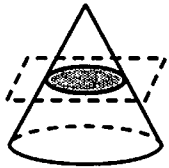
12. Focus: $(-5, 2)$; Directrix: $y = 0$

Vertex $(-5, 1)$
 vert, up
 $p = 1$
 $4p = 4$

Standard Form: $(x + 5)^2 = 4(y - 1)$

CONIC SECTIONS

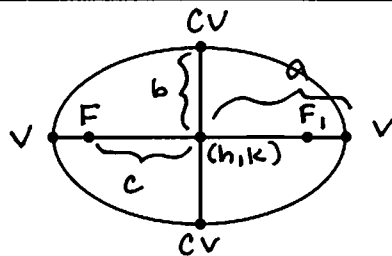
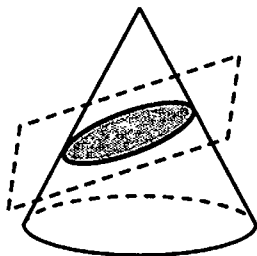
CIRCLES



Standard Form:

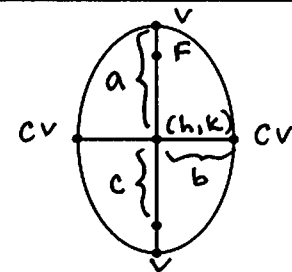
$$(x-h)^2 + (y-k)^2 = r^2$$

ELLIPSES



Standard Form:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

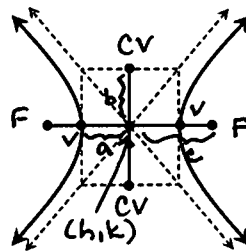
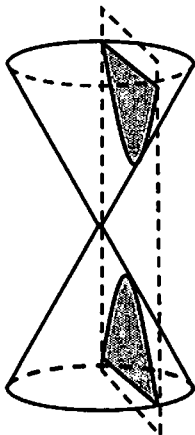


Standard Form:

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

Formula for c: $c^2 = a^2 - b^2$

HYPERBOLAS



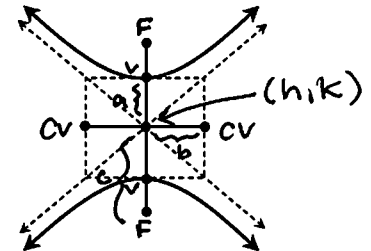
Standard Form:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Asymptotes:

$$y = \pm \frac{b}{a} (x-h) + k$$

Formula for c: $c^2 = a^2 + b^2$



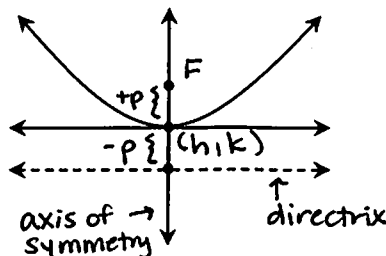
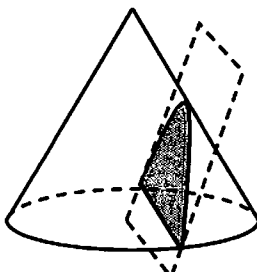
Standard Form:

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Asymptotes:

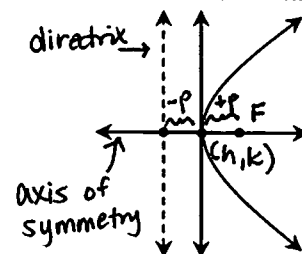
$$y = \pm \frac{a}{b} (x-h) + k$$

PARABOLAS



Standard Form:

$$(x-h)^2 = 4p(y-k)$$



Standard Form:

$$(y-k)^2 = 4p(x-h)$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples				
GENERAL FORM of a Conic Equation	$Ax^2 + Cy^2 + Dx + Ey + F = 0$				
WHICH CONIC IS IT?	➤ Which variables are squared? ○ If just one is squared, it's a <u>parabola</u>. ○ If both are squared, go to the next test. ➤ Look at the coefficients. ○ If they are exactly the same: <u>circle</u> ○ If they are different but have the same sign: <u>ellipse</u> ○ If they have opposite signs: <u>hyperbola</u>				
EXAMPLES	State the conic represented by each equation below. <table border="1"> <tr> <td>1. $\underline{4x^2} + \underline{y^2} - 48x - 14y + 189 = 0$ Ellipse</td><td>2. $\underline{x^2} - 18x + 6y + 99 = 0$ Parabola</td></tr> <tr> <td>3. $\underline{x^2} + \underline{y^2} - 10x - 6y + 18 = 0$ circle</td><td>4. $\underline{-x^2} + \underline{9y^2} + 8x - 54y + 56 = 0$ Hyperbola</td></tr> </table>	1. $\underline{4x^2} + \underline{y^2} - 48x - 14y + 189 = 0$ Ellipse	2. $\underline{x^2} - 18x + 6y + 99 = 0$ Parabola	3. $\underline{x^2} + \underline{y^2} - 10x - 6y + 18 = 0$ circle	4. $\underline{-x^2} + \underline{9y^2} + 8x - 54y + 56 = 0$ Hyperbola
1. $\underline{4x^2} + \underline{y^2} - 48x - 14y + 189 = 0$ Ellipse	2. $\underline{x^2} - 18x + 6y + 99 = 0$ Parabola				
3. $\underline{x^2} + \underline{y^2} - 10x - 6y + 18 = 0$ circle	4. $\underline{-x^2} + \underline{9y^2} + 8x - 54y + 56 = 0$ Hyperbola				

GRAPHING REVIEW

Determine the type of conic, write the equation in standard form, then graph.

5. $x^2 - 8x + y + 14 = 0$

$$x^2 - 8x + \underline{16} = -y - 14 + \underline{16}$$

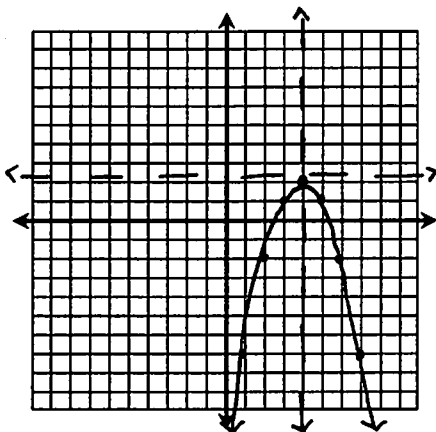
$$(x-4)^2 = -y + 2$$

$$(x-4)^2 = -(y-2)$$

Parabola

$$-1 = 4p$$

$$-\frac{1}{4} = p$$



6. $x^2 - y^2 - 4x - 6y - 9 = 0$

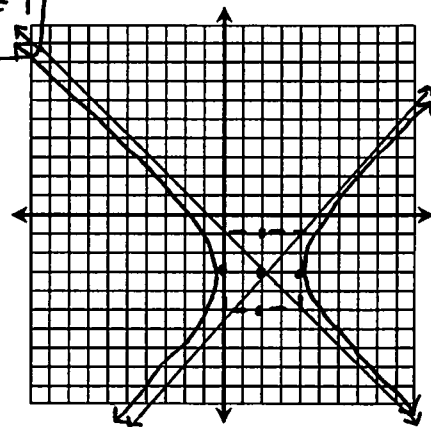
$$x^2 - 4x - y^2 - 6y = 9$$

$$x^2 - 4x + \underline{4} - (y^2 + 6y + \underline{9}) = 9 + \underline{4} - \underline{9}$$

$$(x-2)^2 - (y+3)^2 = 4$$

$$\frac{(x-2)^2}{4} - \frac{(y+3)^2}{4} = 1$$

Hyperbola



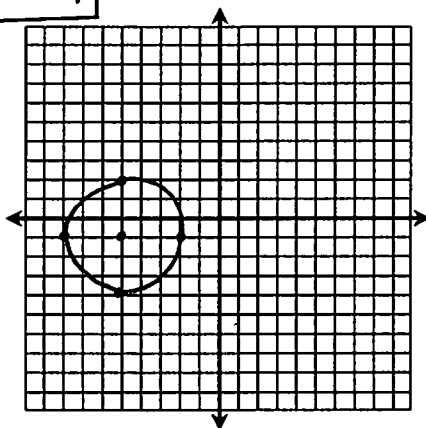
$$7. x^2 + y^2 + 10x + 2y + 17 = 0$$

$$x^2 + 10x + y^2 + 2y = -17$$

$$x^2 + 10x + \underline{25} + y^2 + 2y + \underline{1} = -17 + \underline{25} + \underline{1}$$

$$(x+5)^2 + (y+1)^2 = 9$$

Circle



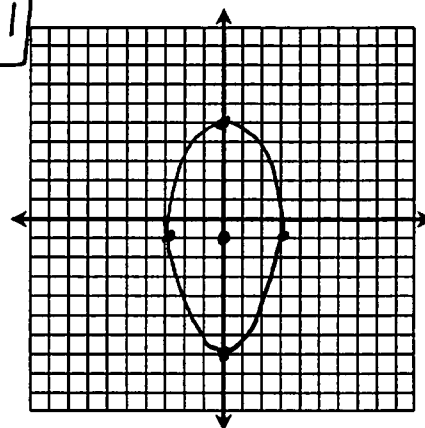
$$8. 4x^2 + y^2 + 2y - 35 = 0$$

$$4x^2 + y^2 + 2y + \underline{1} = 35 + \underline{1}$$

$$4x^2 + (y+1)^2 = 36$$

$$\frac{x^2}{9} + \frac{(y+1)^2}{36} = 1$$

Ellipse



$$9. y^2 - 4x + 8y + 4 = 0$$

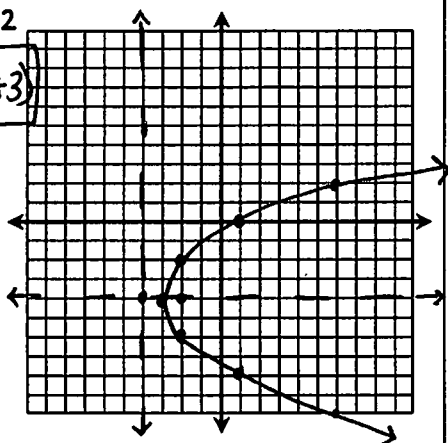
$$y^2 + 8y = 4x - 4$$

$$y^2 + 8y + \underline{16} = 4x - 4 + \underline{16}$$

$$(y+4)^2 = 4x + 12$$

$$(y+4)^2 = 4(x+3)$$

Parabola



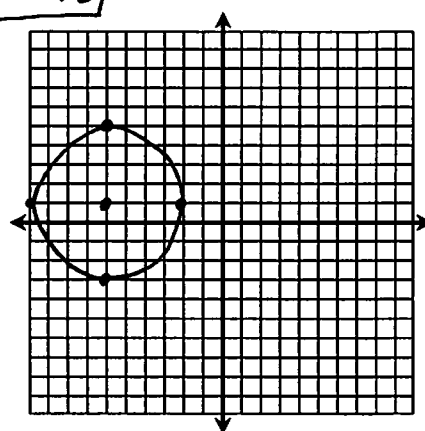
$$10. x^2 + y^2 + 12x - 2y + 21 = 0$$

$$x^2 + 12x + y^2 - 2y = -21$$

$$x^2 + 12x + 36 + y^2 - 2y + 1 = -21 + 36 + 1$$

$$(x+6)^2 + (y-1)^2 = 16$$

Circle



$$11. -16x^2 + 9y^2 - 64x - 18y - 199 = 0$$

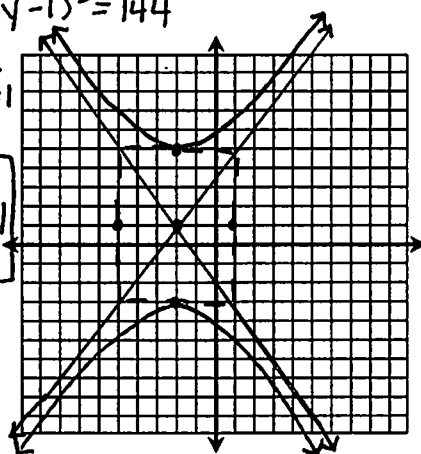
$$-16x^2 - 64x + 9y^2 - 18y = 199$$

$$-16(x^2 + 4x + \underline{4}) + 9(y^2 - 2y + \underline{1}) = 199 - \underline{64} + \underline{9}$$

$$-16(x+2)^2 + 9(y-1)^2 = 144$$

$$-\frac{(x+2)^2}{9} + \frac{(y-1)^2}{16} = 1$$

$$\frac{(y-1)^2}{16} - \frac{(x+2)^2}{9} = 1$$



$$12. x^2 + 9y^2 + 2x + 90y + 190 = 0$$

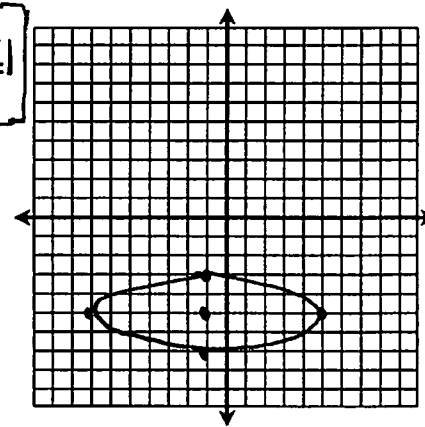
$$x^2 + 2x + 9y^2 + 90y = -190$$

$$x^2 + 2x + \underline{1} + 9(y^2 + 10y + \underline{25}) = -190 + \underline{1} + \underline{225}$$

$$(x+1)^2 + 9(y+5)^2 = 36$$

$$\frac{(x+1)^2}{36} + \frac{(y+5)^2}{4} = 1$$

Ellipse



Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 8: Identifying Conic Sections

** This is a 2-page document! **

State the conic represented by each equation below.

1. $\underline{9x^2} - \underline{4y^2} - 36x + 16y - 16 = 0$

Hyperbola

2. $\underline{x^2} + \underline{y^2} + 4x - 8y + 11 = 0$

Circle

3. $\underline{y^2} - x - 10y + 28 = 0$

Parabola

4. $\underline{49x^2} + \underline{4y^2} - 98x - 147 = 0$

Ellipse

Determine the type of conic, write the equation in standard form, then graph.

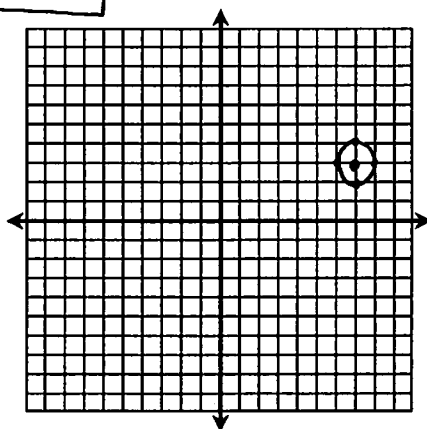
5. $x^2 + y^2 - 14x - 6y + 57 = 0$

Circle

$$x^2 - 14x + y^2 - 6y = -57$$

$$x^2 - 14x + 49 + y^2 - 6y + 9 = -57 + 49 + 9$$

$$(x-7)^2 + (y-3)^2 = 1$$



6. $16x^2 + 25y^2 + 64x - 50y - 311 = 0$

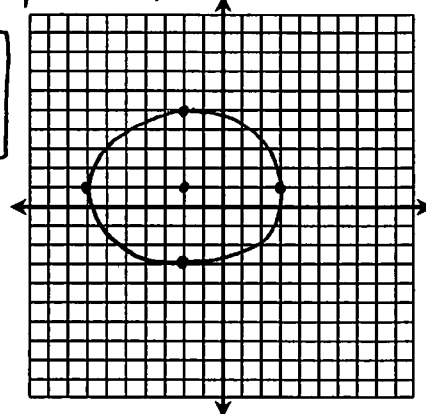
ellipse

$$16x^2 + 64x + 25y^2 - 50y = 311$$

$$16(x^2 + 4x + 4) + 25(y^2 - 2y + 1) = 311 + 64 + 25$$

$$16(x+2)^2 + 25(y-1)^2 = 400$$

$$\frac{(x+2)^2}{25} + \frac{(y-1)^2}{16} = 1$$



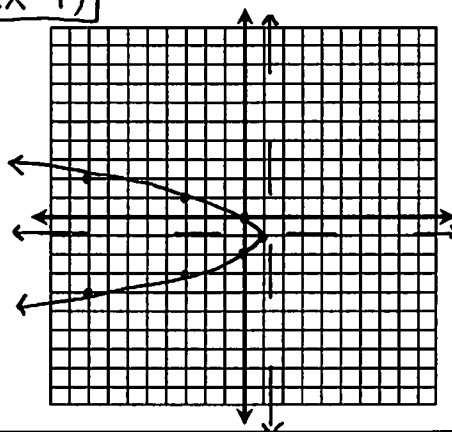
7. $y^2 + x + 2y = 0$

Parabola

$$y^2 + 2y = -x$$

$$y^2 + 2y + 1 = -x + 1$$

$$(y+1)^2 = -(x-1)$$



8. $-x^2 + y^2 + 6x + 6y - 4 = 0$

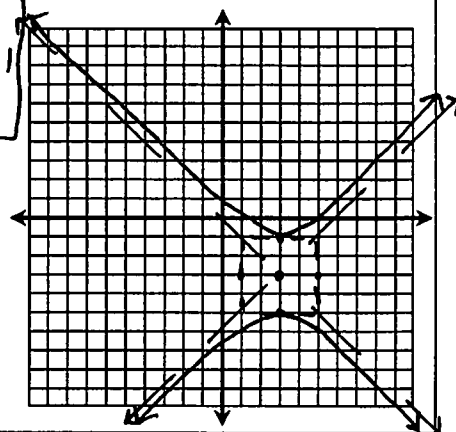
hyperbola

$$-x^2 + 6x + y^2 + 6y = 4$$

$$-(x^2 - 6x + 9) + y^2 + 6y + 9 = 4 - 9 + 9$$

$$-(x-3)^2 + (y+3)^2 = 4$$

$$\frac{(y+3)^2}{4} - \frac{(x-3)^2}{4} = 1$$



9. $2x^2 - 8x - 3y + 11 = 0$

parabola

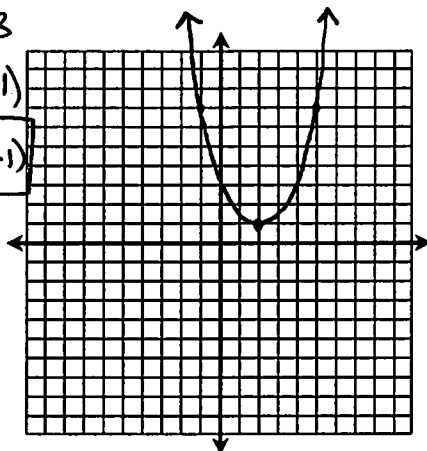
$$2x^2 - 8x = 3y - 11$$

$$2(x^2 - 4x + 4) = 3y - 11 + 8$$

$$2(x-2)^2 = 3y - 3$$

$$2(x-2)^2 = 3(y-1)$$

$$(x-2)^2 = \frac{3}{2}(y-1)$$

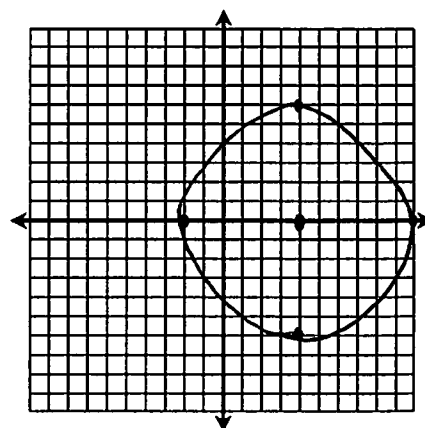


10. $x^2 + y^2 - 8x - 20 = 0$

Circle

$$x^2 - 8x + 16 + y^2 = 20 + 16$$

$$(x-4)^2 + y^2 = 36$$



11. $4x^2 + y^2 - 56x + 6y + 189 = 0$

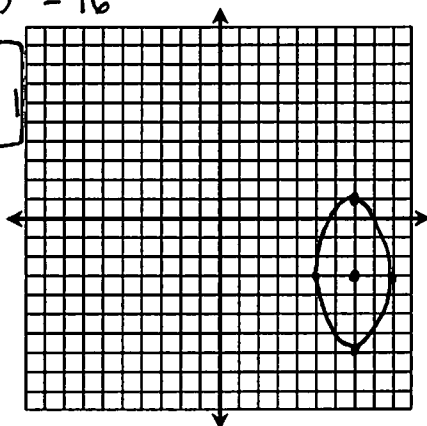
ellipse

$$4x^2 - 56x + y^2 + 6y = -189$$

$$4(x^2 - 14x + 49) + y^2 + 6y + 9 = -189 + 196 + 9$$

$$4(x-7)^2 + (y+3)^2 = 16$$

$$\frac{(x-7)^2}{4} + \frac{(y+3)^2}{16} = 1$$

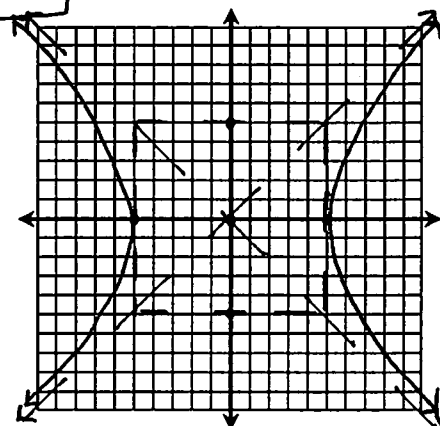


12. $x^2 - y^2 - 25 = 0$

hyperbola

$$x^2 - y^2 = 25$$

$$\frac{x^2}{25} - \frac{y^2}{25} = 1$$



13. $2x^2 + 2y^2 + 12x - 20y + 60 = 0$

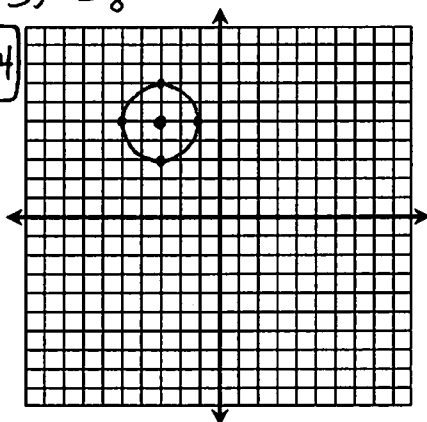
Circle

$$2x^2 + 12x + 2y^2 - 20y = -60$$

$$2(x^2 + 6x + 9) + 2(y^2 - 10y + 25) = -60 + 18 + 50$$

$$2(x+3)^2 + 2(y-5)^2 = 8$$

$$(x+3)^2 + (y-5)^2 = 4$$



14. $y^2 - 12x - 2y - 71 = 0$

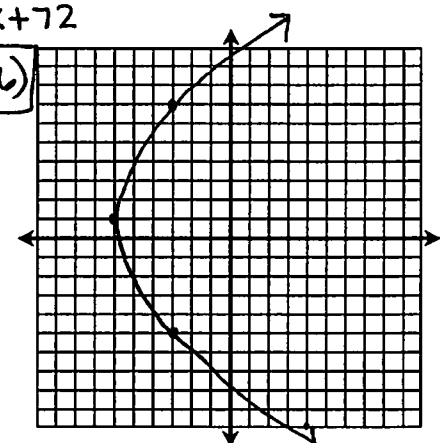
parabola

$$y^2 - 2y = 12x + 71$$

$$y^2 - 2y + 1 = 12x + 71 + 1$$

$$(y-1)^2 = 12x + 72$$

$$(y-1)^2 = 12(x+6)$$



Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

Non-Linear SYSTEM OF EQUATIONS

A system of equations in which one or more equations are not linear.

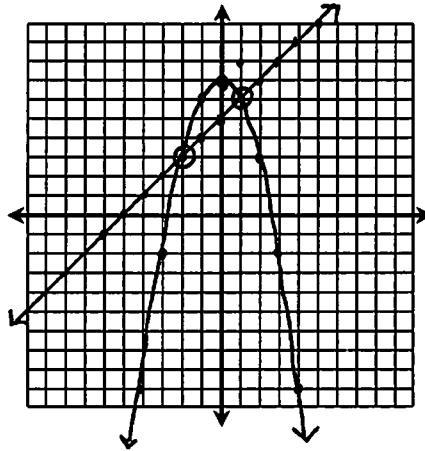
(quadratic/parabola, circle, ellipse, hyperbola, etc.)

SOLUTIONS to Non-Linear Systems

- The **solution set** to a system of equation is the set of points in which the equations intersect.
- There can be **one solution**, **multiple solutions**, or **no real solutions**.

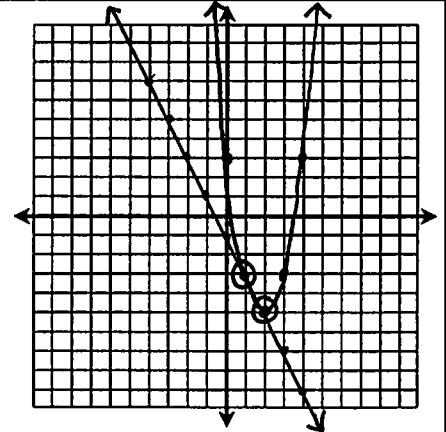
SOLVING A NON-LINEAR SYSTEM *by* GRAPHING

1. $y = -x^2 + 7$
 $y = x + 5$



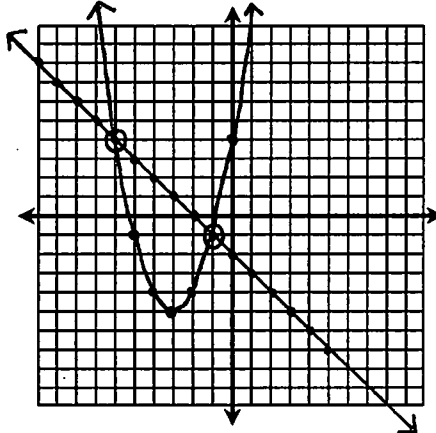
Solutions: $(-2, 3)$ and $(1, 6)$

2. $y = 2x^2 - 8x + 3$
 $y = -2x - 1$



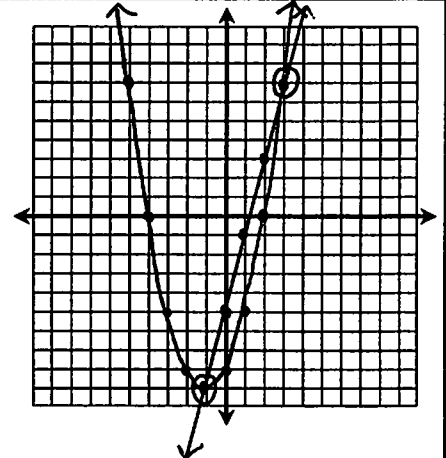
Solutions: $(1, -3)$ and $(2, -5)$

3. $y = x^2 + 6x + 4$
 $x + y = -2$
↓
 $y = -x - 2$



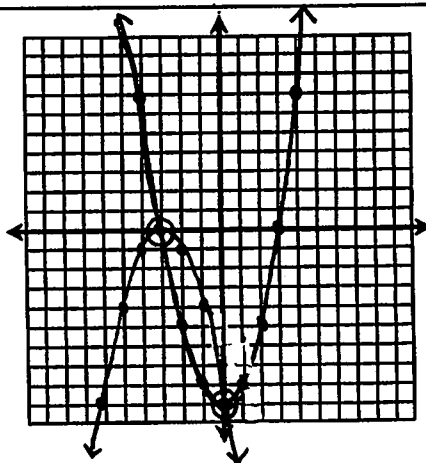
Solutions: $(-6, 4)$ and $(-1, -1)$

4. $y = x^2 + 2x - 8$
 $4x - y = 5$
↓
 $y = 4x - 5$



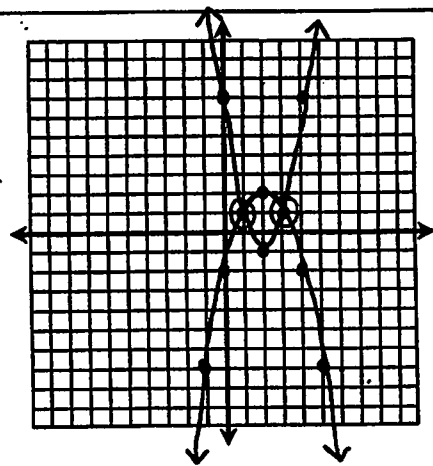
Solutions: $(-1, -9)$ and $(3, 7)$

5. $y = x^2 - 9$
 $y = -x^2 - 6x - 9$



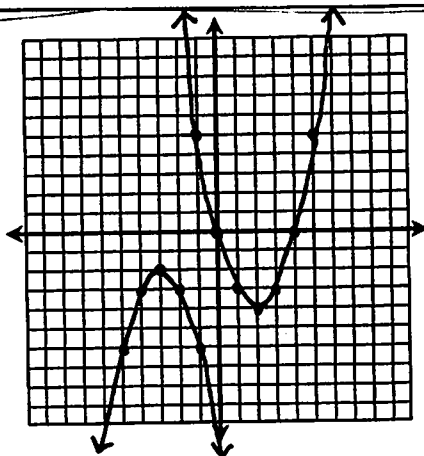
Solutions: $(-3, 0)$ $(0, -9)$

6. $y = 2x^2 - 8x + 7$
 $y = -x^2 + 4x - 2$



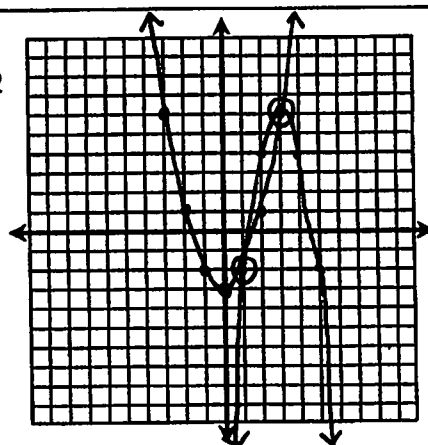
Solutions: $(1, 1)$ and $(3, 1)$

7. $y = -x^2 - 6x - 11$
 $y = x^2 - 4x$



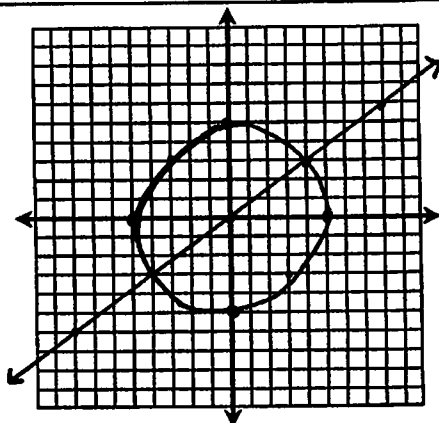
Solutions: No Solution

8. $y = x^2 - 3$
 $y = -2x^2 + 12x - 12$



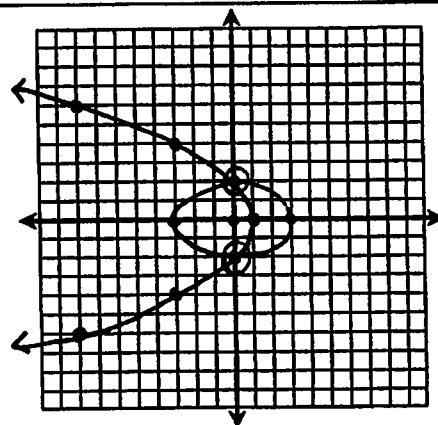
Solutions: $(1, -2)$ and $(3, 6)$

9. $x^2 + y^2 = 25$
 $3x - 4y = 0$
 $\downarrow$
 $y = \frac{3}{4}x$



Solutions: $(-4, -3)$ and $(4, 3)$

10. $y^2 = -4x + 4$
 $4x^2 + 9y^2 = 36$
 $\downarrow$
 $\frac{x^2}{9} + \frac{y^2}{4} = 1$



Solutions: $(0, 2)$ and $(0, -2)$

Name: _____

Unit 9: Conic Sections

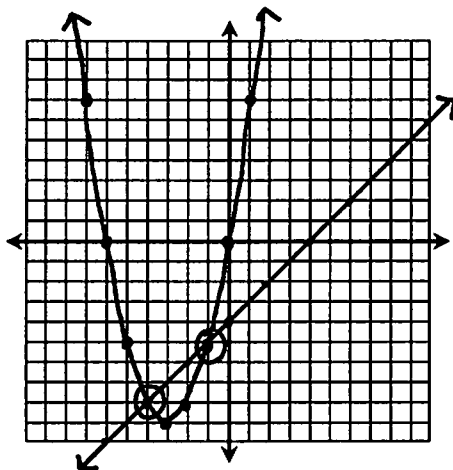
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Homework 9: Non-Linear Systems (Day 1)

**** This is a 2-page document! ****

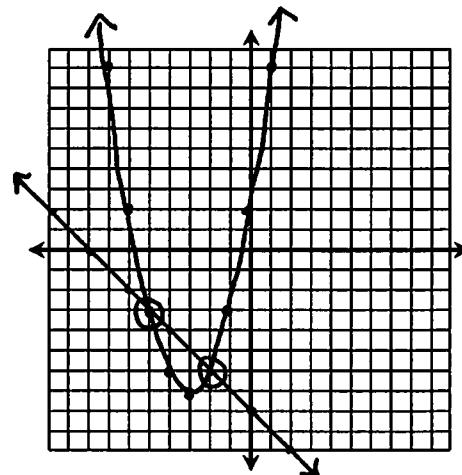
Solve each system of equations by graphing. Identify all possible solutions.

1. $y = x^2 + 6x$
 $y = x - 4$



Solutions: $(-4, -8)$ and $(-1, -5)$

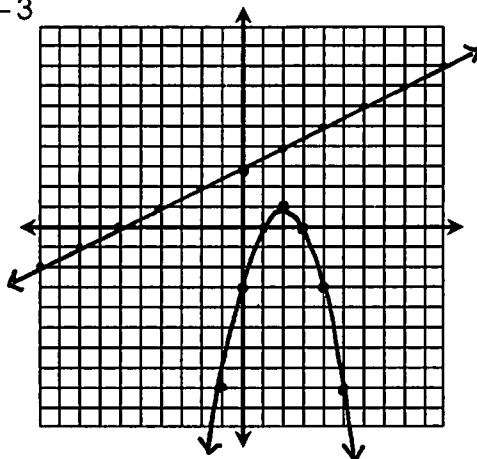
2. $y = x^2 + 6x + 2$
 $y = -x - 8$



Solutions: $(-5, -3)$ and $(-2, -6)$

3. $x - 2y = -6$
 $y = -x^2 + 4x - 3$

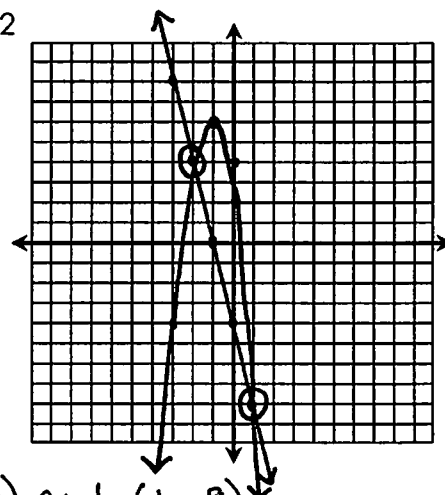
$-2y = -x - 6$
 $y = \frac{1}{2}x + 3$



Solutions: No Solution

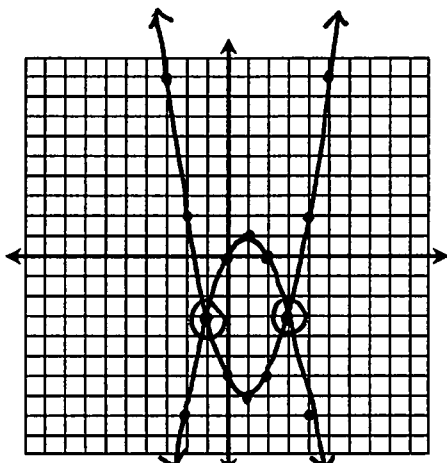
4. $4x + y = -4$
 $y = -3x^2 - 7x + 2$

$y = -4x - 4$



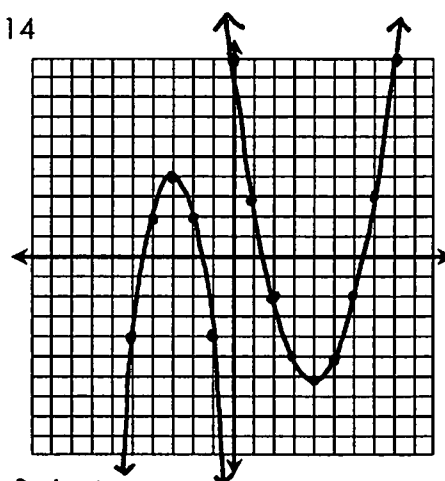
Solutions: $(-2, 4)$ and $(1, -8)$

5. $y = -x^2 + 2x$
 $y = x^2 - 2x - 6$



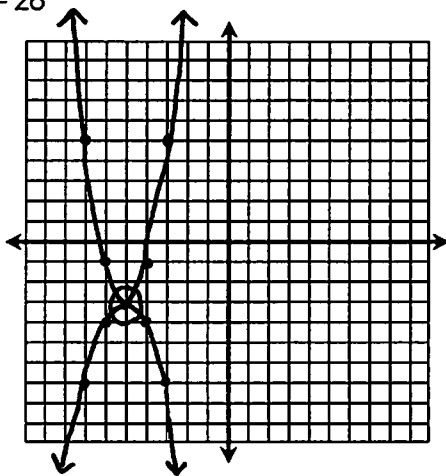
Solutions: $(-1, -3)$ and $(3, -3)$

6. $y = x^2 - 8x + 10$
 $y = -2x^2 - 12x - 14$



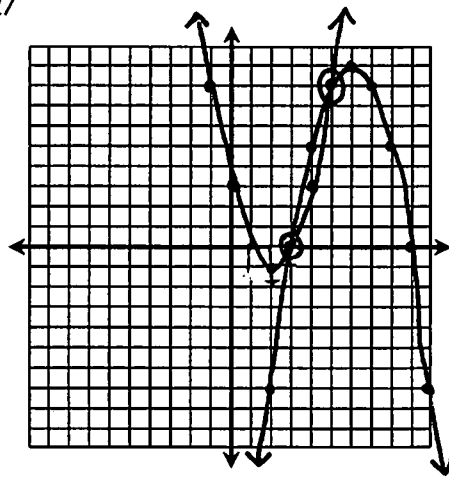
Solutions: No Solution

7. $y = 2x^2 + 20x + 47$
 $y = -x^2 - 10x - 28$



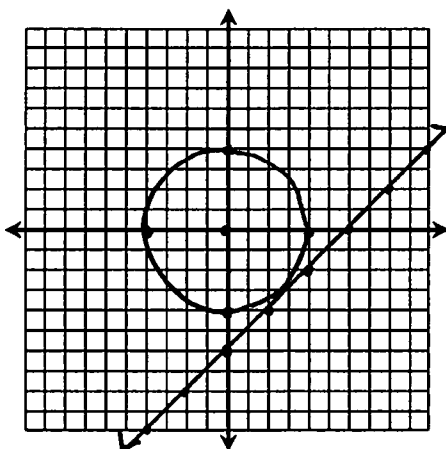
Solutions: $(-5, 3)$

8. $y = x^2 - 4x + 3$
 $y = -x^2 + 12x - 27$



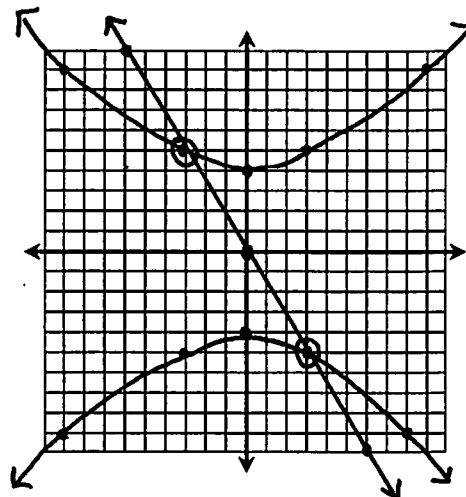
Solutions: $(3, 0)$ and $(5, 8)$

9. $x^2 + y^2 = 16$
 $x - y = 6$
 $\downarrow$
 $y = x - 6$



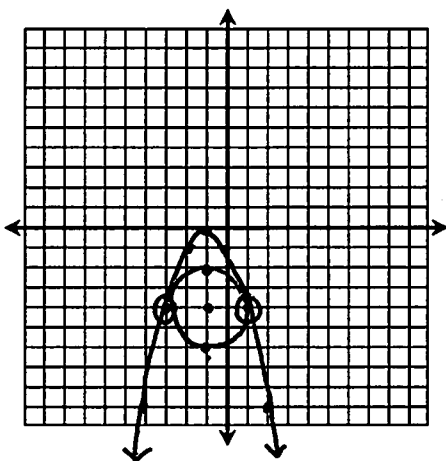
Solutions: No Solution

10. $5x + 3y = 0$
 $y^2 - x^2 = 16$
 $\downarrow$
 $y = -\frac{5}{3}x$



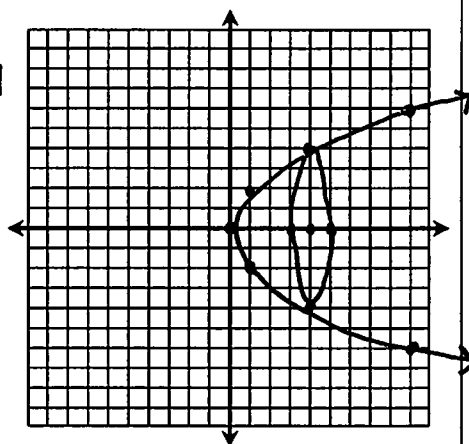
Solutions: $(-3, 5)$ and $(3, -5)$

11. $(x+1)^2 + (y+4)^2 = 4$
 $y = -x^2 - 2x - 1$



Solutions: $(-3, -4)$ and $(1, -4)$

12. $y^2 = 4x$
 $16(x-4)^2 + y^2 = 16$
 $\downarrow$
 $(x-4)^2 + \frac{y^2}{16} = 1$



Solutions: $(4, 4)$ and $(4, -4)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																
SUBSTITUTION <i>Method</i>	$\begin{cases} 4x + y = -20 \rightarrow y = -4x - 20 \\ y = x^2 + 5x - 2 \end{cases}$ $-4x - 20 = x^2 + 5x - 2$ $0 = x^2 + 9x + 18$ $0 = (x+3)(x+6)$ <table style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <tr> <td style="text-align: center; padding: 0 10px;">$x = -3$</td> <td style="text-align: center; padding: 0 10px;">$x = -6$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = -4(-3) - 20$</td> <td style="text-align: center; padding: 0 10px;">$y = -4(-6) - 20$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = -8$</td> <td style="text-align: center; padding: 0 10px;">$y = 4$</td> </tr> </table> $(-3, -8)$ and $(-6, 4)$	$x = -3$	$x = -6$	$\downarrow$	$\downarrow$	$y = -4(-3) - 20$	$y = -4(-6) - 20$	$y = -8$	$y = 4$								
$x = -3$	$x = -6$																
$\downarrow$	$\downarrow$																
$y = -4(-3) - 20$	$y = -4(-6) - 20$																
$y = -8$	$y = 4$																
ELIMINATION <i>Method</i>	$\begin{cases} y = x^2 + 3x + 2 \\ y = x^2 + 5x - 4 \end{cases}$ $y = x^2 + 3x + 2$ $-(y = x^2 + 5x - 4)$ $0 = -2x + 6$ $2x = 6$ $x = 3 \rightarrow$ $y = (3)^2 + 3(3) + 2$ $y = 9 + 9 + 2$ $y = 20$ $(3, 20)$																
EXAMPLES	Solve each system algebraically using your method of choice. 1. $\begin{cases} y = x^2 - 3x - 2 \\ y = x + 3 \end{cases}$ $x + 3 = x^2 - 3x - 2$ $0 = x^2 - 4x - 5$ $0 = (x-5)(x+1)$ <table style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <tr> <td style="text-align: center; padding: 0 10px;">$x = 5$</td> <td style="text-align: center; padding: 0 10px;">$x = -1$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = 5 + 3$</td> <td style="text-align: center; padding: 0 10px;">$y = -1 + 3$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = 8$</td> <td style="text-align: center; padding: 0 10px;">$y = 2$</td> </tr> </table> $(5, 8)$ and $(-1, 2)$ 2. $\begin{cases} y = -x^2 - 8x - 6 \\ 2x + y = -1 \end{cases}$ $\downarrow$ $y = -2x - 1$ $-2x - 1 = -x^2 - 8x - 6$ $0 = -x^2 - 6x - 5$ $0 = x^2 + 6x + 5$ $0 = (x+5)(x+1)$ <table style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <tr> <td style="text-align: center; padding: 0 10px;">$x = -5$</td> <td style="text-align: center; padding: 0 10px;">$x = -1$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> <td style="text-align: center; padding: 0 10px;">$\downarrow$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = -2(-5) - 1$</td> <td style="text-align: center; padding: 0 10px;">$y = -2(-1) - 1$</td> </tr> <tr> <td style="text-align: center; padding: 0 10px;">$y = 9$</td> <td style="text-align: center; padding: 0 10px;">$y = 1$</td> </tr> </table> $(-5, 9)$ and $(-1, 1)$ 3. $\begin{cases} 3x - y = 7 \rightarrow y = 3x - 7 \\ y = 2x^2 + 20x + 46 \end{cases}$ $3x - 7 = 2x^2 + 20x + 46$ $0 = 2x^2 + 17x + 53$ * cannot be factored - imaginary roots * No Solution	$x = 5$	$x = -1$	$\downarrow$	$\downarrow$	$y = 5 + 3$	$y = -1 + 3$	$y = 8$	$y = 2$	$x = -5$	$x = -1$	$\downarrow$	$\downarrow$	$y = -2(-5) - 1$	$y = -2(-1) - 1$	$y = 9$	$y = 1$
$x = 5$	$x = -1$																
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$x = -5$	$x = -1$																
$\downarrow$	$\downarrow$																
$y = -2(-5) - 1$	$y = -2(-1) - 1$																
$y = 9$	$y = 1$																

$$4. \begin{cases} x^2 + y^2 = 18 \\ y = x - 6 \end{cases}$$

$$\begin{aligned} x^2 + (x-6)^2 &= 18 \\ x^2 + x^2 - 12x + 36 &= 18 \\ 2x^2 - 12x + 18 &= 0 \\ 2(x^2 - 6x + 9) &= 0 \\ 2(x-3)(x-3) &= 0 \\ \hline x=3 & \quad | \quad x=3 \end{aligned}$$

$$(3, -3)$$

$$y = 3 - 6 \\ y = -3$$

$$5. \begin{cases} y^2 - x^2 + 75 = 0 \\ 2y - x = 0 \end{cases}$$

$$\downarrow$$

$$x = 2y$$

$$\begin{aligned} y^2 - (2y)^2 + 75 &= 0 \\ y^2 - 4y^2 &= -75 \\ -3y^2 &= -75 \\ y^2 &= 25 \\ y &= \pm 5 \end{aligned}$$

$$x = 2(5) \\ x = 10$$

$$x = 2(-5) \\ x = -10$$

$$(10, 5) \text{ and } (-10, -5)$$

$$6. \begin{cases} y = 2x^2 + 3x - 19 \\ (y = x^2 + 2x - 7) - 1 \end{cases}$$

$$\begin{aligned} &\rightarrow \begin{array}{r} y = 2x^2 + 3x - 19 \\ - y = -x^2 - 2x + 7 \\ \hline 0 = x^2 + x - 12 \\ 0 = (x+4)(x-3) \\ \hline x = -4 \quad | \quad x = 3 \end{array} \end{aligned}$$

$$(-4, 1) \text{ and } (3, 8)$$

$$\begin{aligned} y &= (-4)^2 + 2(-4) - 7 \\ y &= 16 - 8 - 7 \\ y &= 1 \end{aligned}$$

$$\begin{aligned} y &= 3^2 + 2(3) - 7 \\ y &= 9 + 6 - 7 \\ y &= 8 \end{aligned}$$

$$7. \begin{cases} y = 4x^2 - 29 \\ (y = x^2 - 17) - 1 \end{cases}$$

$$\begin{aligned} &\rightarrow \begin{array}{r} y = 4x^2 - 29 \\ - y = -x^2 + 17 \\ \hline 0 = 3x^2 - 12 \end{array} \end{aligned}$$

$$(2, -13) \text{ and } (-2, -13)$$

$$\begin{aligned} 0 &= 3x^2 - 12 \\ 12 &= 3x^2 \\ 4 &= x^2 \\ \pm 2 &= x \end{aligned}$$

$$\begin{aligned} y &= 2^2 - 17 \\ y &= 4 - 17 \\ y &= -13 \\ y &= (-2)^2 - 17 \\ y &= -13 \end{aligned}$$

$$8. \begin{cases} x^2 + 2y^2 = 17 \\ 2x^2 - 3y^2 = 6 \end{cases} - 2$$

$$\begin{aligned} &\rightarrow \begin{array}{r} -2x^2 - 4y^2 = -34 \\ + 2x^2 - 3y^2 = 6 \\ \hline -7y^2 = -28 \end{array} \end{aligned}$$

$$(3, 2), (3, -2), (-3, 2), (-3, -2)$$

$$\begin{aligned} -7y^2 &= -28 \\ y^2 &= 4 \\ y &= \pm 2 \end{aligned}$$

$$\begin{aligned} x^2 + 2(2)^2 &= 17 \\ x^2 + 8 &= 17 \\ x^2 &= 9 \\ x &= \pm 3 \end{aligned}$$

$$\begin{aligned} x^2 + 2(-2)^2 &= 17 \\ x^2 + 8 &= 17 \\ x^2 &= 9 \\ x &= \pm 3 \end{aligned}$$

Name: _____

Unit 9: Conic Sections

Date: _____ Per: _____

Homework 10: Non-Linear Systems (Day 2)

**** This is a 2-page document! ******Solve each system of equations algebraically using substitution or elimination.**

$$1. \begin{cases} y = x^2 - 9x + 12 \\ y = -5x + 9 \end{cases}$$

$$-5x + 9 = x^2 - 9x + 12$$

$$0 = x^2 - 4x + 3$$

$$0 = (x-3)(x-1)$$

$$\begin{array}{c|c} x=3 & x=1 \end{array}$$

$$\downarrow$$

$$y = -5(3) + 9$$

$$y = -6$$

$$\downarrow$$

$$y = -5(1) + 9$$

$$y = 4$$

$(3, -6)$ and $(1, 4)$

$$2. \begin{cases} y = -x^2 - x - 5 \\ x - y = 5 \end{cases}$$

$$\downarrow$$

$$y = x - 5$$

$$x - 5 = -x^2 - x - 5$$

$$0 = -x^2 - 2x$$

$$0 = -x(x+2)$$

$$\begin{array}{c|c} x=0 & x=-2 \end{array}$$

$$\downarrow$$

$$y = 0 - 5$$

$$y = -5$$

$$\downarrow$$

$$y = -2 - 5$$

$$y = -7$$

$(0, -5)$ and $(-2, -7)$

$$3. \begin{cases} y = 2x^2 + 4x \\ x + y = -3 \end{cases}$$

$$\downarrow$$

$$y = -x - 3$$

$$-x - 3 = 2x^2 + 4x$$

$$0 = 2x^2 + 5x + 3$$

$$0 = (2x+3)(x+1)$$

$$\begin{array}{c|c} x = -\frac{3}{2} & x = -1 \end{array}$$

$$\downarrow$$

$$y = -(-\frac{3}{2}) - 3$$

$$y = -\frac{3}{2}$$

$$\downarrow$$

$$y = -(-1) - 3$$

$$y = -2$$

$(-\frac{3}{2}, -\frac{3}{2})$ and $(-1, -2)$

$$4. \begin{cases} y = x^2 - 2x - 3 \\ 2x - y = 7 \end{cases}$$

$$\downarrow$$

$$y = 2x - 7$$

$$2x - 7 = x^2 - 2x - 3$$

$$0 = x^2 - 4x + 4$$

$$0 = (x-2)(x-2)$$

$$\begin{array}{c|c} x=2 & x=2 \end{array}$$

$$\downarrow$$

$$y = 2(2) - 7$$

$$y = 4 - 7$$

$$y = -3$$

$(2, -3)$

$$5. \begin{cases} x^2 + y^2 = 40 \\ x - y = 4 \end{cases}$$

$$\downarrow$$

$$x = y + 4$$

$$(y+4)^2 + y^2 = 40$$

$$y^2 + 8y + 16 + y^2 = 40$$

$$2y^2 + 8y - 24 = 0$$

$$2(y^2 + 4y - 12) = 0$$

$$2(y+6)(y-2) = 0$$

$$\begin{array}{c|c} y = -6 & y = 2 \end{array}$$

$$\downarrow$$

$$x = -6 + 4$$

$$x = -2$$

$$\downarrow$$

$$x = 2 + 4$$

$$x = 6$$

$(-2, -6)$ and $(6, 2)$

$$6. \begin{cases} y^2 = 8x - 3 \\ 2x - y = -1 \end{cases}$$

$$\downarrow$$

$$y = 2x + 1$$

$$(2x+1)^2 = 8x - 3$$

$$4x^2 + 4x + 1 = 8x - 3$$

$$4x^2 - 4x + 4 = 0$$

$$4(x^2 - x + 1) = 0$$

* cannot be factored *

No Solution

$$7. \begin{cases} x^2 = 2y - 9 \\ 4x - y = 3 \end{cases}$$

$$\downarrow$$

$$y = 4x - 3$$

$$x^2 = 2(4x - 3) - 9$$

$$x^2 = 8x - 15$$

$$x^2 - 8x + 15 = 0$$

$$\frac{(x-5)(x-3)=0}{x=5 \quad | \quad x=3}$$

$$y = 4(5) - 3$$

$$y = 17$$

$$y = 4(3) - 3$$

$$y = 9$$

$(5, 17)$ and $(3, 9)$

$$8. \begin{cases} x^2 - 9y^2 = 28 \\ x = 4y \end{cases}$$

$$(4y)^2 - 9y^2 = 28$$

$$16y^2 - 9y^2 = 28$$

$$7y^2 = 28$$

$$y^2 = 4$$

$$y = \pm 2$$

$$x = 4(2)$$

$$x = 8$$

$$x = 4(-2)$$

$$x = -8$$

$(8, 2)$ and $(-8, -2)$

$$9. \begin{cases} y = x^2 + 5x - 13 \\ (y = x^2 - 3x + 3) - 1 \end{cases} \rightarrow \begin{cases} y = x^2 + 5x - 13 \\ -y = -x^2 + 3x - 3 \end{cases}$$

$$0 = 8x - 16$$

$$16 = 8x$$

$$2 = x$$

$$y = 2^2 + 5(2) - 13$$

$$y = 4 + 10 - 13$$

$$y = 1$$

$(2, 1)$

$$10. \begin{cases} y = 3x^2 - 4x - 11 \\ (y = 2x^2 - 7x - 13) - 1 \end{cases} \rightarrow \begin{cases} y = 3x^2 - 4x - 11 \\ -y = -2x^2 + 7x + 13 \end{cases}$$

$$0 = x^2 + 3x + 2$$

$$0 = (x+2)(x+1)$$

$$x = -2 \quad | \quad x = -1$$

$$y = 3(-2)^2 - 4(-2) - 11$$

$$y = 12 + 8 - 11$$

$$y = 9$$

$$y = 3(-1)^2 - 4(-1) - 11$$

$$y = 3 + 4 - 11$$

$$y = -4$$

$(-2, 9)$ and $(-1, -4)$

$$11. \begin{cases} y = x^2 + 9x + 7 \\ -1(y = -x^2 + 3x + 7) \end{cases} \rightarrow \begin{cases} y = x^2 + 9x + 7 \\ -y = x^2 - 3x - 7 \end{cases}$$

$$0 = 2x^2 + 6x$$

$$0 = 2x(x+3)$$

$$x = 0 \quad | \quad x = -3$$

$$y = 0^2 + 9(0) + 7$$

$$y = 7$$

$$y = (-3)^2 + 9(-3) + 7$$

$$y = 9 - 27 + 7$$

$$y = -11$$

$(0, 7)$ and $(-3, -11)$

$$12. \begin{cases} x^2 + y^2 = 13 \\ -1(x^2 + 4y^2 = 25) \end{cases} \rightarrow \begin{cases} x^2 + y^2 = 13 \\ -x^2 - 4y^2 = -25 \end{cases}$$

$$-3y^2 = -12$$

$$y^2 = 4$$

$$y = \pm 2$$

$$x^2 + 2^2 = 13$$

$$x^2 = 9$$

$$x = \pm 3$$

$$x^2 + (-2)^2 = 13$$

$$x^2 = 9$$

$$x = \pm 3$$

$(2, 3), (2, -3), (-2, 3), (-2, -3)$

Name: _____

Algebra 2

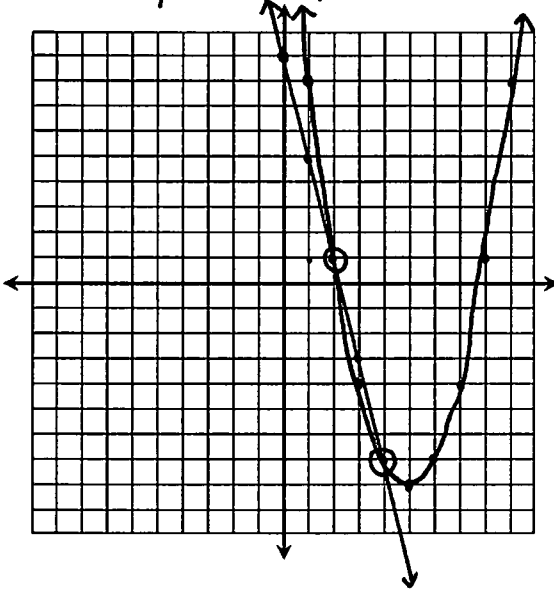
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Unit 9: Conic Sections

Quiz 9-3: Non-Linear Systems of Equations

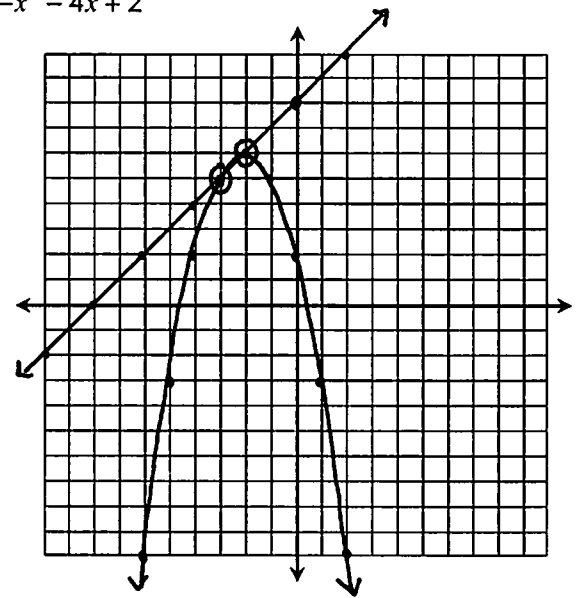
Solve each system by graphing.

1. $\begin{cases} y = x^2 - 10x + 17 \\ 4x + y = 9 \rightarrow y = -4x + 9 \end{cases}$



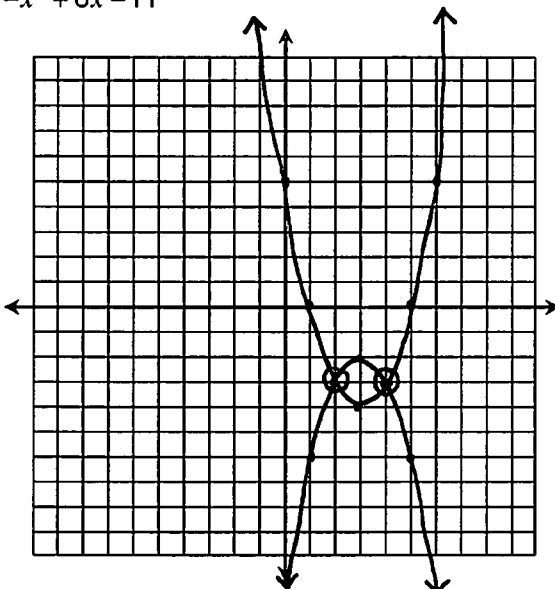
Solution(s): (2, 1) and (4, -7)

2. $\begin{cases} x - y = -8 \rightarrow y = x + 8 \\ y = -x^2 - 4x + 2 \end{cases}$



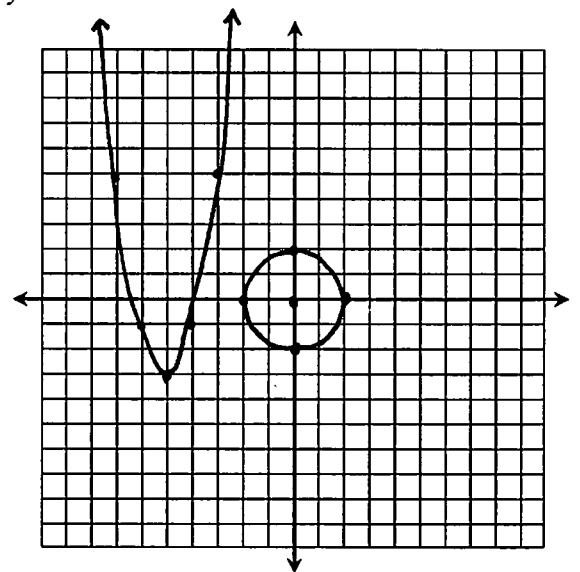
Solution(s): (-2, 6) and (-3, 5)

3. $\begin{cases} y = x^2 - 6x + 5 \\ y = -x^2 + 6x - 11 \end{cases}$



Solution(s): (2, -3) and (4, -3)

4. $\begin{cases} y = 2x^2 + 20x + 47 \\ x^2 + y^2 = 4 \end{cases}$



Solution(s): no solution

Solve each system using substitution or elimination.

$$5. \begin{cases} y = x^2 - 3x - 8 \\ y = x - 3 \end{cases}$$

$$x - 3 = x^2 - 3x - 8$$

$$0 = x^2 - 4x - 5$$

$$0 = (x-5)(x+1)$$

$$\frac{x=5}{x=-1}$$

$$y = 5 - 3$$

$$y = 2$$

$$y = -1 - 3$$

$$y = -4$$

Solution(s): (5, 2) and (-1, -4)

$$7. \begin{cases} x = y + 2 \\ x^2 + y^2 = 10 \end{cases}$$

$$(y+2)^2 + y^2 = 10$$

$$y^2 + 4y + 4 + y^2 = 10$$

$$2y^2 + 4y - 6 = 0$$

$$2(y^2 + 2y - 3) = 0$$

$$2(y+3)(y-1) = 0$$

$$y = -3 + 2$$

$$x = -1$$

$$y = -3$$

$$y = 1$$

$$x = 1 + 2$$

$$x = 3$$

Solution(s): (-1, -3) and (3, 1)

$$6. \begin{cases} y = -x^2 + 9x - 13 \\ 3x - y = 5 \end{cases} \rightarrow y = 3x - 5$$

$$3x - 5 = -x^2 + 9x - 13$$

$$0 = -x^2 + 6x - 8$$

$$0 = x^2 - 6x + 8$$

$$0 = (x-4)(x-2)$$

$$\frac{x=4}{x=2}$$

$$y = 3(4) - 5$$

$$y = 7$$

$$y = 3(2) - 5$$

$$y = 1$$

Solution(s): (4, 7) and (2, 1)

$$8. \begin{cases} y^2 - 2x - 10 = 0 \\ x + y = -1 \end{cases}$$

$$\downarrow$$

$$y = -x - 1$$

$$(-x-1)^2 - 2x - 10 = 0$$

$$x^2 + 2x + 1 - 2x - 10 = 0$$

$$x^2 - 9 = 0$$

$$(x+3)(x-3) = 0$$

$$\frac{x=-3}{x=3}$$

$$y = -(-3) - 1$$

$$y = 2$$

$$y = -3 - 1$$

$$y = -4$$

Solution(s): (-3, 2) and (3, -4)

$$9. \begin{cases} y = x^2 + 4x - 29 \\ (y = x^2 + 2x - 45) - 1 \end{cases} \rightarrow \begin{matrix} y = x^2 + 4x - 29 \\ + -y = -x^2 - 2x + 45 \end{matrix}$$

$$0 = 2x + 16$$

$$-16 = 2x$$

$$-8 = x$$

$$y = (-8)^2 + 4(-8) - 29$$

$$y = 64 - 32 - 29$$

$$y = 3$$

Solution(s): (-8, 3)

$$10. \begin{cases} x^2 + y^2 = 41 \\ x^2 - y^2 = 9 \end{cases}$$

$$2x^2 = 50$$

$$x^2 = 25$$

$$x = \pm 5$$

$$5^2 + y^2 = 41$$

$$25 + y^2 = 41$$

$$y^2 = 16$$

$$y = \pm 4$$

$$(-5)^2 + y^2 = 41$$

$$25 + y^2 = 41$$

$$y^2 = 16$$

$$y = \pm 4$$

Solution(s): (5, 4), (5, -4), (-5, 4), (-5, -4)

Unit 9 Test Study Guide

(Conic Sections)

Name: _____

Date: _____ Per: _____

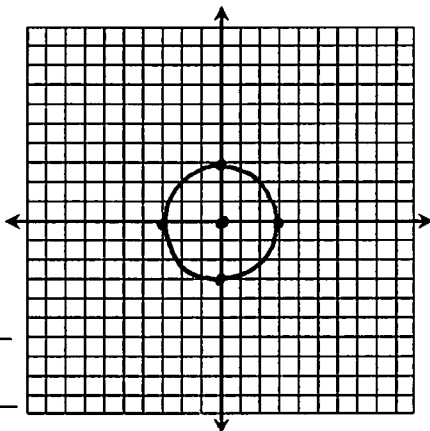
Topic 1: Graphing Circles, Ellipses, Hyperbolas, and Parabolas

1. $\frac{8x^2}{8} + \frac{8y^2}{8} = \frac{72}{8}$

$$x^2 + y^2 = 9$$

Center: $(0, 0)$

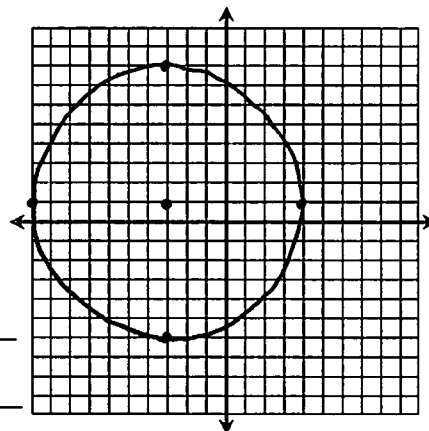
Radius: 3



2. $(x + 3)^2 + (y - 1)^2 = 49$

Center: $(-3, 1)$

Radius: 7



3. $\frac{x^2}{36} + \frac{36y^2}{36} = \frac{36}{36}$

$$\frac{x^2}{36} + y^2 = 1$$

$$c^2 = 36 - 1 = 35$$

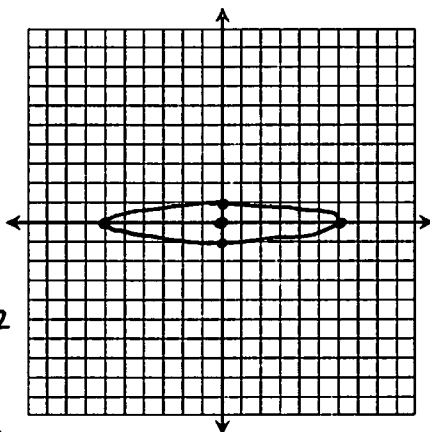
$$c = \sqrt{35} \approx 5.92$$

Center: $(0, 0)$

Vertices: $(\pm 6, 0)$

Co-Vertices: $(0, \pm 1)$

Foci: $(\pm 5.92, 0)$



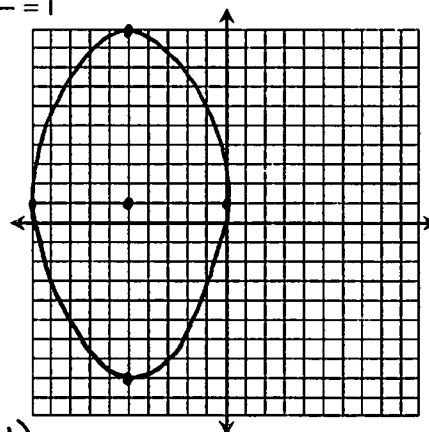
4. $\frac{(x + 5)^2}{25} + \frac{(y - 1)^2}{81} = 1$

Center: $(-5, 1)$

Vertices: $(-5, 10)$ $(-5, -8)$

Co-Vertices: $(10, 1)$ $(0, 1)$

Foci: $(-5, 8.48)$ $(-5, -6.48)$

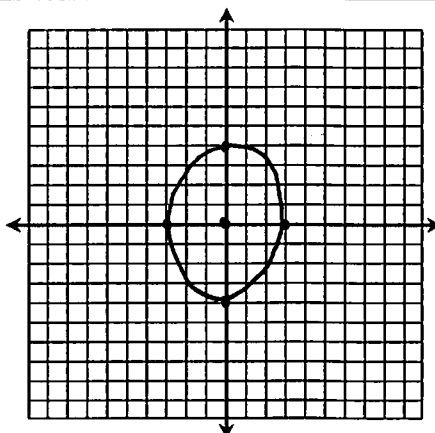


5. $\frac{16x^2}{144} - \frac{9y^2}{144} = \frac{144}{144}$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$c^2 = 9 + 16 = 25$$

$$c = \sqrt{25} = 5$$



Center: $(0, 0)$

Vertices: $(\pm 3, 0)$

Co-Vertices: $(0, \pm 4)$

Foci: $(\pm 5, 0)$

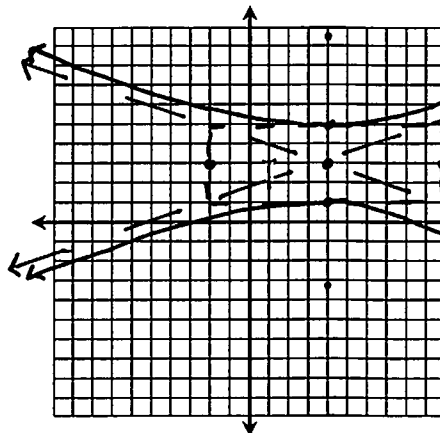
Asymptotes: $y = \frac{4}{3}x$

$y = -\frac{4}{3}x$

$$6. \frac{(y-3)^2}{4} - \frac{(x-4)^2}{36} = 1$$

$$c^2 = 4 + 36 = 40$$

$$c = \sqrt{40} = 2\sqrt{10} \approx 6.3$$



Center: $(4, 3)$

Vertices: $(4, 1)$ $(4, 5)$

Co-Vertices: $(10, 3)$ $(-2, 3)$

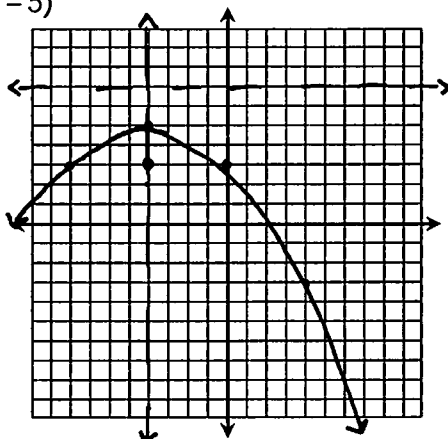
Foci: $(4, 9.3)$ $(4, -3.3)$

Asymptotes: $y = \pm \frac{1}{3}(x-4) + 3 \rightarrow$
 $y = \frac{1}{3}x + \frac{5}{3}$ & $y = -\frac{1}{3}x + \frac{13}{3}$

$$7. (x+4)^2 = -8(y-5)$$

$$-8 = 4p$$

$$-2 = p$$



Vertex: $(-4, 5)$ Axis of Sym: $x = -4$

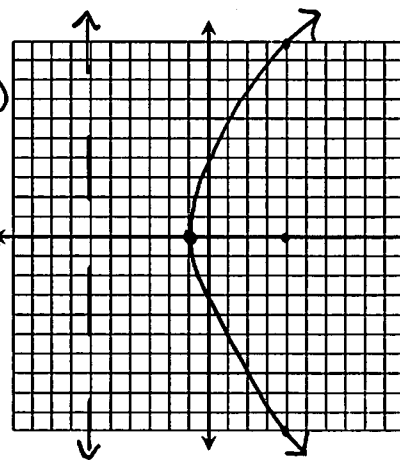
Focus: $(-4, 3)$ Directrix: $y = 7$

$$8. \frac{1}{20}y^2 = x + 1$$

$$y^2 = 20(x+1)$$

$$4p = 20$$

$$p = 5$$



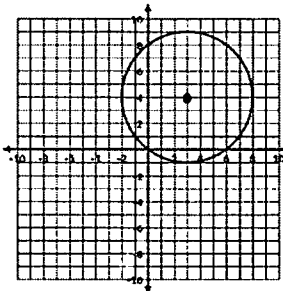
Vertex: $(-1, 0)$ Axis of Sym: $y = 0$

Focus: $(4, 0)$ Directrix: $x = -6$

Topic 2: Writing Equations of Circles, Ellipses, Hyperbolas, and Parabolas

Write the equation of the circle in standard form with the information given.

9.



center: $(3, 4)$
radius: 5

$$(x-3)^2 + (y-4)^2 = 25$$

10. Center: $(-2, 7)$, Radius: $5\sqrt{3}$

$$(x+2)^2 + (y-7)^2 = 75$$

11. Center: $(-4, -1)$, Endpoint: $(2, 3)$

$$\text{rad} = \sqrt{(2+4)^2 + (3+1)^2}$$

$$= \sqrt{6^2 + 4^2}$$

$$= \sqrt{52}$$

$$(x+4)^2 + (y+1)^2 = 52$$

12. Endpoints of Diameter: $(16, -2)$ and $(2, -8)$

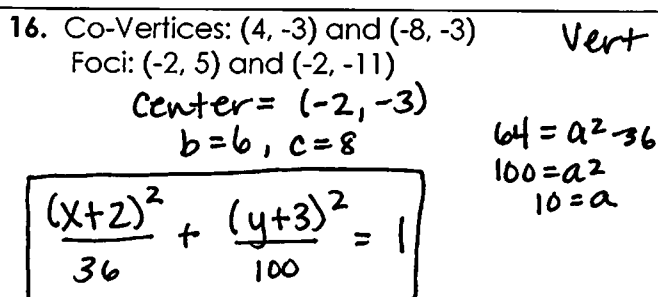
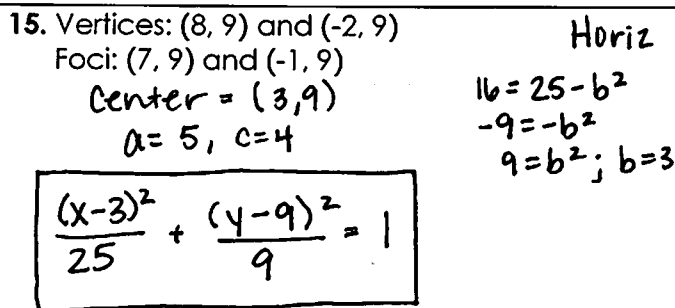
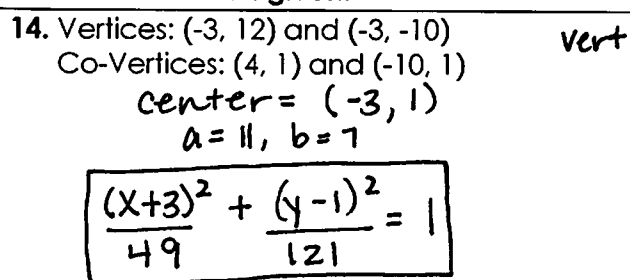
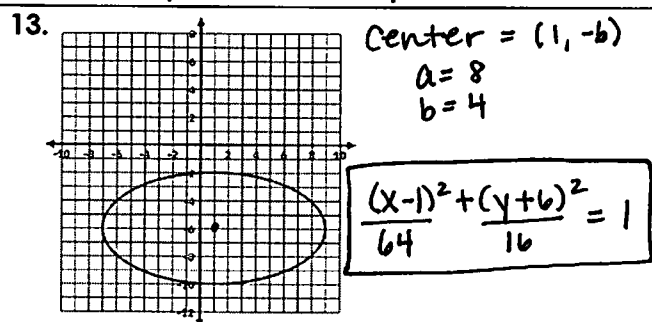
$$\text{center} = \left(\frac{16+2}{2}, \frac{-2-8}{2} \right) = (9, -5)$$

$$\text{radius} = \sqrt{(9-2)^2 + (-5+8)^2}$$

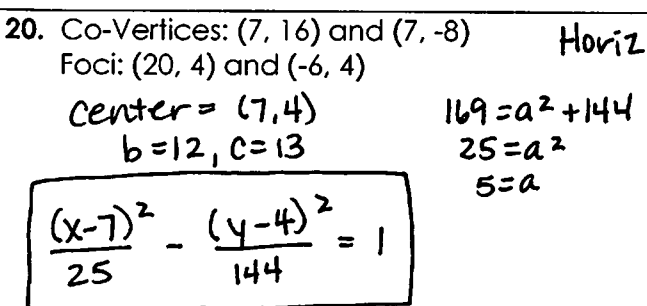
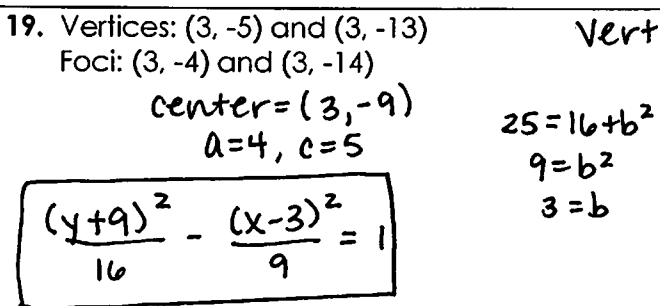
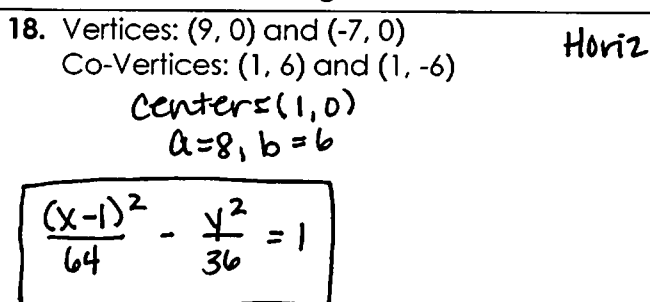
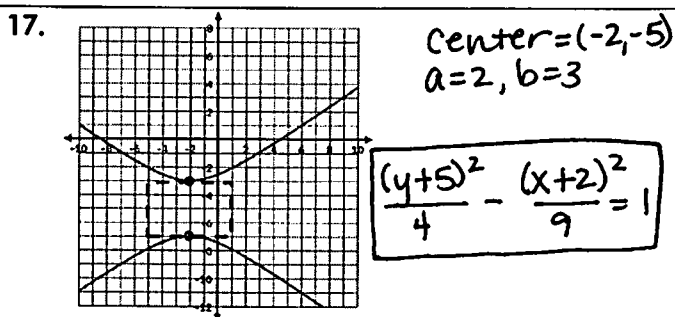
$$= \sqrt{49 + 9} = \sqrt{58}$$

$$(x-9)^2 + (y+5)^2 = 58$$

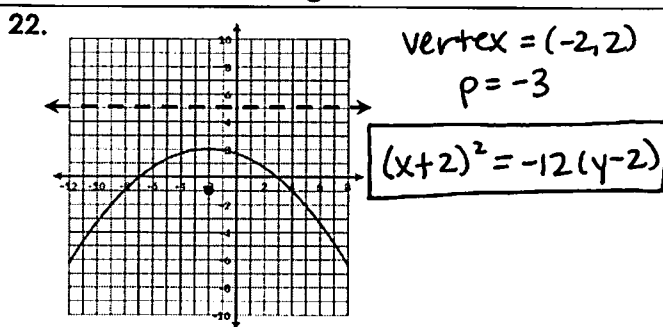
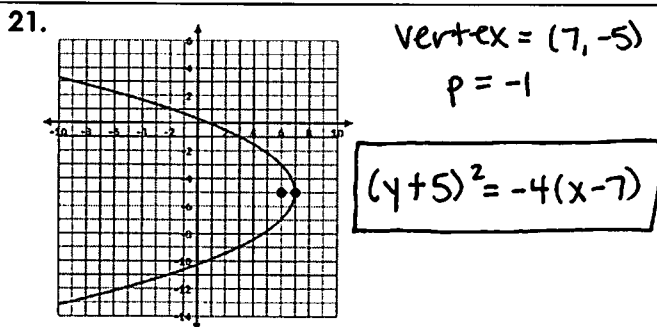
Write the equation of the ellipse in standard form with the information given.



Write the equation of the hyperbola in standard form with the information given.



Write the equation of the parabola in standard form with the information given.



23. Vertex: $(-4, 2)$; Focus: $(2, 2)$ $p = 6$ $(y-2)^2 = 24(x+4)$	24. Vertex: $(1, -3)$; Focus: $(1, -\frac{5}{2})$ $p = \frac{1}{2}$ $(x-1)^2 = 2(y+3)$
25. Vertex: $(6, 0)$; Directrix: $y = 10$ $p = -10$ $(x-6)^2 = -40y$	26. Focus: $(-\frac{11}{2}, -1)$; Directrix: $x = \frac{3}{2}$ Vertex: $(-2, -1)$; $p = -\frac{7}{2}$ $(y+1)^2 = -14(x+2)$

Topic 3: Identifying Conic Sections

Classify the conic, then write the equation in standard form.

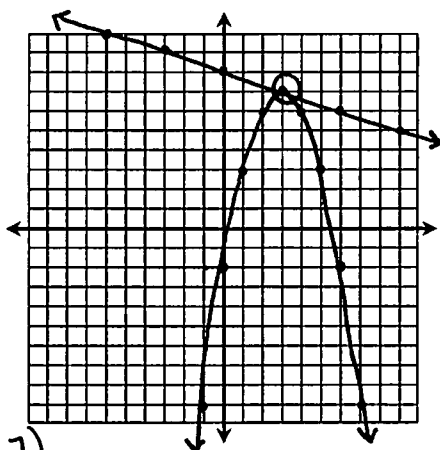
27. $9x^2 + y^2 + 54x - 2y + 46 = 0$ Ellipse $9x^2 + 54x + y^2 - 2y = -46$ $9(x^2 + 6x + 9) + y^2 - 2y + 1 = -46 + 81 + 1$ $9(x+3)^2 + (y-1)^2 = 36$ $\frac{(x+3)^2}{4} + \frac{(y-1)^2}{36} = 1$	28. $x^2 + y^2 - 8x - 4y - 5 = 0$ Circle $x^2 - 8x + y^2 - 4y = 5$ $x^2 - 8x + 16 + y^2 - 4y + 4 = 5 + 16 + 4$ $(x-4)^2 + (y-2)^2 = 25$
29. $x^2 - 10x + 4y + 33 = 0$ Parabola $x^2 - 10x = -4y - 33$ $x^2 - 10x + 25 = -4y - 33 + 25$ $(x-5)^2 = -4y - 8$ $(x-5)^2 = -4(y+2)$	30. $4x^2 - y^2 + 24x - 16y - 32 = 0$ Hyperbola $4x^2 + 24x - y^2 - 16y = 32$ $4(x^2 + 6x + 9) - (y^2 + 16y + 64) = 32 + 36 - 64$ $4(x+3)^2 - (y+8)^2 = 4$ $(x+3)^2 - \frac{(y+8)^2}{4} = 1$
31. $25x^2 + 4y^2 + 100x - 24y + 36 = 0$ Ellipse $25x^2 + 100x + 4y^2 - 24y = -36$ $25(x^2 + 4x + 4) + 4(y^2 - 6y + 9) = -36 + 100 + 36$ $25(x+2)^2 + 4(y-3)^2 = 100$ $\frac{(x+2)^2}{4} + \frac{(y-3)^2}{25} = 1$	32. $y^2 - 10x + 2y + 71 = 0$ Parabola $y^2 + 2y = 10x - 71$ $y^2 + 2y + 1 = 10x - 71 + 1$ $(y+1)^2 = 10x - 70$ $(y+1)^2 = 10(x-7)$

Topic 4: Non-Linear Systems of Equations

Solve by graphing.

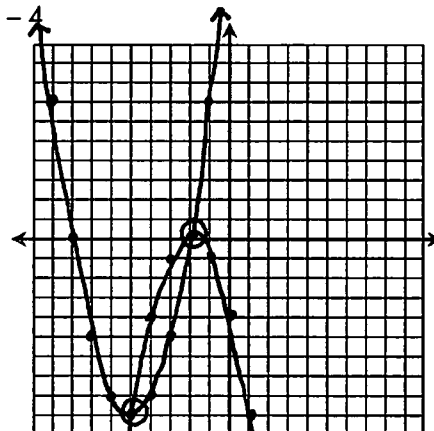
33. $\begin{cases} y = -x^2 + 6x - 2 \\ x + 3y = 24 \end{cases}$

$y = -\frac{1}{3}x + 8$



Solution(s): $(3, 7)$

34. $\begin{cases} y = x^2 + 10x + 16 \\ y = -x^2 - 4x - 4 \end{cases}$



Solution(s): $(-2, 0)$ and $(-5, -9)$

Solve algebraically.

35. $\begin{cases} 3x + y = 1 \\ y = x^2 - 3x \end{cases}$

$3x + x^2 - 3x = 1$

$x^2 = 1$

$x = \pm 1$

$3(1) + y = 1$

$3 + y = 1$

$y = -2$

$3(-1) + y = 1$

$-3 + y = 1$

$y = 4$

Solution(s): $(1, -2)$ and $(-1, 4)$

36. $\begin{cases} y = -x^2 - 6x + 19 \\ y = -x^2 + 2x - 5 \end{cases}$

$-x^2 - 6x + 19 = -x^2 + 2x - 5$

$24 = 8x$

$3 = x$

$y = -(3)^2 - 6(3) + 19$

$y = -9 - 18 + 19$

$y = -8$

Solution(s): $(3, -8)$

37. $\begin{cases} y = 2x^2 + 5x - 3 \\ y = x^2 + 5x - 2 \end{cases} \rightarrow \begin{matrix} y = 2x^2 + 5x - 3 \\ -y = -x^2 - 5x + 2 \\ \hline 0 = x^2 - 1 \end{matrix}$

$0 = x^2 - 1$

$0 = (x+1)(x-1)$

$x = -1 \quad | \quad x = 1$

$y = (-1)^2 + 5(-1) - 2$

$y = 1 - 5 - 2$

$y = -6$

$y = 1^2 + 5(1) - 2$

$y = 1 + 5 - 2$

$y = 4$

Solution(s): $(-1, -6)$ and $(1, 4)$

38. $\begin{cases} x^2 + y^2 = 29 \\ x - y = -7 \rightarrow x = y - 7 \end{cases}$

$(y-7)^2 + y^2 = 29$

$y^2 - 14y + 49 + y^2 = 29$

$2y^2 - 14y + 20 = 0$

$2(y^2 - 7y + 10) = 0$

$2(y-5)(y-2) = 0$

$y = 5 \quad | \quad y = 2$

$x = 5 - 7$

$x = -2$

$x = 2 - 7$

$x = -5$

Solution(s): $(-2, 5)$ and $(-5, 2)$

Name: _____

Date: _____ Per: _____

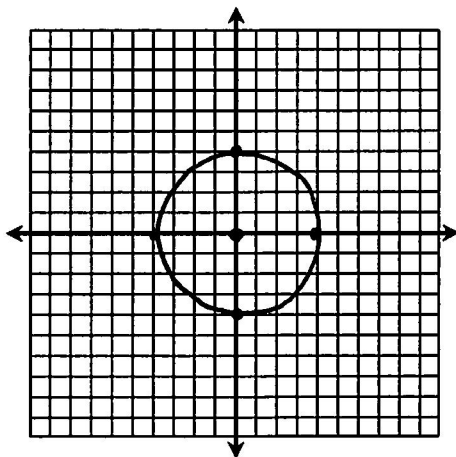
Unit 9 Test

Conic Sections

Graph each equation and identify its key characteristics.

1. $\frac{5x^2}{5} + \frac{5y^2}{5} = \frac{80}{5}$

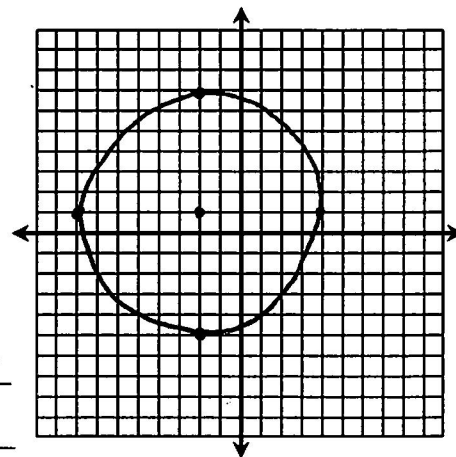
$x^2 + y^2 = 16$



Center: (0,0)

Radius: 4

2. $(x+2)^2 + (y-1)^2 = 36$



Center: (-2,1)

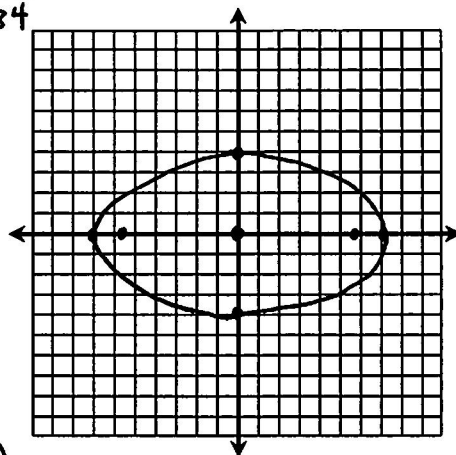
Radius: 6

3. $\frac{16x^2}{784} + \frac{49y^2}{784} = \frac{784}{784}$

$\frac{x^2}{49} + \frac{y^2}{16} = 1$

$a=7$
 $b=4$

$c^2 = 49 - 16 = 33$
 $c = \sqrt{33} \approx 5.74$



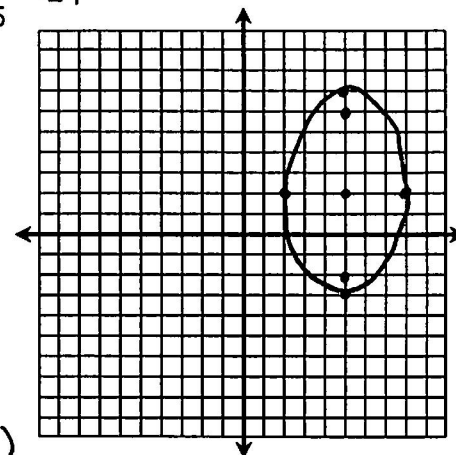
Center: (0,0)

Vertices: (±7,0)

Co-Vertices: (0,±4)

Foci: (±5.74,0)

4. $\frac{(x-5)^2}{9} + \frac{(y-2)^2}{25} = 1$



Center: (5,2)

Vertices: (5,7), (5,-3)

Co-Vertices: (2,2), (8,2)

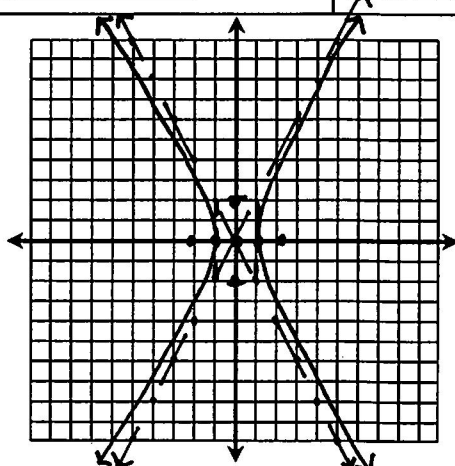
Foci: (5,6), (5,-2)

5. $\frac{4x^2}{4} - \frac{y^2}{4} = \frac{4}{4}$

$x^2 - \frac{y^2}{4} = 1$

$a=1, b=2$

$c^2 = 4 + 1 = 5$
 $c = \sqrt{5} \approx 2.24$



Center: (0,0)

Vertices: (±1,0)

Co-Vertices: (0,±2)

Foci: (±2.24,0)

Asymptotes: $y=2x$ & $y=-2x$

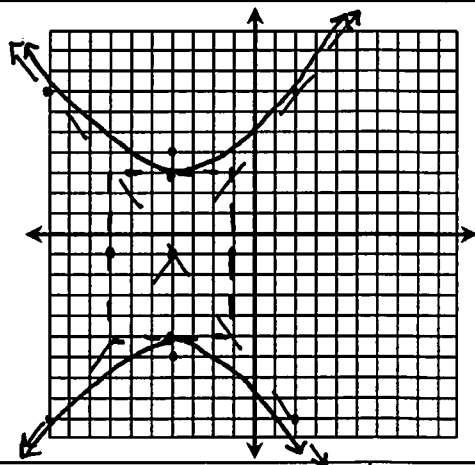
$$6. \frac{(y+1)^2}{16} - \frac{(x+4)^2}{9} = 1$$

$$a=16, b=9$$

$$c^2 = 16+9 = 25$$

$$c = \sqrt{25} = 5$$

$$y = \pm \frac{4}{3}(x+4) - 1$$



Center: $(-4, -1)$

Vertices: $(-4, 3), (-4, -5)$

Co-Vertices: $(-7, -1), (-1, -1)$

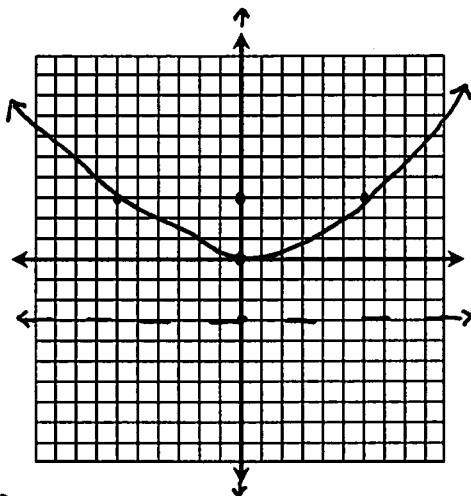
Foci: $(-4, 4), (-4, -6)$

Asymptotes: $y = \frac{4}{3}x + \frac{13}{3}$ &
 $y = -\frac{4}{3}x - \frac{19}{3}$

$$7. x^2 = 12y$$

$$12 = 4p$$

$$3 = p$$



Vertex: $(0, 0)$ Axis of Sym: $x = 0$

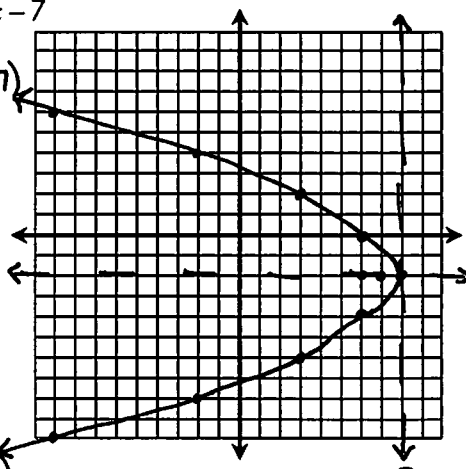
Focus: $(0, 3)$ Directrix: $y = -3$

$$8. -\frac{1}{4}(y+2)^2 = x-7$$

$$(y+2)^2 = -4(x-7)$$

$$-4 = 4p$$

$$-1 = p$$

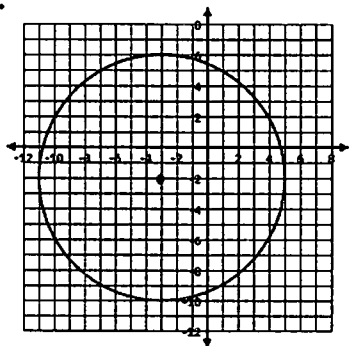


Vertex: $(7, -2)$ Axis of Sym: $y = -2$

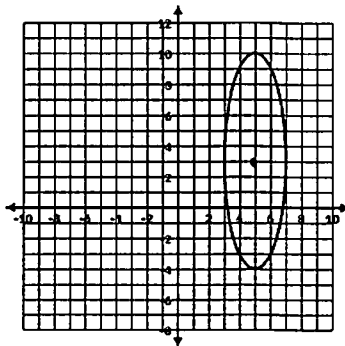
Focus: $(6, -2)$ Directrix: $x = 8$

Write the standard form of the equation given the graph.

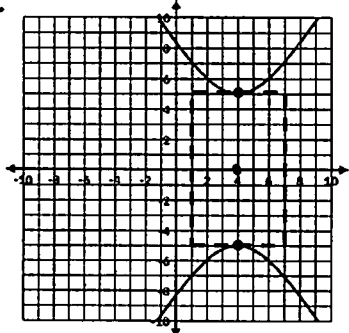
9.



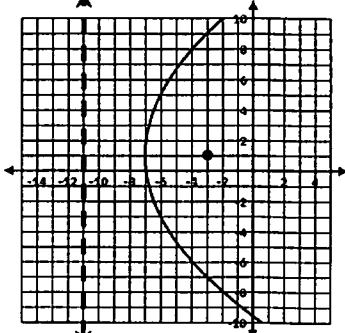
10.



11.



12.



9. $(x+3)^2 + (y+2)^2 = 16$

10. $\frac{(x-5)^2}{16} + \frac{(y-3)^2}{4} = 1$

11. $\frac{y^2}{25} - \frac{(x-4)^2}{9} = 1$

12. $(y-1)^2 = 16(x+7)$

Write the equation of the circle in standard form with the given information.

13. Center: (1, -4), Point on Circle: (-9, -2)

$$\begin{aligned} \text{radius} &= \sqrt{(-9-1)^2 + (-2+4)^2} \\ &= \sqrt{100 + 4} \\ &= \sqrt{104} \end{aligned}$$

Equation: $(x-1)^2 + (y+4)^2 = 104$

14. Endpoints of Diameter: (-7, -4) and (3, -8)

$$\text{Center} = (-2, -6)$$

$$\begin{aligned} \text{radius} &= \sqrt{(-7+2)^2 + (-4+6)^2} \\ &= \sqrt{25 + 4} \\ &= \sqrt{29} \end{aligned}$$

Equation: $(x+2)^2 + (y+6)^2 = 29$

Write the equation of the ellipse in standard form with the given information.

15. Vertices: (-2, -3) and (12, -3)
Co-Vertices: (5, -1) and (5, -5)

$$\text{Center} = (5, -3)$$

$$a=7, b=2$$

Equation: $\frac{(x-5)^2}{49} + \frac{(y+3)^2}{4} = 1$

16. Co-Vertices: (-3, 2) and (-11, 2)
Foci: (-7, 5) and (-7, -1)

$$\text{Center} = (-7, 2)$$

$$b=4, c=3$$

$$\begin{aligned} a^2 &= b^2 + c^2 \\ 25 &= a^2 \\ 5 &= a \end{aligned}$$

Equation: $\frac{(x+7)^2}{16} + \frac{(y-2)^2}{25} = 1$

Write the equation of the hyperbola in standard form with the given information.

17. Vertices: (4, 6) and (-4, 6)
Co-Vertices: (0, -3) and (0, 15)

$$\text{Center} = (0, 6)$$

$$a=4, b=9$$

Equation: $\frac{x^2}{16} - \frac{(y-6)^2}{81} = 1$

18. Vertices: (2, -7) and (2, 9)
Foci: (2, -9) and (2, 11)

$$\text{Center} = (2, 1)$$

$$a=8, c=10$$

$$\begin{aligned} 100 &= 64 + b^2 \\ 36 &= b^2 \\ 6 &= b \end{aligned}$$

Equation: $\frac{(y-1)^2}{64} - \frac{(x-2)^2}{36} = 1$

Write the equation of the parabola in standard form with the given information.

19. Vertex: (-4, 5); Focus: (-7, 5)

$$\begin{aligned} &\text{Horizontal} \\ &p = -3 \end{aligned}$$

Equation: $(y-5)^2 = -12(x+4)$

20. Focus: (-8, 2); Directrix: $y = 0$

$$\begin{aligned} &\text{Vertical} \\ &p = 1 \\ &\text{Vertex} = (-8, 1) \end{aligned}$$

Equation: $(x+8)^2 = 4(y-1)$

Classify each conic and write the equation in standard form.

21. $x^2 - 8x + 2y + 10 = 0$

$$x^2 - 8x = -2y - 10$$

$$x^2 - 8x + 16 = -2y - 10 + 16$$

$$(x-4)^2 = -2y + 6$$

$$(x-4)^2 = -2(y-3)$$

- ☐ Circle
☐ Ellipse
☐ Hyperbola
☒ Parabola

22. $x^2 + y^2 + 4x - 10y - 52 = 0$

$$x^2 + 4x + y^2 - 10y = 52$$

$$x^2 + 4x + 4 + y^2 - 10y + 25$$

$$= 52 + 4 + 25$$

$$(x+2)^2 + (y-5)^2 = 81$$

- ☒ Circle
☐ Ellipse
☐ Hyperbola
☐ Parabola

Equation: $(x-4)^2 = -2(y-3)$

Equation: $(x+2)^2 + (y-5)^2 = 81$

23. $-9x^2 + 16y^2 - 32y - 128 = 0$

$$-9x^2 + 16y^2 - 32y = 128$$

$$-9x^2 + 16(y^2 - 2y + 1) = 128 + 16$$

$$-9x^2 + 16(y-1)^2 = 144$$

$$-\frac{x^2}{16} + \frac{(y-1)^2}{9} = 1$$

$$\frac{(y-1)^2}{9} - \frac{x^2}{16} = 1$$

Equation: $\frac{(y-1)^2}{9} - \frac{x^2}{16} = 1$

24. $4x^2 + y^2 - 16x - 6y + 9 = 0$

$$4x^2 - 16x + y^2 - 6y = -9$$

$$4(x^2 - 4x + 4) + y^2 - 6y + 9 =$$

$$-9 + 16 + 9$$

$$4(x-2)^2 + (y-3)^2 = 16$$

$$\frac{(x-2)^2}{4} + \frac{(y-3)^2}{16} = 1$$

- ☐ Circle
☒ Ellipse
☐ Hyperbola
☐ Parabola

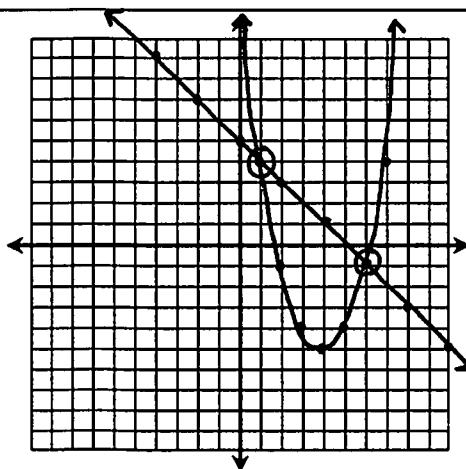
Equation: $\frac{(x-2)^2}{4} + \frac{(y-3)^2}{16} = 1$

Solve the system by graphing.

25. $\begin{cases} y = x^2 - 8x + 11 \\ x + y = 5 \end{cases}$

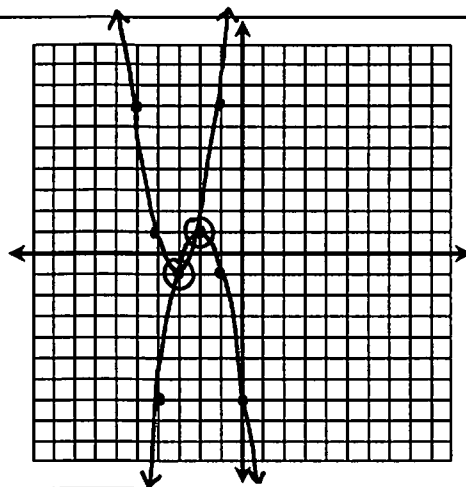
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$$y = -x + 5$$



Solution(s):
 $(1, 4)$ and
 $(6, -1)$

26. $\begin{cases} y = 2x^2 + 12x + 17 \\ y = -2x^2 - 8x - 7 \end{cases}$



Solution(s):
 $(-3, -1)$
 and
 $(-2, 1)$

Solve the system algebraically.

27. $\begin{cases} y = -x^2 + 7x - 15 \\ 2x - y = 11 \end{cases} \rightarrow y = 2x - 11$

$$2x - 11 = -x^2 + 7x - 15$$

$$0 = -x^2 + 5x - 4$$

$$0 = x^2 - 5x + 4$$

$$0 = (x-4)(x-1)$$

$$\begin{array}{c|c} x=4 & x=1 \end{array}$$

$$y = 2(4) - 11$$

$$y = -3$$

$$y = 2(1) - 11$$

$$y = -9$$

Solution(s): $(4, -3)$ and $(1, -9)$

28. $\begin{cases} y = x^2 - 7x - 11 \\ y = x^2 - 4x - 35 \end{cases} \rightarrow \begin{array}{l} y = x^2 - 7x - 11 \\ - (y = x^2 - 4x - 35) \end{array} \rightarrow \begin{array}{l} y = x^2 - 7x - 11 \\ -y = -x^2 + 4x + 35 \\ \hline 0 = -3x + 24 \end{array}$

$$3x = 24$$

$$x = 8$$

$$y = 8^2 - 7(8) - 11$$

$$y = 64 - 56 - 11$$

$$y = -3$$

Solution(s): $(8, -3)$

29. $\begin{cases} y = 2x^2 - 23 \\ y = x^2 + x - 11 \end{cases} \rightarrow \begin{array}{l} y = 2x^2 - 23 \\ - (y = x^2 + x - 11) \end{array} \rightarrow \begin{array}{l} y = 2x^2 - 23 \\ -y = -x^2 - x + 11 \\ \hline 0 = x^2 - x - 12 \end{array}$

$$0 = (x-4)(x+3)$$

$$\begin{array}{c|c} x=4 & x=-3 \end{array}$$

$$y = 2(4)^2 - 23$$

$$y = 2(16) - 23$$

$$y = 9$$

$$y = 2(-3)^2 - 23$$

$$y = 2(9) - 23$$

$$y = -5$$

Solution(s): $(4, 9)$ and $(-3, -5)$

30. $\begin{cases} x^2 + y^2 = 53 \\ x - y = -5 \end{cases} \rightarrow x = y - 5$

$$(y-5)^2 + y^2 = 53$$

$$y^2 - 10y + 25 + y^2 = 53$$

$$2y^2 - 10y - 28 = 0$$

$$2(y^2 - 5y - 14) = 0$$

$$2(y-7)(y+2) = 0$$

$$\begin{array}{c|c} y=7 & y=-2 \end{array}$$

$$x = 7 - 5$$

$$x = 2$$

$$x = -2 - 5$$

$$x = -7$$

Solution(s): $(2, 7)$ and $(-7, -2)$

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!